

PROF. MUHAMMAD SHOAIB GHUMMAN

Exercise 7.1

Chapter No. 7

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نیز گزرا ہل 511-B سٹاٹ: ڈائن کرنا ہل

①

$$y = \frac{(x-2)^2}{x^2}$$

$$x^2 y = (x-2)^2$$

$$x^2 y = x^2 + 4 - 4x$$

$$x^2 y - x^2 - 4 + 4x = 0$$

Horizontal Asymptotes:-

Equate the coefficient of highest power of x to zero.

$$\text{i.e. } y - 1 = 0$$

$$y = 1$$

is an asymptote parallel to x -axis

Vertical Asymptotes

Equate the

coefficient of highest power of y to zero.

$$\text{i.e. } x^2 = 0$$

$$x = 0 \text{ or } x = 0$$

is an asymptote parallel to y -axis

②

$$x^2 y^2 = 12(x-3)$$

$$x^2 y^2 = 12x - 36$$

$$x^2 y^2 - 12x + 36 = 0$$

Horizontal Asymptotes

Equate the

coefficient of highest power of x to zero.

$$\text{i.e. } y^2 = 0$$

$$\text{or } y = 0 \text{ or } y = 0$$

are asymptotes parallel to x -axis.

Vertical Asymptotes:-

Equate the coefficient of highest power of y to zero

$$\text{i.e. } x^2 = 0$$

$$x = 0 \text{ or } x = 0$$

are the asymptotes parallel to y -axis.

$$2xy = x^2 + 3$$

$$2xy - x^2 - 3 = 0$$

Horizontal Asymptotes:-

Equate the

coefficient of highest power of x to zero.

$$\text{i.e. } -1 = 0 \text{ (meaningless)}$$

-1 cannot be equated to zero.

therefore, there is no horizontal asymptote.

Vertical Asymptotes:-

Equate the

coefficient of highest power of y to zero.

$$\text{i.e. } 2x = 0$$

$$x = 0$$

is an asymptote || to y -axis

Oblique / Inclined Asymptotes

Let $y = mx + c$ be the

equation of inclined asymptotes.

$$\text{Let } \phi_2(x, y) = 2xy - x^2$$

$$\phi_1(x, y) = 0$$

$$\text{put } x = 1, y = m$$

$$\phi_2(m) = 2m - 1$$

$$\phi_2'(m) = 2$$

$$\phi_1'(m) = 0$$

To find m :- put $\phi_2(m) = 0$

$$2m - 1 = 0$$

$$2m = 1$$

$$m = \frac{1}{2}$$

To find c :- $c \phi_2'(m) + \phi_1(m) = 0$

$$2c + 0 = 0$$

$$c = 0$$

Hence $y = mx + c$

$$y = \frac{1}{2}x + 0$$

$$y = \frac{1}{2}x$$

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Exercise 7.1

Chapter No. 7

(3)

$$y = \frac{(x-2)^2}{x^2}$$

$$x^2 y = (x-2)^2$$

$$x^2 y = x^2 + 4 - 4x$$

$$x^2 y - x^2 - 4 + 4x = 0$$

Horizontal Asymptotes:-

Equate the coefficient of highest power of x to zero.

$$\text{i.e. } y - 1 = 0$$

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is an asymptote parallel to x -axis

Vertical Asymptotes

Equate the

coefficient of highest power of y to zero.

$$\text{i.e. } x^2 = 0$$

$$x = 0 \text{ or } x = 0$$

is an asymptote parallel to y -axis

(2)

$$x^2 y^2 = 12(x-3)$$

$$x^2 y^2 = 12x - 36$$

$$x^2 y^2 - 12x + 36 = 0$$

Horizontal Asymptotes

Equate the

coefficient of highest power of x to zero.

$$\text{i.e. } y^2 = 0$$

$$\text{or } y = 0 \text{ or } y = 0$$

are asymptotes parallel to x -axis.

Vertical Asymptotes:-

Equate the coefficient of highest power of y to zero

$$\text{i.e. } x^2 = 0$$

$$x = 0 \text{ or } x = 0$$

are the asymptotes parallel to y -axis.

$$2xy = x^2 + 3$$

$$2xy - x^2 - 3 = 0$$

Horizontal Asymptotes:-

Equate the

coefficient of highest power of x to zero.

$$\text{i.e. } -1 = 0 \text{ (meaningless)}$$

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Vertical Asymptotes:-

Equate the

coefficient of highest power of y to zero.

$$\text{i.e. } 2x = 0$$

$$x = 0$$

is an asymptote || to y -axis is oblique / inclined Asymptotes

Let $y = mx + c$ be the

equation of inclined asymptotes.

$$\text{let } \phi_2(x, y) = 2xy - x^2$$

$$\phi_1(x, y) = 0$$

$$\text{put } x = 1, y = m$$

$$\phi_2(m) = 2m - 1$$

$$\phi_2'(m) = 2$$

$$\phi_1'(m) = 0$$

To find m :- put $\phi_2(m) = 0$

$$2m - 1 = 0$$

$$2m = 1$$

$$m = \frac{1}{2}$$

To find c :- $c \phi_2'(m) + \phi_1(m) = 0$

$$2c + 0 = 0$$

$$c = 0$$

Hence $y = mx + c$

$$y = \frac{1}{2}x + 0$$

$$y = \frac{1}{2}x$$

$$\begin{aligned} (4) \quad & x^2(x-y)^2 + a^2(x^2-y^2) - a^2xy = 0 \\ & x^2(x^2+y^2-2xy) + a^2x^2 - a^2y^2 - a^2xy = 0 \\ & x^4 + x^2y^2 - 2x^3y + a^2x^2 - a^2y^2 - a^2xy = 0 \end{aligned}$$

Horizontal Asymptotes:- Equate the Coefficient of highest power of x to zero.

$$1 = 0 \text{ (Meaningless)}$$

$\therefore$, there is no horizontal asymptote.

Vertical Asymptotes:- Equate the Coeff. of highest power of y to zero.

$$\begin{aligned} \text{i.e.} \quad & x^2 - a^2 = 0 \\ & x^2 = a^2 \\ & x = \pm a \end{aligned}$$

are asymptotes || to y -axis

Inclined asymptotes:-

let $y = mx + c$ be the eq. of oblique asymptotes.

$$\text{let } \phi_4(x, y) = x^4 + x^2y^2 - 2x^3y$$

$$\phi_3(x, y) = 0$$

$$\phi_2(x, y) = a^2x^2 - a^2y^2 - a^2xy$$

$$\text{put } x=1, y=m$$

$$\phi_4(m) = 1 + m^2 - 2m$$

$$\phi_3(m) = 0$$

$$\phi_2(m) = a^2 - a^2m^2 - a^2m$$

$$\phi_4'(m) = 2m - 2$$

$$\phi_4''(m) = 2$$

$$\phi_3'(m) = 0$$

$$\begin{aligned} \text{To find } m:- \quad & \text{put } \phi_4(m) = 0 \\ & m^2 - 2m + 1 = 0 \\ & (m-1)^2 = 0 \end{aligned}$$

$$m-1=0 \text{ or } m-1=0$$

$$m=1, m=1$$

$$\text{To find } c:- \frac{c^2}{2} \phi_4'(m) + c \phi_3'(m) + \phi_2(m) = 0$$

$$\frac{c^2}{2} \cdot (2) + c(0) + a^2 - a^2m^2 - a^2m = 0$$

$$\text{put } m=1$$

$$c^2 + a^2 - a^2 - a^2 = 0$$

$$c^2 = a^2$$

$$c = \pm a$$

$$\therefore y = mx + c$$

$$y = x \pm a$$

$$\begin{aligned} (5) \quad & (x-y)^2(x^2+y^2) - 10(x-y)x^2 + \\ & 12y^2 + 2x + y = 0 \end{aligned}$$

$$\begin{aligned} & (x^2+y^2-2xy)(x^2+y^2) - 10x^3 + 10x^2y + \\ & 12y^2 + 2x + y = 0 \end{aligned}$$

$$\begin{aligned} & x^4 + x^2y^2 + x^2y^2 + y^4 - 2x^3y - 2xy^3 - 10x^3 \\ & + 10x^2y + 12y^2 + 2x + y = 0 \end{aligned}$$

Horizontal Asymptotes:- Equate the Coeff. of highest power of x to zero.

$$\text{i.e. } 1 = 0 \text{ (meaningless)}$$

$\therefore$, there is no horizontal asymptotes.

Vertical Asymptotes:- Equate the Coeff. of highest power of y to zero.

$$\text{i.e. } 1 = 0 \text{ (Meaningless)}$$

$\therefore$ there is no vertical asymptotes.

Inclined Asymptotes:-

let $y = mx + c$ be the equation of inclined asymptotes.

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$$\text{let } \phi_4(x, y) = x^4 + x^2y^2 - 2x^3y + x^2y^2 + y^4 - 2xy^3$$

$$\phi_3(x, y) = -10x^3 + 10x^2y$$

$$\phi_2(x, y) = 12y^2$$

put $x=1, y=m$

$$\phi_4(m) = 1 + m^2 - 2m + m^2 + m^4 - 2m^3$$

$$\phi_4(m) = 1 + 2m^2 - 2m + m^4 - 2m^3$$

$$\phi_4'(m) = 4m - 2 + 4m^3 - 6m^2$$

$$\phi_4''(m) = 4 + 12m^2 - 12m$$

$$\phi_3(m) = -10 + 10m$$

$$\phi_3'(m) = 10$$

$$\phi_2(m) = 12m^2$$

to find m : - put $\phi_4(m) = 0$

$$1 + m^2 - 2m + m^2 + m^4 - 2m^3 = 0$$

$$(1 + m^2 - 2m) + m^2(1 + m^2 - 2m) = 0$$

$$(1 + m^2 - 2m)(1 + m^2) = 0$$

$$(1 - m)^2(1 + m^2) = 0$$

$$(1 - m)^2 = 0 \quad \text{or} \quad 1 + m^2 = 0$$

$$m^2 = -1$$

$$(1 - m)(1 - m) = 0 \quad (\text{neglect})$$

$$1 - m = 0 \quad \text{or} \quad 1 - m = 0$$

$$m = 1, \quad m = 1$$

To find C : - $\frac{C^2}{2} \phi_4''(m) + C \phi_3'(m) + \phi_2(m) = 0$

$$\frac{C^2}{2} (12m^2 - 12m + 4) + C(10) + 12m^2 = 0$$

put $m = 1$

$$\frac{C^2}{2} (4) + 10C + 12 = 0$$

$$2C^2 + 10C + 12 = 0$$

$$C^2 + 5C + 6 = 0$$

$$(C + 2)(C + 3) = 0$$

$$C + 2 = 0 \quad \text{or} \quad C + 3 = 0$$

$$C = -2$$

$$C = -3$$

now $y = mx + C$

when $C = -2, m = 1 \Rightarrow y = x - 2$

and

when $C = -3, m = 1 \Rightarrow y = x - 3$

⑥ $x^2y + xy^2 + xy + y^2 + 3x = 0$

Horizontal Asymptotes: - Equate the coefficient of highest power of x to zero.

i.e. $y = 0$

is an asymptote || to x -axis

Vertical Asymptotes: - Equate the coefficient of highest power of y to zero.

i.e. $x + 1 = 0$

$$x = -1$$

is an asymptote || to y -axis

Inclined Asymptotes

let $y = mx + C$ be the equation of oblique asymptotes.

let $\phi_3(x, y) = x^2y + xy^2$

$$\phi_2(x, y) = xy + y^2$$

$$\phi_1(x, y) = 3x$$

put $x=1, y=m$

$$\phi_3(m) = m + m^2$$

$$\phi_3'(m) = 1 + 2m$$

$$\phi_2(m) = m + m^2$$

$$\phi_1(m) = 3$$

To find m : - put $\phi_3(m) = 0$

$$m^2 + m = 0$$

$$m(m + 1) = 0$$

$$m = 0 \quad \text{or} \quad m + 1 = 0$$

(neglect)

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$$m+1=0$$

$$m=-1$$

To find C: $C \phi'_3(m) + \phi_2(m) = 0$

$$C(1+2m) + m + m^2 = 0$$

put $m=-1$

$$C(1-2) + -1 + (1) = 0$$

$$C = 0$$

$$\therefore y = mx + C$$

$$y = -x$$

⑦

$$(x-y+1)(x-y-2)(x+y) = 8x-1$$

$$(x^2 - xy - 2x - xy + y^2 + 2y - x - y - 2)(x+y)$$

$$(x+y) = 8x-1$$

$$(x^2 - 2xy - x + y^2 - y - 2)(x+y) = 8x-1$$

$$x^3 - 2x^2y - x^2 + xy^2 - xy - 2x +$$

$$x^2y - 2xy^2 - xy + y^3 - y^2 - 2y - 8x + 1 = 0$$

$$x^3 - x^2y - xy^2 - x^2 - 2xy - 10x$$

$$- 2y + 1 - y^2 + y^3 = 0$$

Horizontal Asymptotes:- Equate

the coefficient of highest power

of x to zero.

i.e. $1=0$ (meaningless)

$\therefore$ there is no asymptote

parallel to x -axis

Vertical Asymptotes:- Equate the

coefficient of highest power of

y to zero.

i.e. $1=0$ (meaningless)

$\therefore$ there is no asymptote

parallel to y -axis

Inclined Asymptotes:-

Let $y = mx + C$ be the equation of oblique asymptotes.

$$\phi_3(x, y) = x^3 - x^2y - xy^2 + y^3$$

$$\phi_2(x, y) = -x^2 - 2xy + y^2$$

$$\phi_1(x, y) = -10x - 2y$$

Put $x=1, y=m$

$$\phi_3(m) = 1 - m - m^2 + m^3$$

$$\phi'_3(m) = -1 - 2m + 3m^2$$

$$\phi''_3(m) = -2 + 6m$$

$$\phi_2(m) = -1 - 2m + m^2$$

$$\phi'_2(m) = -2 + 2m$$

$$\phi_1(m) = -10 - 2m$$

To find m :- Put $\phi_3(m) = 0$

$$m^3 - m^2 - m + 1 = 0$$

$$m^2(m-1) - 1(m-1) = 0$$

$$(m-1)(m^2-1) = 0$$

$$m-1=0 \quad \text{or} \quad m^2-1=0$$

$$m=1 \quad m^2=1$$

$$m = \pm 1$$

To find C:- when $m=-1$

$$C \phi'_3(m) + \phi_2(m) = 0$$

$$C(3m^2 - 2m - 1) + (-1 - 2m + m^2) = 0$$

put $m=-1$

$$C(3+2-1) + (-1+2+1) = 0$$

$$4C + 2 = 0$$

$$4C = -2$$

$$C = -\frac{1}{2}$$

hence $y = mx + C$

$$y = -x - \frac{1}{2}$$

To find C:- when $m=1, 1$

$$\frac{C^2}{2} \phi''_3(m) + C \phi'_2(m) + \phi_1(m) = 0$$

$$\frac{C^2}{2} (6m-2) + C(-2+2m) - 10-2m = 0$$

put $m=1$

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$$\frac{c^2}{2}(6m-2) + c(0) - 10 - 2 = 0$$

$$2c^2 - 12 = 0$$

$$2c^2 = 12$$

$$c^2 = 6$$

$$c = \pm\sqrt{6}$$

hence $y = mx + c$

$$y = x \pm \sqrt{6}$$

3) $y^3 + x^2y + 2xy^2 - y + 1 = 0$

Horizontal Asymptotes Equate

the coefficient of highest power of x to zero.

i.e. $y = 0$

is an asymptote || to x -axis

Vertical Asymptotes:- Equate

the coefficient of highest power of y to zero.

$1 = 0$ (meaningless)

$\therefore$ there is no asymptote || to y -axis.

Inclined Asymptotes:- Let

$y = mx + c$ be the equation of asymptotes.

Let $\phi_3(x, y) = y^3 + x^2y + 2xy^2$

$$\phi_2(x, y) = 0$$

$$\phi_1(x, y) = -y$$

put $x=1, y=m$

$$\phi_3(m) = m^3 + m + 2m^2$$

$$\phi_3'(m) = 3m^2 + 1 + 4m$$

$$\phi_3''(m) = 6m + 4$$

$$\phi_2(m) = 0$$

$$\phi_1(m) = -m$$

To find m :- put $\phi_3(m) = 0$

$$m^3 + 2m^2 + m = 0$$

$$m(m^2 + 2m + 1) = 0$$

$m = 0$ or $m^2 + 2m + 1 = 0$

(neglect)

$$(m+1)^2 = 0$$

$$(m+1)(m+1) = 0$$

$$m+1 = 0 \text{ or } m+1 = 0$$

$$m = -1 \text{ or } m = -1$$

To find c :-

$$\frac{c^2}{2}\phi_3'(m) + c\phi_2'(m) + \phi_1'(m) = 0$$

$$\frac{c^2}{2}(6m+4) + c(0) - m = 0$$

put $m = -1$

$$-c^2 + 12 = 0$$

$$-c^2 = -1$$

$$c^2 = 1$$

$$c = \pm 1$$

hence $y = mx + c$

$$y = -x \pm 1$$

9) $y(x-y)^2 = x+y$

$$y(x^2 + y^2 - 2xy) = x + y$$

$$x^2y + y^3 - 2xy^2 - x - y = 0$$

Horizontal Asymptotes:- Equate the coefficient of highest power of x to zero.

i.e. $y = 0$

is an asymptote || to x -axis

Vertical Asymptotes:- Equate the

coefficient of highest power of

y to zero i.e. $1 = 0$ (meaningless)

$\therefore$ there is no asymptote || to y -axis.

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Inclined Asymptotes :-

Let $y = mx + c$ be the equation of oblique asymptotes.

Let $\phi_3(x, y) = x^2y + y^3 - 2xy^2$

$$\phi_2(x, y) = 0$$

$$\phi_1(x, y) = -x - y$$

Put $x=1, y=m$

$$\phi_3(m) = m + m^3 - 2m^2$$

$$\phi_3'(m) = 1 + 3m^2 - 4m$$

$$\phi_3''(m) = 6m - 4$$

$$\phi_2(m) = 0$$

$$\phi_1(m) = -1 - m$$

To find m :- $\phi_3(m) = 0$

$$m^3 - 2m^2 + m = 0$$

$$m(m^2 - 2m + 1) = 0$$

$m = 0$ or $m^2 - 2m + 1 = 0$
(neglect) $(m-1)^2 = 0$

$$(m-1)(m-1) = 0$$

$$m-1 = 0 \text{ or } m-1 = 0$$

$$m = 1, 1$$

To find c :-

$$\frac{c^2}{2} \phi_3''(m) + c \phi_2'(m) + \phi_1(m) = 0$$

$$\frac{c^2}{2} (6m - 4) + c(0) - 1 - m = 0$$

Put $m = 1$

$$c^2 = 2$$

$$c = \pm \sqrt{2}$$

hence $y = mx + c$

$$y = x \pm \sqrt{2}$$

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$$\textcircled{10} \quad x^2y^2(x^2-y^2)^2 - (x^2+y^2)^3 = 0$$

$$x^2y^2(x^4+y^4-2x^2y^2) - (x^6+y^6+3x^2y^2x^2y^2) = 0$$

$$x^6y^2 + x^2y^6 - 2x^4y^4 - x^6 - y^6 - 3x^4y^2 - 3x^2y^4 = 0$$

Horizontal Asymptotes :- Equate the coefficient of highest power of  $y$  to zero.

i.e.  $y^2 - 1 = 0$

$$y^2 = 1$$

$$y = \pm 1$$

are the asymptotes parallel to  $x$ -axis.

Vertical Asymptotes :- Equate the coefficient of highest power of  $x$  to zero.

i.e.  $x^2 - 1 = 0$

$$x^2 = 1$$

$$x = \pm 1$$

are the asymptotes parallel to  $y$ -axis.

Oblique Asymptotes :-

Let  $y = mx + c$  be the eq of oblique asymptotes.

Let  $\phi_8(x, y) = x^6y^2 + x^2y^6 - 2x^4y^4$

$$\phi_7(x, y) = 0$$

$$\phi_6(x, y) = -x^6 - y^6 - 3x^4y^2 - 3x^2y^4$$

Put  $x=1, y=m$

$$\phi_8(m) = m^2 + m^6 - 2m^4$$

$$\phi_8'(m) = 2m + 6m^5 - 8m^3$$

$$\phi_8''(m) = 2 + 30m^4 - 24m^2$$

$$\phi_7(m) = 0$$

$$\phi_6(m) = -1 - m^6 - 3m^2 - 3m^4$$



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To find  $m$ :- Put  $\phi_8(m) = 0$

$$m^6 - 2m^4 + m^2 = 0$$

$$m^2(m^4 - 2m^2 + 1) = 0$$

$$m^2 = 0 \text{ or } m^4 - 2m^2 + 1 = 0$$

$$\text{neglect } (m^2 - 1)^2 = 0$$

$$(m^2 - 1)(m^2 - 1) = 0$$

$$m^2 - 1 = 0 \text{ or } m^2 - 1 = 0$$

$$m^2 = 1 \quad m^2 = 1$$

$$m = \pm 1 \quad m = \pm 1$$

To find  $c$ :- When  $m = -1, -1$

$$\frac{c^2}{2} \phi_8''(m) + c \phi_7'(m) + \phi_6(m) = 0$$

$$\frac{c^2}{2} (30m^4 - 24m^2 + 2) + 0 + (-m^6 - 3m^4 - 3m^2 - 1) = 0$$

$$\text{Put } m = -1$$

$$\frac{c^2}{2} (30 - 24 + 2) + (-1 - 3 - 3 - 1) = 0$$

$$\frac{c^2}{2} (8) + (-8) = 0$$

$$4c^2 - 8 = 0$$

$$c^2 - 2 = 0$$

$$c^2 = 2$$

$$c = \pm \sqrt{2}$$

$$\text{hence } y = mx + c$$

$$y = -x \pm \sqrt{2}$$

To find  $c$ :- When  $m = 1, 1$

$$\frac{c^2}{2} \phi_8''(m) + c \phi_7'(m) + \phi_6(m) = 0$$

$$\frac{c^2}{2} (30m^4 - 24m^2 + 2) + 0 + (-m^6 - 3m^4 - 3m^2 - 1) = 0$$

$$\text{Put } m = 1$$

$$\frac{c^2}{2} (30 - 24 + 2) + (-1 - 3 - 3 - 1) = 0$$

$$4c^2 - 8 = 0$$

$$4c^2 = 8$$

$$c^2 = 2$$

$$c = \pm \sqrt{2}$$

$$\text{hence } y = mx + c$$

$$y = x \pm \sqrt{2}$$

$$(11) \quad xy^2 = (x+y)^2$$

$$xy^2 = x^2 + y^2 + 2xy$$

$$xy^2 - x^2 - y^2 - 2xy = 0$$

Horizontal Asymptotes:- Equate the coefficient of highest power of  $x$  to zero.

$$\text{i.e. } -1 = 0 \text{ (meaningless)}$$

$\therefore$ , there is no asymptote  $\parallel$  to  $x$ -axis.

Vertical Asymptotes:- Equate

Coefficient of highest power of  $y$  to zero.

$$\text{i.e. } x - 1 = 0$$

$$x = 1$$

is an asymptote  $\parallel$  to  $y$ -axis

Inclined Asymptotes:-

Let  $y = mx + c$  be the eqn of oblique asymptote

$$\text{Let } \phi_3(x, y) = xy^2$$

$$\phi_2(x, y) = -x^2 - y^2 - 2xy$$

$$\phi_1(x, y) = 0$$

$$\text{Put } x = 1, y = m$$

$$\phi_3(m) = m^2$$

$$\phi_2(m) = -1 - m^2 - 2m$$

$$\phi_1(m) = 0$$

$$\phi_3'(m) = 2m$$

$$\phi_3''(m) = 2$$

$$\phi_2'(m) = -2m - 2$$

$$\text{To find } m:- \text{ Put } \phi_3(m) = 0$$

$$m^2 = 0$$

$$m = 0, 0$$

$$\text{To find } c:- \frac{c^2}{2} \phi_3(m) + c \phi_2'(m) + \phi_1(m)$$

$$= 0$$

$$\text{Put } m = 0$$

$$c^2 - 2c = 0$$

$$c(c - 2) = 0$$

$$c = 0 \text{ or } c - 2 = 0$$

$$c = 2$$

$$\text{hence } y = mx + c$$

$$\text{When } c = 0, y = 0$$

$$\text{When } c = 2, y = 2$$

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(12)  $xy^2 - x^2y - 3x^2 - 2xy + y^2 + x - 2y + 1 = 0$

Horizontal Asymptotes:- Equate the coefficient of highest power of  $x$  to zero.

$$\text{i.e. } -y - 3 = 0$$

$$y + 3 = 0$$

$$y = -3$$

is an asymptote || to  $x$ -axis

Vertical Asymptotes:- Equate the

coefficient of highest power of

$y$  to zero.

$$\text{i.e. } x + 1 = 0$$

$$x = -1$$

is an asymptote || to  $y$ -axis

Inclined Asymptotes:-

Let  $y = mx + c$  be the equation of oblique asymptotes.

$$\text{Let } \phi_3(x, y) = xy^2 - x^2y$$

$$\phi_2(x, y) = -3x^2 - 2xy + y^2$$

$$\phi_1(x, y) = x - 2y$$

$$\text{put } x = 1, y = m$$

$$\phi_3(m) = m^2 - m$$

$$\phi_3'(m) = 2m - 1$$

$$\phi_2(m) = -3 - 2m + m^2$$

$$\phi_1(m) = 1 - 2m$$

$$\text{To find } m:- \text{ put } \phi_3(m) = 0$$

$$m^2 - m = 0$$

$$m(m - 1) = 0$$

$$m = 0 \text{ or } m - 1 = 0$$

$$(\text{we may neglect}) \quad m = 1$$

$$\text{To find } c:- c\phi_3'(m) + \phi_2(m) = 0$$

$$c(2m - 1) + (-3 - 2m + m^2) = 0$$

$$\text{put } m = 1$$

$$c = 4$$

$$\text{hence } y = mx + c$$

$$y = x + 4$$

(13)  $r = \frac{a}{\theta}$

$$\text{as } r \rightarrow \infty, \text{ when } \theta = 0$$

$$\text{i.e. } \alpha = 0$$

$$\text{To find } \frac{d\theta}{dr}$$

$$\frac{dr}{d\theta} = -\frac{a}{\theta^2}$$

$$\frac{d\theta}{dr} = -\frac{\theta^2}{a}$$

$$p = \lim_{\theta \rightarrow 0} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow 0} \frac{a^2}{\theta^2} \left(-\frac{\theta^2}{a}\right)$$

$$p = -a$$

$\therefore$  Eq of asymptote is

$$p = r \sin(\alpha - \theta)$$

$$-a = r \sin(0 - \theta)$$

$$-a = -r \sin \theta$$

$$a = r \sin \theta$$

(14)  $r = \frac{a}{\sqrt{\theta}}$

$$\text{as } r \rightarrow \infty, \text{ when } \theta = 0$$

$$\text{i.e. } \alpha = 0$$

$$\text{To find } \frac{d\theta}{dr} = ?$$

$$r = a \theta^{-1/2}$$

$$\frac{dr}{d\theta} = a \cdot -\frac{1}{2} \theta^{-3/2}$$

$$\frac{dr}{d\theta} = -\frac{a}{2} \theta^{-3/2}$$

$$\frac{dr}{d\theta} = -\frac{a}{2\theta^{3/2}}$$

$$\frac{d\theta}{dr} = \frac{2\theta^{3/2}}{-a}$$

$$p = \lim_{\theta \rightarrow 0} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow 0} \frac{a^2}{\theta} \cdot \frac{2\theta^{3/2}}{-a}$$

$$= \lim_{\theta \rightarrow 0} -2a\theta^{1/2}$$

$$p = 0$$

$\therefore$  Eq of asymptote is

$$p = r \sin(\alpha - \theta)$$

$$0 = r \sin(0 - \theta)$$

$$0 = -r \sin \theta$$

$$r \sin \theta = 0$$

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(15)

$$r = a \cos \theta + b$$

$$r = \frac{a}{\sin \theta} + b$$

$$\text{When } \theta = 0, \pi, r \rightarrow \infty$$

$$r = a \cos \theta + b$$

$$\frac{dr}{d\theta} = -a \cos \theta \cot \theta$$

$$= -a \cdot \frac{1}{\sin \theta} \cdot \frac{\cos \theta}{\sin \theta}$$

$$\frac{dr}{d\theta} = -\frac{a \cos \theta}{\sin^2 \theta}$$

$$\frac{d\theta}{dr} = \frac{\sin^2 \theta}{-a \cos \theta}$$

$$i) p = \lim_{\theta \rightarrow 0} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow 0} \left( \frac{a}{\sin \theta} + b \right)^2 \cdot -\frac{\sin^2 \theta}{a \cos \theta}$$

$$= \lim_{\theta \rightarrow 0} \left( \frac{a + b \sin \theta}{\sin \theta} \right)^2 \cdot -\frac{\sin^2 \theta}{a \cos \theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{(a + b \sin \theta)^2}{\sin^2 \theta} \cdot -\frac{\sin^2 \theta}{a \cos \theta}$$

$$p = -a$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$-a = r \sin(0 - \theta)$$

$$-a = -r \sin \theta$$

$$a = r \sin \theta$$

$$ii) p = \lim_{\theta \rightarrow \pi} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow \pi} \left( \frac{a}{\sin \theta} + b \right)^2 \cdot -\frac{\sin^2 \theta}{a \cos \theta}$$

$$= \lim_{\theta \rightarrow \pi} \left( \frac{a + b \sin \theta}{\sin \theta} \right)^2 \cdot -\frac{\sin^2 \theta}{a \cos \theta}$$

$$= \lim_{\theta \rightarrow \pi} \frac{(a + b \sin \theta)^2}{\sin^2 \theta} \cdot -\frac{\sin^2 \theta}{a \cos \theta}$$

$$p = a$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$a = r \sin(\pi - \alpha)$$

$$a = r \sin \theta$$

$$(16) r = 2a \sin \theta \tan \theta$$

$$\text{when } \theta = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots, r \rightarrow \infty$$

$$\text{As } r = 2a \cdot \sin \theta \cdot \frac{\sin \theta}{\cos \theta}$$

$$r = 2a \frac{\sin^2 \theta}{\cos \theta}$$

$$\frac{dr}{d\theta} = 2a \cdot \frac{\cos \theta \cdot 2 \sin \theta \cos \theta - \sin^2 \theta \cdot (-\sin \theta)}{\cos^2 \theta}$$

$$= 2a \cdot \frac{2 \sin \theta \cos^2 \theta + (1 - \cos^2 \theta) \sin \theta}{\cos^2 \theta}$$

$$= 2a \cdot \frac{2 \sin \theta \cos^2 \theta + \sin \theta - \sin \theta \cos^2 \theta}{\cos^2 \theta}$$

$$= 2a \cdot \frac{\sin \theta \cos^2 \theta + \sin \theta}{\cos^2 \theta}$$

$$\frac{dr}{d\theta} = 2a \cdot \frac{\sin \theta (\cos^2 \theta + 1)}{\cos^2 \theta}$$

$$\frac{d\theta}{dr} = \frac{\cos^2 \theta}{2a \sin \theta (\cos^2 \theta + 1)}$$

$$i) p = \lim_{\theta \rightarrow \pi/2} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow \pi/2} \frac{4a^2 \sin^4 \theta}{\cos^2 \theta} \cdot \frac{\cos^2 \theta}{2a \sin \theta (\cos^2 \theta + 1)}$$

$$p = 2a$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$2a = r \sin(\pi/2 - \theta)$$

$$2a = r \cos \theta$$

$$ii) p = \lim_{\theta \rightarrow -\pi/2} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow -\pi/2} \frac{4a^2 \sin^4 \theta}{\cos^2 \theta} \cdot \frac{\cos^2 \theta}{2a \sin \theta (\cos^2 \theta + 1)}$$

$$= \lim_{\theta \rightarrow -\pi/2} \frac{2a \sin^3 \theta}{(\cos^2 \theta + 1)}$$

$$p = -2a$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$-2a = r \sin(-\pi/2 - \theta)$$

$$-2a = -r \sin(\pi/2 + \theta)$$

$$-2a = -r \cos \theta$$

$$2a = r \cos \theta$$



(17)  $r \sin 2\theta = a \cos 3\theta$

$$r = \frac{a \cos 3\theta}{\sin 2\theta}$$

when  $\theta = 0, \pi/2, \pi, \dots$   $r \rightarrow \infty$

$$\frac{dr}{d\theta} = a \cdot \frac{(\sin 2\theta)(-3\sin 3\theta) - \cos 3\theta \cdot 2\cos 2\theta}{(\sin 2\theta)^2}$$

$$= a \cdot \frac{-3\sin 2\theta \sin 3\theta - 2\cos 3\theta \cos 2\theta}{\sin^2 2\theta}$$

$$\frac{1}{r} \frac{dr}{d\theta} = -a \cdot \frac{(3\sin 2\theta \sin 3\theta + 2\cos 3\theta \cos 2\theta)}{\sin^2 2\theta}$$

$$\frac{d\theta}{dr} = \frac{\sin^2 2\theta}{-a(3\sin 2\theta \sin 3\theta + 2\cos 3\theta \cos 2\theta)}$$

i)  $p = \lim_{\theta \rightarrow 0} r^2 \frac{d\theta}{dr}$

$$= \lim_{\theta \rightarrow 0} - \frac{\sin^2 2\theta}{\sin^2 2\theta} \cdot \frac{a^2 \cos^2 3\theta}{a^2 (3\sin 2\theta \sin 3\theta + 2\cos 3\theta \cos 2\theta)}$$

$$p = -\frac{a^2}{2a}$$

$$p = -\frac{a}{2}$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$-\frac{a}{2} = r \sin(0 - \theta)$$

$$-\frac{a}{2} = -r \sin \theta$$

$$\frac{a}{2} = r \sin \theta$$

ii)  $p = \lim_{\theta \rightarrow \pi/2} r^2 \frac{d\theta}{dr} = \lim_{\theta \rightarrow \pi/2} - \frac{\sin^2 2\theta}{\sin^2 2\theta} \cdot \frac{a^2 \cos^2 3\theta}{a^2 (3\sin 2\theta \sin 3\theta + 2\cos 3\theta \cos 2\theta)}$

$$p = \frac{0}{0} \text{ (Indeterminate form)}$$

$\therefore$  no asymptote.

iii)

$$p = \lim_{\theta \rightarrow \pi/2} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow \pi/2} - \frac{a^2 \cos^2 3\theta}{\sin^2 2\theta} \cdot \frac{\sin^2 2\theta}{a^2 (3\sin 2\theta \sin 3\theta + 2\cos 3\theta \cos 2\theta)}$$

$$p = \frac{-a^2}{-2a}$$

$$p = \frac{a}{2}$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$\frac{a}{2} = r \sin(\pi - \theta)$$

$$\frac{a}{2} = r \sin \theta$$

(18)  $r = \frac{a}{1 - \cos \theta}$

when  $\theta = 0$ ,  $r \rightarrow \infty$

$$r = a(1 - \cos \theta)^{-1}$$

$$\frac{dr}{d\theta} = a \cdot -1(1 - \cos \theta)^{-1-1} (\sin \theta)$$

$$\frac{dr}{d\theta} = -\frac{a \sin \theta}{(1 - \cos \theta)^2}$$

$$\frac{d\theta}{dr} = \frac{(1 - \cos \theta)^2}{-a \sin \theta}$$

$$p = \lim_{\theta \rightarrow 0} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow 0} \frac{a^2}{(1 - \cos \theta)^2} \cdot \frac{(1 - \cos \theta)^2}{-a \sin \theta}$$

$$p = \infty$$

$\therefore$  no asymptote.

(19)  $r \sin n\theta = a$

$$r = \frac{a}{\sin n\theta}$$

when  $n\theta = k\pi$  ( $k = 0, \pm 1, \pm 2, \dots$ ),  $r \rightarrow \infty$

$$\theta = \frac{k\pi}{n}$$

now  $r = a \csc n\theta$

$$\frac{dr}{d\theta} = -a \csc n\theta \cot n\theta \cdot n$$

$$= -a \cdot \frac{1}{\sin n\theta} \cdot \frac{\cos n\theta}{\sin n\theta} \cdot n$$

$$\frac{dr}{d\theta} = -na \cdot \frac{\cos n\theta}{\sin^2 n\theta}$$

$$\frac{d\theta}{dr} = \frac{\sin^2 n\theta}{-na \cos n\theta}$$

$$p = \lim_{\theta \rightarrow \frac{\pi}{n}} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow \frac{\pi}{n}} \left( \frac{a}{\sin n\theta} \right)^2 \cdot \frac{-\sin n\theta}{n a \cos n\theta}$$

$$= \lim_{\theta \rightarrow \frac{\pi}{n}} \frac{a^2}{\sin^2 n\theta} \cdot \frac{-\sin n\theta}{n a \cos n\theta}$$

$$= \lim_{\theta \rightarrow \frac{\pi}{n}} -\frac{a}{n} \sec n\theta$$

$$p = -\frac{a}{n} \sec n\pi$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$-\frac{a}{n} \sec n\pi = r \sin\left(\frac{\pi}{n} - \theta\right)$$

$$\frac{a}{n} \sec n\pi = +r \sin\left(\theta - \frac{\pi}{n}\right)$$

$$(20) \quad r(e^\theta - 1) = a(e^\theta + 1)$$

$$r = \frac{a(e^\theta + 1)}{(e^\theta - 1)}$$

$$\text{when } \theta = 0 \quad r \rightarrow \infty$$

$$\frac{dr}{d\theta} = a \cdot \frac{(e^\theta - 1)(e^\theta) - (e^\theta + 1)(e^\theta)}{(e^\theta - 1)^2}$$

$$= a \cdot \frac{\cancel{2\theta} \cdot \cancel{e} - \cancel{2\theta} \cdot \cancel{e}}{(e^\theta - 1)^2}$$

$$\frac{dr}{d\theta} = \frac{-2ae^\theta}{(e^\theta - 1)^2}$$

$$\frac{d\theta}{dr} = \frac{(e^\theta - 1)^2}{-2ae^\theta}$$

$$p = \lim_{\theta \rightarrow 0} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow 0} \frac{a^2(e^\theta + 1)^2}{(e^\theta - 1)^2} \cdot \frac{(e^\theta - 1)^2}{-2ae^\theta}$$

$$p = -\frac{4a}{2}$$

$$p = -2a$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$-2a = r \sin(0 - \theta)$$

$$-2a = -r \sin \theta$$

$$2a = r \sin \theta$$

$$(21) \quad r^n \sin n\theta = a^n$$

$$r^n = \frac{a^n}{\sin n\theta}$$

$$\text{when } n\theta = k\pi \quad (k=0, \pm 1, \pm 2, \dots)$$

$$r \rightarrow \infty$$

$$\text{now, } r^n \sin n\theta = a^n$$

$$\ln(r^n \sin n\theta) = \ln a^n$$

$$\ln r^n + \ln \sin n\theta = \ln a^n$$

$$n \ln r + \ln \sin n\theta = \ln a^n$$

Diff. w.r.t.  $\theta$

$$n \cdot \frac{1}{r} \frac{dr}{d\theta} + \frac{n \cos n\theta}{\sin n\theta} = 0$$

$$\cancel{n} \cdot \frac{1}{r} \frac{dr}{d\theta} = -\cancel{n} \cdot \frac{\cos n\theta}{\sin n\theta}$$

$$\frac{dr}{d\theta} = -r \frac{\cos n\theta}{\sin n\theta}$$

$$p = \lim_{\theta \rightarrow \frac{\pi}{n}} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow \frac{\pi}{n}} r^2 \cdot \frac{-\sin n\theta}{r \cos n\theta}$$

$$= \lim_{\theta \rightarrow \frac{\pi}{n}} \frac{a}{(\sin n\theta)^{1/n}} \cdot \frac{-\sin n\theta}{\cos n\theta}$$

$$= \lim_{\theta \rightarrow \frac{\pi}{n}} -a (\sin n\theta)^{1-1/n} \cdot \frac{1}{\cos n\theta}$$

$$p = 0$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$0 = r \sin\left(\frac{\pi}{n} - \theta\right)$$

$$\sin\left(\frac{\pi}{n} - \theta\right) = 0$$

$$\frac{\pi}{n} - \theta = 0$$

$$\frac{\pi}{n} = \theta$$

(2.2)

$$r^2 \sin \theta = a^2 \cos 2\theta$$

$$r^2 = \frac{a^2 \cos 2\theta}{\sin \theta}$$

when  $\theta = 0, \pi, 2\pi, \dots$   $r \rightarrow \infty$

now,  $r^2 = \frac{a^2 \cos 2\theta}{\sin \theta}$

Diff. w.r.t  $\theta$

$$2r \frac{dr}{d\theta} = a^2 \cdot \frac{\sin \theta (-2 \sin 2\theta) - \cos 2\theta \cdot \cos \theta}{\sin^2 \theta}$$

$$\frac{dr}{d\theta} = -a^2 \cdot \frac{2 \sin \theta \sin 2\theta + \cos 2\theta \cdot \cos \theta}{2r \sin^2 \theta}$$

$$\frac{d\theta}{dr} = \frac{2r \sin^2 \theta}{-a^2 (2 \sin \theta \sin 2\theta + \cos 2\theta \cdot \cos \theta)}$$

$$\therefore p = \lim_{\theta \rightarrow 0} r^2 \frac{d\theta}{dr}$$

$$= \lim_{\theta \rightarrow 0} \frac{a^2 \cos 2\theta}{\sin \theta} \cdot 2r \frac{\sin^2 \theta}{-a^2 (2 \sin \theta \sin 2\theta + \cos 2\theta \cdot \cos \theta)}$$

$$= \lim_{\theta \rightarrow 0} \frac{a^2 \cos 2\theta}{\sin \theta} \cdot 2 \cdot a \sqrt{\frac{\cos 2\theta}{\sin \theta}} \frac{\sin^2 \theta}{-a^2 (2 \sin \theta \sin 2\theta + \cos 2\theta \cdot \cos \theta)}$$

$$= \lim_{\theta \rightarrow 0} \frac{2a \cos 2\theta \sqrt{\cos 2\theta} \cdot \sqrt{\sin \theta}}{-(2 \sin \theta \sin 2\theta + \cos 2\theta \cos \theta)}$$

$$p = 0$$

$$\therefore p = r \sin(\alpha - \theta)$$

$$0 = r \sin(0 - \theta)$$

$$0 = -r \sin \theta$$

$$-r \sin \theta = 0$$

$$\sin \theta = 0$$

$$\theta = 0$$

$$(ii) p = \lim_{\theta \rightarrow \pi} r^2 \frac{d\theta}{dr}$$

$$p = \lim_{\theta \rightarrow \pi} \frac{2a \cos 2\theta \sqrt{\cos 2\theta} \cdot \sqrt{\sin \theta}}{-(2 \sin \theta \sin 2\theta + \cos 2\theta \cos \theta)}$$

$$p = 0$$

$$\therefore p = r \sin(\pi - \theta)$$

$$0 = r \sin(\pi - \theta)$$

$$0 = r \sin \theta$$

$$\sin \theta = 0$$

$$\theta = \pi$$

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$$① f(x) = 2x^3 - 15x^2 + 36x + 10$$

$$f'(x) = 6x^2 - 30x + 36$$

$$f''(x) = 12x - 30$$

For the critical points

$$f'(x) = 0$$

$$6x^2 - 30x + 36 = 0$$

$$x^2 - 5x + 6 = 0$$

$$(x-2)(x-3) = 0$$

$$x = 2, 3$$

Therefore  $x = 2, 3$  are the critical points where the function has extreme values.

To check where  $f$  has relative maxima or relative minima, we use the 2nd derivative test

$$\text{For } x = 2, f''(x) = 12(2) - 30 = 24 - 30 = -6 < 0$$

$\therefore f(x)$  has relative max. at  $x = 2$

$$\text{For } x = 3, f''(x) = 12(3) - 30 = 36 - 30 = 6 > 0$$

$\therefore f(x)$  has relative min. at  $x = 3$

$$② f(x) = 3x^4 - 4x^3 + 5$$

$$f'(x) = 12x^3 - 12x^2$$

$$f''(x) = 36x^2 - 24x$$

For the critical points

$$f'(x) = 0$$

$$12x^3 - 12x^2 = 0$$

$$12x^2(x-1) = 0$$

$$12x^2 = 0 \text{ or } x - 1 = 0$$

$$x = 0$$

$$x = 1$$

Therefore  $x = 0, 1$  are the critical points where the function has extreme values.

To check where  $f$  has relative maxima or relative minima, we use the 2nd derivative test.



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For  $x=0$

$$f''(x) = 0$$

So 2nd derivative test fails.

Also, we see that  $f'(x)$  does not change sign before and after  $x=0$ .

So  $f(x)$  has no relative extrema at  $x=0$

For  $x=1$

$$f''(x) = 36 - 24 = 12 > 0$$

$f(x)$  has relative min at  $x=1$

$$f(x) = (x-1)(x-2)(x-3)$$

$$f(x) = x^3 - 6x^2 + 11x - 6$$

$$f'(x) = 3x^2 - 12x + 11$$

$$f''(x) = 6x - 12$$

For the critical points

$$f'(x) = 0$$

$$3x^2 - 12x + 11 = 0$$

$$a=3 \quad b=-12 \quad c=11$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{12 \pm \sqrt{144 - 4 \cdot 3 \cdot 11}}{2 \cdot 3}$$

$$x = \frac{12 \pm \sqrt{144 - 132}}{6}$$

$$x = \frac{12 \pm \sqrt{12}}{6}$$

$$x = \frac{12 \pm 2\sqrt{3}}{6}$$

$$x = \frac{6 \pm \sqrt{3}}{3}$$

Therefore  $x = \frac{6 \pm \sqrt{3}}{3}$  are the critical points where the function has extreme values.

To check where  $f$  has relative maxima or relative minima, we use the 2nd derivative test.

$$\text{For } x = \frac{6+\sqrt{3}}{3}, \quad f''(x) = 6\left(\frac{6+\sqrt{3}}{3}\right) - 12 = 2\sqrt{3} > 0$$

$\therefore f(x)$  has relative min at  $x = \frac{6+\sqrt{3}}{3}$

$$\text{For } x = \frac{6-\sqrt{3}}{3}, \quad f''(x) = 6\left(\frac{6-\sqrt{3}}{3}\right) - 12 = -2\sqrt{3} < 0$$

$\therefore f(x)$  has relative max. at  $x = \frac{6-\sqrt{3}}{3}$

$$(5) \quad f(x) = \sin x \cos 2x$$

$$= \sin x (1 - 2\sin^2 x)$$

$$f(x) = \sin x - 2\sin^3 x$$

$$f'(x) = \cos x - 6\sin^2 x \cos x$$

$$f'(x) = \cos x (1 - 6\sin^2 x)$$

$$f''(x) = \cos x (-12\sin x \cos x) + (1 - 6\sin^2 x)(-\sin x)$$

$$f''(x) = -12\sin x \cos^2 x - \sin x + 6\sin^3 x$$

For critical points

$$f'(x) = 0$$

$$\cos x (1 - 6\sin^2 x) = 0$$

$$\cos x = 0$$

$$x = \cos^{-1}(0)$$

$$x = \pm \pi/2$$

$$1 - 6\sin^2 x = 0$$

$$+ 6\sin^2 x = 1$$

$$\sin^2 x = \frac{1}{6}$$

$$\sin x = \pm \frac{1}{\sqrt{6}}$$

$$\text{For } x = \pi/2, \quad f''(x) = 0 - 1 + 6 = 5 > 0$$

$\therefore f$  has relative min at  $x = \pi/2$

$$\text{For } x = -\pi/2$$

$$f''(x) = 1 - 6 = -5 < 0$$

$\therefore f$  has relative max. at  $x = -$

$$\text{For } \sin x = \frac{1}{\sqrt{6}}$$

$$f''(x) = -12 \cdot \frac{1}{\sqrt{6}} \left(1 - \frac{1}{6}\right) - \frac{1}{\sqrt{6}} + 6 \cdot \frac{1}{6\sqrt{6}}$$

$$= -\frac{12}{\sqrt{6}} \cdot \frac{5}{6} = -\frac{10}{\sqrt{6}} < 0$$

$\therefore f$  has a relative max at  $x = \sin^{-1}\left(\frac{1}{\sqrt{6}}\right)$

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For  $\sin x = -\frac{1}{\sqrt{6}}$

$$f''(x) = -12\left(-\frac{1}{\sqrt{6}}\right)\left(1 - \frac{1}{6}\right) - \left(-\frac{1}{\sqrt{6}}\right) - 6 \cdot \frac{1}{6\sqrt{6}}$$

$$= 12 \cdot \frac{1}{\sqrt{6}} \cdot \frac{5}{6} > 0$$

$\therefore f$  has relative min  
at  $x = \sin^{-1}\left(-\frac{1}{\sqrt{6}}\right)$

⑥  $f(x) = a \sec x + b \csc x$

$$f'(x) = a \sec x \tan x - b \csc x \cot x$$

$$= a \cdot \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} - b \cdot \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}$$

$$= \frac{a \sin x}{\cos^2 x} - \frac{b \cos x}{\sin^2 x}$$

$$f'(x) = \frac{a \sin^3 x - b \cos^3 x}{\cos^2 x \sin^2 x}$$

$$f''(x) = \frac{\cos^2 x \sin^2 x (3a \sin^2 x \cos x + 3b \cos^2 x \sin x)}{(\cos^2 x \sin^2 x)^2}$$

$$- \frac{(a \sin^3 x - b \cos^3 x)(\sin^2 x (-2 \cos x \sin x) + \cos^2 x (\frac{2 \sin x}{\cos^2 x}))}{(\cos^2 x \sin^2 x)^2}$$

$$\frac{\sin^3 x \cos^3 x (3a \sin x + 3b \cos x) - (a \sin^3 x - b \cos^3 x) \cos^4 x \sin^4 x}{\cos^4 x \sin^4 x}$$

For critical points

$$f'(x) = 0$$

$$\frac{a \sin^3 x - b \cos^3 x}{\cos^2 x \sin^2 x} = 0$$

$$a \sin^3 x - b \cos^3 x = 0$$

$$a \sin^3 x = b \cos^3 x$$

$$\frac{\sin^3 x}{\cos^3 x} = \frac{b}{a}$$

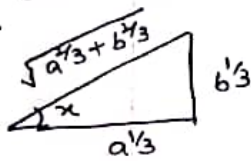
$$\tan^3 x = \frac{b}{a}$$

$$\tan x = \frac{b^{1/3}}{a^{1/3}}$$

now, we find form a right triangle.

$$\sin x = \pm \frac{b^{1/3}}{\sqrt{a^{2/3} + b^{2/3}}}$$

$$\cos x = \pm \frac{a^{1/3}}{\sqrt{a^{2/3} + b^{2/3}}}$$



when  $\sin x$  and  $\cos x$  are both +ve, then

$$f''(x) > 0$$

so  $f$  has relative min

$$\text{at } x = \sin^{-1}\left(\frac{b^{1/3}}{\sqrt{a^{2/3} + b^{2/3}}}\right)$$

$$\text{or } x = \cos^{-1}\left(\frac{a^{1/3}}{\sqrt{a^{2/3} + b^{2/3}}}\right)$$

when  $\sin x$  and  $\cos x$  are both -ve then

$$f''(x) < 0$$

so  $f$  has relative max.

$$\text{at } x = \sin^{-1}\left(-\frac{b^{1/3}}{\sqrt{a^{2/3} + b^{2/3}}}\right)$$

$$\text{or } x = \cos^{-1}\left(-\frac{b^{1/3}}{\sqrt{a^{2/3} + b^{2/3}}}\right)$$

⑦  $f(x) = \sin x \cos^2 x$

$$f'(x) = \sin x (-2 \sin x \cos x) + \cos^2 x$$

$$f'(x) = -2 \cos x \sin^2 x + \cos^3 x$$

$$= \cos x (-2 \sin^2 x + \cos^2 x)$$

$$f'(x) = \cos x (-2 \sin^2 x + 1 - \sin^2 x)$$

$$f'(x) = \cos x (1 - 3 \sin^2 x)$$

$$f''(x) = \cos x (-6 \sin x \cos x) + (1 - 3 \sin^2 x)(-\sin x)$$

$$f''(x) = -6 \sin x \cos^2 x - \sin x + 3 \sin^3 x$$

$$= -6 \sin x (1 - \sin^2 x) - \sin x + 3 \sin^3 x$$

$$= -6 \sin x + 6 \sin^3 x - \sin x + 3 \sin^3 x$$

$$f''(x) = 9 \sin^3 x - 7 \sin x$$

For critical points

$$f'(x) = 0$$

$$\cos x (1 - 3 \sin^2 x) = 0$$

$$\cos x = 0$$

$$x = \cos^{-1}(0)$$

$$x = \pm \pi/2$$

$$1 - 3 \sin^2 x = 0$$

$$3 \sin^2 x = 1$$

$$\sin^2 x = 1/3$$

$$\sin x = \pm \frac{1}{\sqrt{3}}$$

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For  $x = \pi/2$

$$f''(x) > 0$$

$\therefore f$  has relative min  
at  $x = \pi/2$

For  $x = -\pi/2$

$$f''(x) < 0$$

$\therefore f$  has relative max  
at  $x = -\pi/2$

For  $\sin x = \frac{1}{\sqrt{3}}$

$$f''(x) = \frac{9}{3\sqrt{3}} - \frac{7}{\sqrt{3}} \\ = \frac{3}{\sqrt{3}} - \frac{7}{\sqrt{3}} = -\frac{4}{\sqrt{3}} < 0$$

$\therefore f$  has relative max.  
at  $x = \sin^{-1}(\frac{1}{\sqrt{3}})$

For  $\sin x = -\frac{1}{\sqrt{3}}$

$$f''(x) = 9(-\frac{1}{3\sqrt{3}}) - 7(-\frac{1}{\sqrt{3}}) \\ = -\frac{3}{\sqrt{3}} + \frac{7}{\sqrt{3}} \\ = \frac{4}{\sqrt{3}} > 0$$

$\therefore f$  has relative min  
at  $x = \sin^{-1}(-\frac{1}{\sqrt{3}})$

(B)  $f(x) = e^x \cos(x-a)$

$$f'(x) = e^x \cos(x-a) - e^x \sin(x-a)$$

$$f'(x) = e^x [\cos(x-a) - \sin(x-a)]$$

$$f''(x) = e^x [-\sin(x-a) - \cos(x-a)] \\ + [\cos(x-a) - \sin(x-a)] e^x$$

$$f''(x) = -2e^x \sin(x-a)$$

For critical points

$$f'(x) = 0$$

$$e^x [\cos(x-a) - \sin(x-a)] = 0$$

$$e^x \neq 0$$

$$\cos(x-a) - \sin(x-a) = 0$$

$$\sin(x-a) = \cos(x-a)$$

$$\tan(x-a) = 1$$

$$x-a = \pi/4, \pi + \pi/4$$

$$x-a = \frac{\pi}{4}, \quad x-a = \pi + \frac{\pi}{4}$$

$$x = a + \pi/4, \quad x = a + \frac{5\pi}{4}$$

For  $x = a + \pi/4$

$$f''(x) = -2e^x \sin \pi/4 < 0$$

$\therefore f$  has relative max.  
at  $x = a + \pi/4$

For  $x = a + \frac{5\pi}{4}$

$$f''(x) = -2e^x \sin \frac{5\pi}{4} > 0$$

$\therefore f$  has relative min  
at  $x = a + \frac{5\pi}{4}$

(9)

$$f(x) = x^x$$

Taking ln both sides

$$\ln f(x) = \ln x^x$$

$$\ln f(x) = x \ln x$$

Diff. w.r.t x

$$\frac{f'(x)}{f(x)} = x \cdot \frac{1}{x} + \ln x \cdot 1$$

$$f'(x) = f(x) (1 + \ln x)$$

$$f'(x) = x^x (1 + \ln x) \quad \text{from (1)}$$

Diff. w.r.t x

$$f''(x) = x^x (\frac{1}{x}) + (1 + \ln x) \cdot x^x (1 + \ln x)$$

$$f''(x) = x^x (\frac{1}{x} + (1 + \ln x)^2)$$

For critical points

$$f'(x) = 0$$

$$x^x (1 + \ln x) = 0$$

$$x^x \neq 0$$

$$1 + \ln x = 0$$

$$\ln x = -1$$

$$\ln x = -\ln e$$

$$\ln x = \ln \frac{1}{e}$$

$$x = \frac{1}{e}$$

For  $x = \frac{1}{e}$

$$f''(x) > 0$$

$\therefore f$  has relative min  
at  $x = \frac{1}{e}$



(10)

$$f(x) = \frac{\ln x}{x}$$

$$f'(x) = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$$

$$f'(x) = \frac{1 - \ln x}{x^2}$$

$$f''(x) = \frac{x^2(-1/x) - (1 - \ln x)(2x)}{(x^2)^2}$$

$$= \frac{-x - 2x + 2x \ln x}{x^4}$$

$$= \frac{-3x + 2x \ln x}{x^4}$$

$$f''(x) = \frac{-3 + 2 \ln x}{x^3}$$

For critical points

$$f'(x) = 0$$

$$\frac{1 - \ln x}{x^2} = 0$$

$$1 - \ln x = 0$$

$$-\ln x = -1$$

$$\ln x = \ln e$$

$$x = e$$

For  $x = e$ ,

$$f''(x) = \frac{-3 + 2 \ln e}{e^3}$$

$$= -\frac{1}{e^3} < 0$$

$\therefore f$  has relative max.

at  $x = e$

(12) Let  $P(a, b)$  be the point on the line

$$2x - 7y + 5 = 0$$

that is closest to the origin.

$$2a - 7b + 5 = 0$$

$$-7b = -2a - 5$$

$$b = \frac{2a + 5}{7}$$

$P(a, b)$

The distance

$d$  from the origin

$$d = \sqrt{a^2 + b^2}$$

$$d^2 = a^2 + b^2 \text{ from (1)}$$

$$F = a^2 + \left(\frac{2a+5}{7}\right)^2$$

$$= a^2 + \frac{4a^2 + 20a + 25}{49}$$

$$F = \frac{1}{49} [53a^2 + 20a + 25]$$

Diff. w.r.t 'a'

$$F' = \frac{1}{49} [106a + 20]$$

$$F'' = \frac{1}{49} (106)$$

For critical pt

$$F' = 0$$

$$\frac{1}{49} [106a + 20] = 0$$

$$106a + 20 = 0$$

$$106a = -20$$

$$a = -10/53$$

For  $a = -10/53$ ,  $F'' > 0$

$d$  is min at  $a = -10/53$

Then from (1)

$$b = \frac{2a + 5}{7} = \frac{2(-10/53) + 5}{7}$$

$$= \frac{-20 + 265}{7 \times 53}$$

$$= \frac{245}{7 \times 53}$$

$$b = \frac{35}{53}$$

$$\therefore P(a, b) = \left(-\frac{10}{53}, \frac{35}{53}\right)$$

(13)

$$\frac{c^4}{r^2} = \frac{a^2}{\sin^2 \theta} + \frac{b^2}{\cos^2 \theta}$$

$$c^4 r^{-2} = a^2 \csc^2 \theta + b^2 \sec^2 \theta$$

Diff. w.r.t  $\theta$

$$c^4 (-2) r^{-3} \frac{dr}{d\theta} = -2a^2 \csc^2 \theta \cot \theta + 2b^2 \sec^2 \theta \tan \theta$$

$$\frac{c^4}{r^3} \frac{dr}{d\theta} = \frac{2a^2 \cot \theta}{\sin^3 \theta} - \frac{b^2 \tan \theta}{\cos^3 \theta}$$

$$\frac{dr}{d\theta} = \frac{r^3}{c^4} \left( \frac{2a^2 \cot \theta}{\sin^3 \theta} - \frac{b^2 \tan \theta}{\cos^3 \theta} \right)$$

— (1)

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For critical points

$$\frac{dr}{d\theta} = 0$$

$$\frac{r^3}{c^4} \left( \frac{a^2 \cos^4 \theta - b^2 \sin^4 \theta}{\sin^3 \theta \cos^3 \theta} \right) = 0$$

$$a^2 \cos^4 \theta - b^2 \sin^4 \theta = 0$$

$$a^2 \cos^4 \theta = b^2 \sin^4 \theta$$

$$\tan^4 \theta = \frac{a^2}{b^2}$$

$$\tan^2 \theta = \frac{a}{b}$$

$$\tan \theta = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\sin \theta = \frac{\sqrt{a}}{\sqrt{a+b}}, \quad \cos \theta = \frac{\sqrt{b}}{\sqrt{a+b}}$$

$$\sin^2 \theta = \frac{a}{a+b}, \quad \cos^2 \theta = \frac{b}{a+b}$$

Diff. w.r.t  $\theta$

$$= \frac{r^3 \left[ \sin^3 \theta \cos^3 \theta \cdot \frac{d}{d\theta} (a^2 \cos^4 \theta - b^2 \sin^4 \theta) \right]}{c^4 (\sin^3 \theta \cos^3 \theta)^2}$$

$$\rightarrow \frac{-(a^2 \cos^4 \theta - b^2 \sin^4 \theta) \frac{d}{d\theta} (\sin^3 \theta \cos^3 \theta)}{c^4 (\sin^3 \theta \cos^3 \theta)^2}$$

$$\text{For } \sin^2 \theta = \frac{a}{a+b}, \quad \cos^2 \theta = \frac{b}{a+b}$$

$$\text{we have } \frac{dr}{d\theta} < 0$$

So  $r$  is max. for this value

of  $\theta$  and max. value of

$r$  is given by

$$\frac{c^4}{r^2} = \frac{a^2}{\frac{a}{a+b}} + \frac{b^2}{\frac{b}{a+b}}$$

$$\frac{c^4}{r^2} = \frac{a^2(a+b)}{a} + \frac{b^2(a+b)}{b}$$

$$= a(a+b) + b(a+b)$$

$$= a^2 + ab + ba + b^2$$

$$= (a^2 + 2ab + b^2)$$

$$\frac{c^4}{r^2} = (a+b)^2$$

$$\frac{c^2}{r} = (a+b)$$

$$r_{\max} = \frac{c^2}{a+b}$$

2nd Method

$$\text{Q. No. 13 } \frac{c^4}{r^2} = \frac{a^2}{\sin^2 \theta} + \frac{b^2}{\cos^2 \theta}$$

$$\frac{c^4}{r^2} = a^2 \sec^2 \theta + b^2 \sec^2 \theta$$

$$= a^2(1 + \tan^2 \theta) + b^2(1 + \tan^2 \theta)$$

$$\frac{c^4}{r^2} = a^2 + a^2 \tan^2 \theta + b^2 + b^2 \tan^2 \theta$$

$$= (a^2 + b^2) + a^2 \tan^2 \theta + b^2 \tan^2 \theta$$

$$= (a^2 + b^2 + 2ab) + a^2 \tan^2 \theta + b^2 \tan^2 \theta - 2ab$$

$$= (a+b)^2 + a^2 \tan^2 \theta + b^2 \tan^2 \theta$$

$$- 2ab(\cot \theta)(\tan \theta)$$

$$\frac{c^4}{r^2} = (a+b)^2 + (a \cot \theta - b \tan \theta)^2$$

$$r^2 = \frac{c^4}{(a+b)^2 + (a \cot \theta - b \tan \theta)^2}$$

$$r = \frac{c^2}{\sqrt{(a+b)^2 + (a \cot \theta - b \tan \theta)^2}}$$

Radius vector ' $r$ ' will be max.

when den. of R.H.S will min.

i.e. min value is

$$\text{when } a \cot \theta - b \tan \theta = 0$$

$$r_{\max} = \frac{c^2}{\sqrt{(a+b)^2}}$$

$$= \frac{c^2}{a+b}$$

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$$y = \frac{x^3 - x}{3x^2 + 1} \dots \dots \dots 1$$

Diff. w.r.t x

$$\frac{dy}{dx} = \frac{(3x^2+1)(3x^2-1) - (x^3-x)(6x)}{(3x^2+1)^2}$$

$$= \frac{9x^4 - 1 - 6x^4 + 6x^2}{(3x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{3x^4 + 6x^2 - 1}{(3x^2+1)^2}$$

Diff. w.r.t x

$$\frac{d^2y}{dx^2} = \frac{(3x^2+1)^2(12x^3+12x) - (3x^4+6x^2-1) \cdot 2(3x^2+1) \cdot 6x}{(3x^2+1)^4}$$

$$(3x^2+1)[(3x^2+1)(12x^3+12x) - 12x(3x^4+6x^2-1)]$$

$$\frac{36x^5 + 36x^3 + 12x^3 + 12x - 36x^5 - 72x^3 + 12x}{(3x^2+1)^3}$$

$$\frac{d^2y}{dx^2} = \frac{-24x^3 + 24x}{(3x^2+1)^3}$$

Diff. w.r.t x

$$\frac{d^3y}{dx^3} = \frac{(3x^2+1)^3(-72x^2+24) - 3(3x^2+1)^2(6x)(-24x^3+24x)}{(3x^2+1)^6}$$

$$(3x^2+1)^3(-72x^2+24) - 18x(-24x^3+24x)(3x^2+1)^2$$

For possible points of inflections

$$\frac{d^2y}{dx^2} = 0$$

$$\frac{-24x^3 + 24x}{(3x^2+1)^3} = 0$$

$$-24x^3 + 24x = 0$$

$$x^3 - x = 0$$

$$x(x^2 - 1) = 0$$

$$x = 0 \text{ or } x^2 - 1 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

① ⇒

$$\text{For } x=0, y=0$$

$$x=1, y=0$$

$$x=-1, y=0$$

So (0,0), (1,0), (-1,0) are possible pts of inflection.

At (0,0)  $\frac{d^3y}{dx^3}$  is finite, (0,0) is pt of inflection

At (1,0)  $\frac{d^3y}{dx^3}$  is finite, (1,0) is pt of inflection

At (-1,0)  $\frac{d^3y}{dx^3}$  is finite, (-1,0) is pt of inflection

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$$f(x) = ax^3 + bx^2$$

$$y = ax^3 + bx^2$$

Since (1,6) is a pt of inflection

So (1,6) lies on the curve.

$$6 = a + b$$

$$a + b = 6 \dots \dots \dots 1,$$

Again

$$y = ax^3 + bx^2$$

$$y' = 3ax^2 + 2bx$$

$$y'' = 6ax + 2b$$

Since (1,6) is a pt of inflection.

So  $y'' = 0$  at (1,6)

$$0 = 6a + 2b$$

$$3a + b = 0 \dots \dots \dots 2,$$

Solve 1, & 2)

$$a + b = 6$$

$$3a + b = 0$$

$$-2a = 6$$

$$a = -3$$

Put in (2)

$$3(-3) + b = 0$$

$$-9 + b = 0$$

$$b = 9$$

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$$y = 3x^5 - 40x^3 + 3x - 20 \dots \dots \dots 1$$

$$\frac{dy}{dx} = 15x^4 - 120x^2 + 3$$

$$\frac{d^2y}{dx^2} = 60x^3 - 240x$$

$$\frac{d^3y}{dx^3} = 180x^2 - 240$$

$$\text{put } \frac{d^2y}{dx^2} = 0$$

$$60x^3 - 240x = 0$$

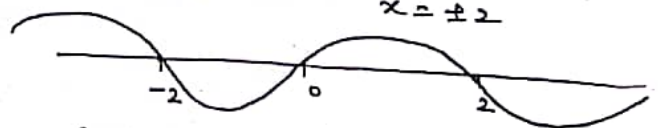
$$60x(x^2 - 4) = 0$$

$$60x = 0 \text{ or } x^2 - 4 = 0$$

$$x = 0$$

$$x^2 = 4$$

$$x = \pm 2$$



Consider the intervals

$]-\infty, -2[$ ,  $]-2, 0[$ ,  $]0, 2[$ ,  $]2, \infty[$



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i, In the interval  $]-\infty, -2[$

$$\frac{d^2y}{dx^2} < 0$$

So the curve faces down.

ii, In the interval  $]-2, 0[$

$$\frac{d^2y}{dx^2} > 0$$

So the curve faces up.

iii, In the interval  $]0, 2[$

$$\frac{d^2y}{dx^2} < 0$$

So the curve faces down.

iv, In the interval  $]2, \infty[$

$$\frac{d^2y}{dx^2} > 0$$

So the curve faces up.

2nd part

At  $x=0$ ,  $x=2$ ,  $x=-2$

$\Rightarrow y=-20$ ,  $y=238$ ,  $y=198$

possible points of inflection are

$(0, -20)$ ,  $(2, 238)$ ,  $(-2, 198)$

At  $(0, -20)$ ,  $(2, 238)$ ,  $(-2, 198)$

$$\frac{d^3y}{dx^3} \text{ exists}$$

So  $(0, -20)$ ,  $(2, 238)$ ,  $(-2, 198)$  are pts of inflection.

$$y = (x^2 + 4x + 5)e^{-x}$$

$$\frac{dy}{dx} = (x^2 + 4x + 5)(-e^{-x}) + e^{-x}(2x + 4)$$

$$= -e^{-x}(-x^2 - 4x - 5 + 2x + 4)$$

$$= -e^{-x}(-x^2 - 2x - 1)$$

$$= -e^{-x}(x^2 + 2x + 1)$$

$$\frac{dy}{dx} = -e^{-x}(x+1)^2$$

$$\frac{d^2y}{dx^2} = -2e^{-x}(x+1) + e^{-x}(x+1)^2$$

$$= -e^{-x}(-2x - 2 + x^2 + 1 + 2x)$$

$$\frac{d^2y}{dx^2} = -e^{-x}(x^2 - 1)$$

$$\frac{d^3y}{dx^3} = -e^{-x}(2x) + (x^2 - 1)(-e^{-x})$$

$$= -e^{-x}(-x^2 + 2x + 1)$$

$$\text{Put } \frac{d^2y}{dx^2} = 0$$

$$-e^{-x}(x^2 - 1) = 0$$

$$-e^{-x} \neq 0 \quad x^2 - 1 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

Consider the intervals

$]-\infty, -1[$ ,  $]-1, 1[$ ,  $]1, \infty[$



i, In the interval  $]-\infty, -1[$

$$\frac{d^2y}{dx^2} > 0$$

So the curve faces up.

ii, In the interval  $]-1, 1[$

$$\frac{d^2y}{dx^2} < 0$$

So the curve faces down.

iii, In the interval  $]1, \infty[$

$$\frac{d^2y}{dx^2} > 0$$

So the curve faces up.

2nd part

At  $x=1$   $\Rightarrow y = 10/e$

$x=-1$   $\Rightarrow y = 2e$

So  $(1, 10/e)$ ,  $(-1, 2e)$  are possible pts of inflection.

At  $(1, 10/e)$ ,  $(-1, 2e)$

$$\frac{d^3y}{dx^3} \text{ exists}$$

So  $(1, 10/e)$ ,  $(-1, 2e)$  are pts of inflection.

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(21)

$$f(x) = 5x^2 - 20x + 81$$

$$f'(x) = 10x - 20$$

$$f''(x) = 10$$

For Critical points

$$f'(x) = 0$$

$$10x - 20 = 0$$

$$10x = 20$$

$$x = 2$$

For  $x = 2$

$$f''(x) > 0$$

$f$  has relative min at  $x = 2$

The Curve is Concave up.

Hence  $f(x) > 0$  for all  $x$ .

(22)

$$f(x) = x^4 - 4x^3 + 12x^2 + 40$$

$$f'(x) = 4x^3 - 12x^2 + 24x$$

$$f''(x) = 12x^2 - 24x + 24$$

For Critical points

$$f'(x) = 0$$

$$4x^3 - 12x^2 + 24x = 0$$

$$4x(x^2 - 3x + 6) = 0$$

$$4x = 0$$

$$\text{or } x^2 - 3x + 6 = 0$$

$$x = 0$$

(neglect)

For  $x = 0$

$$f''(x) > 0$$

$f$  has relative min at  $x = 0$

The Curve is Concave up.

Hence  $f(x) > 0$  for all  $x$ .

$$\text{Area} = (2x)(2y)$$

$$= 4xy$$

$$= 4x\sqrt{r^2 - x^2} \quad \text{from } \Delta$$

$$= 16x^2(r^2 - x^2)$$

$$f(x) = 16r^2x^2 - 16x^4$$

$$f'(x) = 32r^2x - 64x^3$$

$$\text{Put } f'(x) = 0$$

$$32r^2x - 64x^3 = 0$$

$$32x(r^2 - 2x^2) = 0$$

$$32x = 0$$

or

$$r^2 - 2x^2 = 0$$

$$x = 0$$

(neglect)

$$2x^2 = r^2$$

$$x^2 = \frac{r^2}{2}$$

$$x = \frac{r}{\sqrt{2}}$$

$$f''(x) = 32r^2 - 192x^2$$

$$\text{at } x = \frac{r}{\sqrt{2}}$$

$$= 32r^2 - 192 \cdot \frac{r^2}{2}$$

$$= 32r^2 - 96r^2$$

$$= -64r^2 < 0$$

Area will be max. at  $x = \frac{r}{\sqrt{2}}$

$$\therefore \text{Length} = 2x = 2 \cdot \frac{r}{\sqrt{2}} = \sqrt{2}r$$

$$\text{Breadth} = 2y = 2\sqrt{r^2 - x^2}$$

$$= 2\sqrt{r^2 - \frac{r^2}{2}}$$

$$= 2\sqrt{\frac{r^2}{2}} = \sqrt{2}r$$

(23)

Let Length of rectangle =  $2x$

Breadth of rectangle =  $2y$

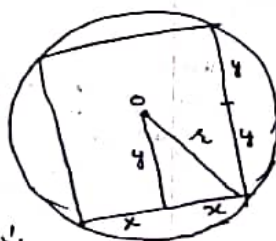
and Given  $r$  be the radius of Circle.

From Fig.

$$r^2 = x^2 + y^2$$

$$y^2 = r^2 - x^2$$

$$y = \sqrt{r^2 - x^2}$$



(24)

Let  $2x$  and  $2y$  be the sides of window.

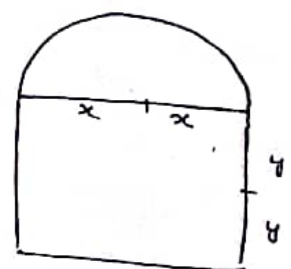
Then radius of semi-circle is  $x$ .

Perimeter of window =  $m$

$$2x + 2y + 2x + 2y + \pi x = m$$

$$4x + 4y + \pi x = m$$

$$4y = m - 4x - \pi x$$



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Area enclosed by window is  $= (2x)(2y) + \frac{\pi x^2}{2}$

$$A = 4xy + \frac{\pi x^2}{2}$$

$$= x(4y) + \frac{\pi x^2}{2} \quad \text{from (1)}$$

$$A = x(m - 4x - \pi x) + \frac{\pi x^2}{2}$$

$$A = mx - 4x^2 - \pi x^2 + \frac{\pi x^2}{2}$$

$$A = mx - 4x^2 - \frac{\pi x^2}{2}$$

Diff. w.r.t.  $x$

$$\frac{dA}{dx} = m - 8x - \pi x$$

Diff. w.r.t.  $x$

$$\frac{d^2A}{dx^2} = -8 - \pi$$

For extreme values.

$$\frac{dA}{dx} = 0$$

$$m - 8x - \pi x = 0$$

$$m = 8x + \pi x$$

$$m = (8 + \pi)x$$

$$x = \frac{m}{8 + \pi}$$

$$\frac{d^2A}{dx^2} < 0 \quad \text{at } x = \frac{m}{8 + \pi}$$

Area will be maximum

$$\text{Put } x = \frac{m}{8 + \pi} \text{ in (1)}$$

$$4y = m - 4x - \pi x$$

$$= m - (4 + \pi)x$$

$$= m - (4 + \pi) \cdot \frac{m}{(8 + \pi)}$$

$$4y = \frac{4m}{\pi + 8}$$

$$y = \frac{m}{\pi + 8}$$

So req. dimension is

$$2x = \frac{2m}{8 + \pi}$$

$$\text{and } 2y = \frac{2m}{8 + \pi}$$

(25) Let

$x$  = radius of cylinder

$z$  = radius of cone

$\alpha$  = semi-vertical angle.

Let

$om = h_1$

$mm' = h$

$bc$  the height of cylinder

In  $\triangle Aom$

$$\tan \alpha = \frac{x}{h_1}$$

$$h_1 = \frac{x}{\tan \alpha}$$

$$h_1 = x \cot \alpha$$

In  $\triangle Bom'$

$$\tan \alpha = \frac{z}{h + h_1}$$

$$h + h_1 = \frac{z}{\tan \alpha}$$

$$h + h_1 = z \cot \alpha$$

$$h = z \cot \alpha - h_1$$

$$(h_1 = x \cot \alpha)$$

$$h = z \cot \alpha - x \cot \alpha$$

$$h = (z - x) \cot \alpha$$

Area of Curved Surface /  $= 2\pi x \cdot h$

$$S = (2\pi x) \cdot (z - x) \cot \alpha$$

$$S = 2\pi \cot \alpha (zx - x^2)$$

Diff. w.r.t.  $x$

$$\frac{dS}{dx} = 2\pi \cot \alpha (z - 2x)$$

$$\frac{d^2S}{dx^2} = 2\pi \cot \alpha (-2) = -4\pi \cot \alpha$$

For extrema of  $S$

$$\frac{dS}{dx} = 0$$

$$2\pi \cot \alpha (z - 2x) = 0$$

$$z - 2x = 0$$

$$z = 2x$$

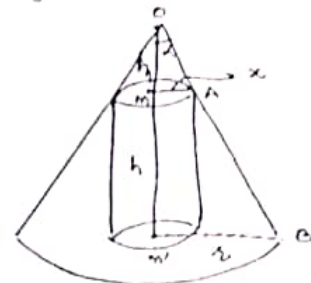
$$x = \frac{z}{2}$$

$$\frac{d^2S}{dx^2} = -4\pi \cot \alpha < 0 \quad \text{at } x = \frac{z}{2}$$

So 'S' will be max

$$\text{at } x = \frac{z}{2}$$

Showing that curved surface will be maximum when radius of cylinder is half of the radius of cone.



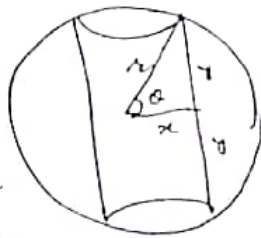


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Let  $x$  = radius of cylinder  
 $2y$  = height of cylinder

Required surface of cylinder is

$S$  = Area of upper face  
 + Area of lower face  
 + Area of curved surface



$$S = \pi x^2 + \pi x^2 + 2\pi x \cdot 2y$$

$$S = 2\pi x^2 + 4\pi xy$$

Put  $x = r \cos \theta$ ,  $y = r \sin \theta$

$$S = 2\pi r^2 \cos^2 \theta + 4\pi r^2 \cos \theta \sin \theta$$

$$S = 2\pi r^2 (\cos^2 \theta + 2 \sin \theta \cos \theta)$$

$$S = 2\pi r^2 (\cos^2 \theta + \sin 2\theta)$$

$$\frac{dS}{d\theta} = 2\pi r^2 (-2 \sin 2\theta + 2 \cos 2\theta)$$

$$= 2\pi r^2 (-\sin 2\theta + 2 \cos 2\theta)$$

$$\frac{d^2S}{d\theta^2} = 2\pi r^2 (-2 \cos 2\theta - 4 \sin 2\theta)$$

$$= -4\pi r^2 (\cos 2\theta + 2 \sin 2\theta)$$

For extrema of  $S$

$$\frac{dS}{d\theta} = 0$$

$$2\pi r^2 (-\sin 2\theta + 2 \cos 2\theta) = 0$$

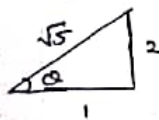
$$-\sin 2\theta + 2 \cos 2\theta = 0$$

$$2 \cos 2\theta = \sin 2\theta$$

$$2 = \tan 2\theta$$

$$\sin 2\theta = \frac{2}{\sqrt{5}}$$

$$\cos 2\theta = \frac{1}{\sqrt{5}}$$



$$\frac{d^2S}{d\theta^2} = -4\pi r^2 \left( \frac{1}{\sqrt{5}} + \frac{4}{\sqrt{5}} \right)$$

$$= -4\pi r^2 \cdot \frac{5}{\sqrt{5}} < 0$$

at  $\sin 2\theta = \frac{2}{\sqrt{5}}$  &  $\cos 2\theta = \frac{1}{\sqrt{5}}$

So ' $S$ ' will be maximum.

max value of  $S$

$$S = 2\pi r^2 (\cos^2 \theta + \sin 2\theta)$$

$$= 2\pi r^2 \left( \frac{1}{2} (1 + \cos 2\theta) + \sin 2\theta \right)$$

$$= 2\pi r^2 \left( \frac{1}{2} \left( 1 + \frac{1}{\sqrt{5}} \right) + \frac{2}{\sqrt{5}} \right)$$

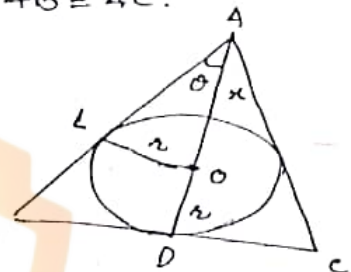
$$= 2\pi r^2 \left( \frac{\sqrt{5}+1}{2\sqrt{5}} + \frac{2}{\sqrt{5}} \right)$$

$$= \pi r^2 \frac{(\sqrt{5}+1+4)}{\sqrt{5}}$$

$$= \pi r^2 (\sqrt{5}+5)/\sqrt{5}$$

$$= \pi r^2 (1+\sqrt{5})$$

27 Let  $\triangle ABC$  is an isosceles triangle such that  $AB = AC$ .



Let  $OA = x$

Then in  $\triangle AOL$

$$AL^2 + OL^2 = OA^2$$

$$AL^2 + r^2 = x^2$$

$$AL^2 = x^2 - r^2$$

$$AL = \sqrt{x^2 - r^2}$$

$$AD = x + r$$

From  $\triangle ABD$

$$\tan \theta = \frac{BD}{AD}$$

$$BD = AD \tan \theta$$

$$= AD \cdot \frac{OL}{AL}$$

$$= (x+r) \cdot \frac{r}{\sqrt{x^2 - r^2}}$$

now Perimeter of  $\triangle ABC$  is

$$P = AB + BC + AC$$

$$= AB + BC + AB \quad (AC = AB)$$

$$P = 2AB + BC$$

$$= 2(AL + LB) + 2BD$$

$$= 2(AL + BD) + 2BD \quad (AL = BD)$$

$$P = 2AL + 4BD$$

$$= 2\sqrt{x^2 - r^2} + \frac{4r(x+r)}{\sqrt{x^2 - r^2}}$$

$$= \frac{2(x^2 - r^2) + 4r(x+r)}{\sqrt{x^2 - r^2}}$$

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$$P = \frac{2x^2 - 2x + 4x + 4x^2}{\sqrt{x^2 - x^2}}$$

$$P = \frac{2x^2 + 4x + 2x^2}{\sqrt{x^2 - x^2}}$$

$$P = \frac{2(x^2 + 2x + x^2)}{\sqrt{x^2 - x^2}}$$

$$P = \frac{2(x+x)^2}{\sqrt{x^2 - x^2}} \quad \text{--- } \downarrow$$

Diff. w.r.t. x

$$= \frac{\sqrt{x^2 - x^2} \cdot 2 \cdot 2(x+x) - 2(x+x)^2 \cdot \frac{2x}{2\sqrt{x^2 - x^2}}}{(\sqrt{x^2 - x^2})^2}$$

$$= \frac{4(x+x)\sqrt{x^2 - x^2} - \frac{2x(x+x)^2}{\sqrt{x^2 - x^2}}}{(x^2 - x^2)}$$

$$= \frac{4(x+x)(x^2 - x^2) - 2x(x+x)^2}{\sqrt{x^2 - x^2}(x^2 - x^2)}$$

$$= \frac{2(x+x)^2[2(x-x) - x]}{(x^2 - x^2)^{3/2}}$$

$$= \frac{2(x+x)^2(x-2x)}{(x^2 - x^2)^{3/2}}$$

Diff. w.r.t. x

$$\frac{(x^2 - x^2)^{3/2} [2(x+x)^2 + 2(x-2x)2(x+x)] - [2(x+x)^2(x-2x)] \cdot \frac{3}{2}(x^2 - x^2)^{1/2}(2x)}{[(x^2 - x^2)^{3/2}]^2}$$

$$\rightarrow \frac{2(x+x)^2(x-2x) \cdot \frac{3}{2}(x^2 - x^2)^{1/2}(2x)}{[(x^2 - x^2)^{3/2}]^2}$$

For extreme values

$$\frac{dP}{dx} = 0$$

$$\frac{2(x+x)^2(x-2x)}{(x^2 - x^2)^{3/2}} = 0$$

$$2(x+x)^2(x-2x) = 0$$

$$x - 2x = 0$$

$$x = 2x$$

$$\frac{d^2P}{dx^2} > 0 \quad \text{at } x = 2x$$

∴ perimeter is minimum at  $x = 2x$

Minimum value of P

Put  $x = 2x$  in (1)

$$P = \frac{2(2x+x)^2}{\sqrt{4x^2 - x^2}} = \frac{2 \cdot 9x^2}{\sqrt{3x^2}}$$

$$P = \frac{2 \cdot 9x^2}{\sqrt{3x^2}} = \frac{2 \cdot 3 \cdot 3x}{\sqrt{3}} = 6\sqrt{3}x$$

(28)

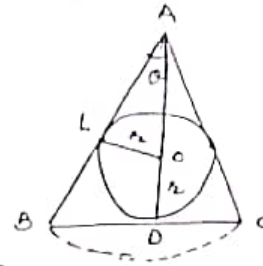
Let  $OA = x$

$$AD = x + x$$

$$BD = AD \tan \theta$$

$$= AD \cdot \frac{OL}{AL}$$

$$= (x+x) \cdot \frac{x}{\sqrt{x^2 - x^2}}$$



Volume of cone is

$$V = \frac{1}{3} \pi (BD)^2 (AD)$$

$$= \frac{1}{3} \pi (x+x)^2 \cdot \frac{x}{\sqrt{x^2 - x^2}} \cdot (x+x)$$

$$V = \frac{1}{3} \pi x^2 \frac{(x+x)^2}{(x-x)}$$

Diff. w.r.t. x

$$\frac{dV}{dx} = \frac{1}{3} \pi x^2 \left[ \frac{(x-x)2(x+x) - (x+x)^2 \cdot 1}{(x-x)^2} \right]$$

$$= \frac{\pi x^2}{3} \cdot \frac{(x+x)(2x-2x-x-x)}{(x-x)^2}$$

$$\frac{dV}{dx} = \frac{\pi x^2}{3} \cdot \frac{(x+x)(x-3x)}{(x-x)^2}$$

Diff. w.r.t. x

$$\frac{d^2V}{dx^2} = \frac{\pi x^2}{3} \left[ \frac{(x-x)^2 [(x+x) + (x-3x)] - (x+x)(x-3x) \cdot 2(x-x)}{(x-x)^4} \right]$$

For extreme values

$$\frac{dV}{dx} = 0$$

$$\frac{\pi x^2}{3} \cdot \frac{(x+x)(x-3x)}{(x-x)^2} = 0$$

$$(x+x)(x-3x) = 0$$

$$x - 3x = 0$$

$$x = 3x$$

$$\frac{d^2V}{dx^2} > 0 \quad \text{at } x = 3x$$

So volume is min at  $x = 3x$ .

$$\text{Req. altitude of Cone is } AD = x + x = 3x + x = 4x$$

From figure :-

$$\sin \theta = \frac{OL}{OA}$$

$$\sin \theta = \frac{x}{x}$$

$$\sin \theta = \frac{x}{3x}$$

$$\sin \theta = \frac{1}{3}$$

$$\theta = \sin^{-1}\left(\frac{1}{3}\right)$$

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(29) Let length of one side =  $x$   
length of another side =  $y$

Perimeter = 1000

$x + y + y = 1000$

$x + 2y = 1000$

$x = 1000 - 2y$

$A = xy$

$= (1000 - 2y)y$

$A = 1000y - 2y^2$

$\frac{dA}{dy} = 1000 - 4y$

$\frac{d^2A}{dy^2} = -4$

for extreme values

$\frac{dA}{dy} = 0$

$1000 - 4y = 0$

$y = 250$

$\frac{d^2A}{dy^2} = -4 < 0$

Area will be maximum

at  $y = 250$

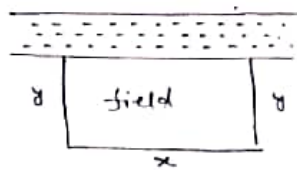
from (i)  $x = 1000 - 2(250)$

$x = 500$

max. area is cover

$x = 500$

$y = 250$



Cost of 4 sides = (Area of 4 sides)  $\times$  (rate of sides)  
 $= 4xy \cdot 2$   
 $= 8xy$

Total cost

$C = \text{Cost of base} + \text{Cost of 4 sides}$

$C = 3x^2 + 8xy$

Volume = 1296

$x \cdot x \cdot y = 1296$

$x^2 y = 1296$

$y = \frac{1296}{x^2}$

ii)  $\Rightarrow$

$C = 3x^2 + 8x \left( \frac{1296}{x^2} \right)$

$C = 3x^2 + \frac{8(1296)}{x}$

Diff. w.r.t  $x$

$\frac{dC}{dx} = 6x + 8(1296) \left( -\frac{1}{x^2} \right)$

$\frac{d^2C}{dx^2} = 6 + 8(1296) \left( \frac{2}{x^3} \right)$

For extreme values

$\frac{dC}{dx} = 0$

$6x - 8(1296) \cdot \frac{1}{x^2} = 0$

$6x = 8(1296) \cdot \frac{1}{x^2}$

$6x^3 = 8(1296)$

$x^3 = 1728$

$x^3 = (12)^3$

$x = 12$

$\frac{d^2C}{dx^2} > 0$  at  $x = 12$

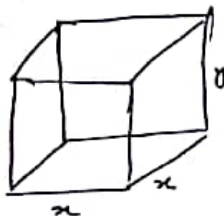
So cost is min at  $x = 12$

ii)  $\Rightarrow y = \frac{1296}{(12)^2} = 9$

So dimension of box is 12cm, 12cm, 9cm.

(30) Let length of the base =  $x$  cm

Rate of Cost of base = Rs 3 per sq. cm



So Cost of Base = Area of Base  $\times$  rate of Base  
 $= x^2 \cdot 3$   
 $= 3x^2$

Let length of side =  $y$  cm

Rate of Cost of sides = Rs 2 per sq. cm



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length of  
card board / = 8 dm

Breadth of  
card board / = 5 dm

Let  $x$  be the side of square  
cut from each corner.  
Then the edges of box  
formed by bending the  
sides are  $8-2x$ ,  $5-2x$ ,  $x$

Volume of the box is

$$V = x(8-2x)(5-2x)$$

$$V = x(40-16x-10x+4x^2)$$

$$V = x(40-26x+4x^2)$$

$$V = 4x^3 - 26x^2 + 40x$$

$$\frac{dV}{dx} = 12x^2 - 52x + 40$$

$$\frac{d^2V}{dx^2} = 24x - 52$$

For extreme values

$$\frac{dV}{dx} = 0$$

$$12x^2 - 52x + 40 = 0$$

$$3x^2 - 13x + 10 = 0$$

$$(3x-10)(x-1) = 0$$

$$3x-10=0 \quad \text{or} \quad x-1=0$$

$$3x=10$$

$$x=1$$

$$x = \frac{10}{3}$$

(neglect)

$$\frac{d^2V}{dx^2} < 0 \quad \text{at} \quad x=1$$

So volume is max. when  $x=1$

Hence edge of square is 1 dm.

32) passengers RS

|                          |                                |
|--------------------------|--------------------------------|
| 1200                     | 2.                             |
| on decreasing one person |                                |
| (1200+10)                | (2 - $\frac{1}{100}$ )         |
| on decreasing two person |                                |
| (1200+2(10))             | (2 - 2( $\frac{1}{100}$ ))     |
| ⋮                        | ⋮                              |
| on decreasing $x$ person |                                |
| (1200+x(10))             | (2 - $x \cdot \frac{1}{100}$ ) |

Total Revenue

$$R = (1200 + 10x)(2 - \frac{x}{100})$$

$$R = 2400 - 12x + 20x - \frac{x^2}{10}$$

$$R = 2400 + 8x - \frac{x^2}{10}$$

$$\frac{dR}{dx} = 8 - \frac{2x}{10}$$

$$\frac{d^2R}{dx^2} = -\frac{2}{10}$$

For extreme values

$$\frac{dR}{dx} = 0$$

$$8 - \frac{2x}{10} = 0$$

$$-\frac{2x}{10} = -8$$

$$\frac{x}{5} = +8$$

$$x = 40$$

$$\text{At } \frac{d^2R}{dx^2} < 0 \quad \text{at } x=40$$

So revenue will be max. at  $x=40$   
Hence the fare should be lowered  
by 40 paise to earn the max. revenue

33) Let cattle are sold after  $x$  weeks  
weight gained by the cattle / =  $5x$   
in  $x$  weeks

Total weight =  $(200+5x)$  quantity

Rate of profit after  $x$  weeks / =  $(500-10x)$  Rs

So profit is

$$P = (200+5x)(500-10x)$$

$$P = 100000 - 2000x + 2500x - 50x^2$$

$$\frac{dP}{dx} = 500 - 100x$$

$$\frac{d^2P}{dx^2} = -100$$

For extreme values

$$\frac{dP}{dx} = 0$$

$$500 - 100x = 0$$

$$x = 5$$

So profit is max. when  $x=5$

So cattle should be sold  
after 5 weeks to get  
maximum profit.

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So possible multiple points are  
 $(0,0), (3, \frac{9}{2})$

out of these only  $(0,0)$   
 satisfies the given equation.  
 So  $(0,0)$  is a multiple pt.

At  $(0,0)$

$$f_{xx} = 0 \quad f_{xy} = 0 \\ f_{yy} = 2$$

now  $(f_{xy})^2 - f_{xx} \cdot f_{yy}$   
 $= 0 - 0(2) = 0$

So  $(0,0)$  is CUSP.

②  $y^3 = x^3 + ax^2$

$$f(x,y) = x^3 + ax^2 - y^3 = 0$$

$$f_x = 3x^2 + 2ax$$

$$f_{xx} = 6x + 2a$$

$$f_y = -3y^2$$

$$f_{yy} = -6y$$

$$f_{xy} = 0$$

now  $f_x = 0$  &  $f_y = 0$

$$3x^2 + 2ax = 0 \quad \text{--- 1,} \\ -3y^2 = 0 \quad \text{--- 2,}$$

③  $y^2 = 0$       ④  $3x^2 + 2ax = 0$   
 $y = 0$        $x(3x + 2a) = 0$   
 $x \geq 0$  or  $3x + 2a = 0$   
 $3x = -2a$   
 $x = -\frac{2a}{3}$

So possible multiple

points are  $(0,0), (-\frac{2a}{3}, 0)$

out of these only  $(0,0)$   
 satisfies given equation.

So  $(0,0)$  is only multiple point.

At  $(0,0)$   $f_{xx} = 2a, f_{xy} = 0, f_{yy} = 0$

now  $(f_{xy})^2 - f_{xx} \cdot f_{yy}$   
 $= 0 - 0 = 0$

So  $(0,0)$  is CUSP.

①  $x^4 + y^3 - 2x^3 + 3y^2 = 0$

$$f(x,y) = x^4 + y^3 - 2x^3 + 3y^2 = 0$$

$$f_x = 4x^3 - 6x^2$$

$$f_{xx} = 12x^2 - 12x$$

$$f_y = 3y^2 + 6y$$

$$f_{yy} = 6y + 6$$

$$f_{xy} = 0$$

now  $f_x = 0, f_y = 0$

$$4x^3 - 6x^2 = 0 \quad \text{--- 1,}$$

$$3y^2 + 6y = 0 \quad \text{--- 2,}$$

①  $\Rightarrow 2x^2(2x-3) = 0$

$$2x^2 = 0 \quad \text{or} \quad 2x-3 = 0$$

$$x = 0 \quad \text{or} \quad 2x = 3 \\ x = \frac{3}{2}$$

②  $\Rightarrow 3y(y+2) = 0$

$$3y = 0 \quad \text{or} \quad y+2 = 0$$

$$y = 0 \quad \text{or} \quad y = -2$$

So possible multiples are

$(0,0), (0,-2), (\frac{3}{2}, 0), (\frac{3}{2}, -2)$

out of these only  $(0,0)$

satisfies the given eq.

So  $(0,0)$  is only multiple point.

At  $(0,0)$   $f_{xx} = 0, f_{xy} = 0$   
 $f_{yy} = 6$

now  $(f_{xy})^2 - f_{xx} \cdot f_{yy}$   
 $= 0 - 0 = 0$

So  $(0,0)$  is CUSP.

⑧

$$f(x, y) = x^3 + 2x^2 + 2xy - y^2 + 5x - 2y = 0$$

$$f_x = 3x^2 + 4x + 2y + 5$$

$$f_{xx} = 6x + 4$$

$$f_{xy} = 2$$

$$f_{yy} = 2x - 2y - 2$$

$$f_{yy} = -2$$

$$\text{now } f_x = 0, \quad f_y = 0$$

$$3x^2 + 4x + 2y + 5 = 0 \quad \text{--- 1,}$$

$$2x - 2y - 2 = 0 \quad \text{--- 2,}$$

$$\text{③ } \Rightarrow x - y - 1 = 0$$

$$x - 1 = y$$

$$\text{put in 1,} \quad \text{--- 3,}$$

$$3x^2 + 4x + 2(x-1) + 5 = 0$$

$$3x^2 + 4x + 2x - 2 + 5 = 0$$

$$3x^2 + 6x + 3 = 0$$

$$x^2 + 2x + 1 = 0$$

$$(x+1)^2 = 0$$

$$x+1 = 0$$

$$x = -1$$

$$\text{put in (3)}$$

$$y = -2$$

The possible multiple point is  $(-1, -2)$ . satisfies the given equation.

$$\text{At } (-1, -2)$$

$$f_{xx} = -2 \quad f_{xy} = 2 \quad f_{yy} = -2$$

$$\text{now } f_{xx} = -2$$

$$f_{xy} = 2$$

$$f_{yy} = -2$$

$$\text{now } (f_{xy})^2 - f_{xx}f_{yy}$$

$$= (2)^2 - (-2)(-2)$$

$$= 4 - 4 = 0$$

So  $(-1, -2)$  is a cusp.

$$\text{Let } f(x, y) =$$

$$\text{① } (2y + x + 1)^2 - 4(1-x)^5 = 0$$

$$f_x = 2(2y + x + 1) + 20(1-x)^4 = 0$$

$$f_{xx} = 2 - 80(1-x)^3$$

$$f_y = 2(2y + x + 1)(2)$$

$$f_y = 4(2y + x + 1)$$

$$f_{yy} = 8$$

$$f_{xy} = 4$$

$$\text{now } f_x = 0 \quad \& \quad f_y = 0$$

$$2(2y + x + 1) + 20(1-x)^4 = 0$$

$$2y + x + 1 + 10(1-x)^4 = 0$$

$$4(2y + x + 1) = 0$$

$$2y + x + 1 = 0 \quad \text{--- 2,}$$

$$\text{put (2) in (1)}$$

$$10(1-x)^4 = 0$$

$$(1-x)^4 = 0$$

$$1-x = 0$$

$$x = 1$$

$$\text{put in (2)}$$

$$2y + 1 + 1 = 0$$

$$2y + 2 = 0$$

$$2y = -2$$

$$y = -1$$

The possible multiple point is  $(1, -1)$ . satisfies the equation.

$$\text{At } (1, -1)$$

$$f_{xx} = 2, \quad f_{yy} = 8$$

$$f_{xy} = 4$$

$$\text{now } (f_{xy})^2 - f_{xx}f_{yy}$$

$$= (4)^2 - (2)(8)$$

$$= 16 - 16 = 0$$

So  $(1, -1)$  is a cusp.



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⑩  $f(x, y) = (y^2 - a^2)^3 + x^4(2x + 3a)^2$

$$f_x = x^3 \cdot 2(2x + 3a)(2) + (2x + 3a)^2 \cdot 4x^3$$

$$= 4x^3(2x + 3a) + 4x^3(2x + 3a)^2$$

$$= 4x^3(2x + 3a)(x + 2x + 3a)$$

$$f_x = 4x^3(2x + 3a)(3x + 3a)$$

$$f_x = 24x^5 + 36a^2x^3 + 60a^2x^4$$

$$f_y = 6y(y^2 - a^2)^2$$

$$f_x = 0 \quad f_y = 0$$

$$\begin{cases} 24x^5 + 36a^2x^3 + 60a^2x^4 = 0 \\ 6y(y^2 - a^2)^2 = 0 \end{cases}$$

$$\Rightarrow 0, x = -\frac{3a}{2}, x = -a \quad \begin{cases} y = 0 \\ y^2 - a^2 = 0 \\ y^2 = a^2 \\ y = \pm a \end{cases}$$

Possible points are

$$(0, 0), (0, a), (0, -a), (-\frac{3a}{2}, 0), (-\frac{3a}{2}, -a)$$

$$(-\frac{3a}{2}, a), (-a, 0), (-a, a), (-a, -a)$$

$$\text{only } (0, \pm a), (-\frac{3a}{2}, \pm a), (-a, 0)$$

satisfying the equation.

So there are the multiple points.

$$\text{now } f_{xx} = 36x^2(2x + 3a)(x + a)$$

$$+ 24x^3(x + a) + 12x^3(2x + 3a)$$

$$f_{yy} = 6(y^2 - a^2)^2 + 24y^2(y^2 - a^2)$$

$$f_{xy} = 6(y^2 - a^2)(5y^2 - a^2)$$

$$\text{At } (0, a) \quad f_{xx} = 0 \quad f_{xy} = 0$$

$$\text{using } (f_{xy})^2 - f_{xx}f_{yy} = 0 - 0 = 0$$

Thus  $(0, a)$  is a CUSP.

$$\text{At } (0, -a) \quad f_{xx} = -12a^4 \\ f_{yy} = 0 \\ f_{xy} = 0$$

$$\text{using } (f_{xy})^2 - f_{xx}f_{yy}$$

$$= 0 - 0$$

$$= 0$$

Hence  $(0, -a)$  is a CUSP.

$$\text{At } (-\frac{3a}{2}, 0)$$

$$f_{xx} = \frac{81a^4}{2} \quad f_{yy} = 0 \quad f_{xy} = 0$$

$$\text{using } (f_{xy})^2 - f_{xx}f_{yy}$$

$$= 0 - (\frac{81a^4}{2}) \cdot 0$$

$$= 0$$

So  $(-\frac{3a}{2}, 0)$  is a CUSP.

$$\text{At } (-\frac{3a}{2}, -a)$$

$$f_{xx} = \frac{81a^4}{2}$$

$$f_{xy} = 0 \quad f_{yy} = 0$$

$$\text{using } (f_{xy})^2 - f_{xx}f_{yy}$$

$$= 0 - 0$$

$$= 0$$

$(-\frac{3a}{2}, -a)$  is a CUSP.

$$\text{At } (-a, 0)$$

$$f_{xx} = -12a^4 \quad f_{yy} = 6a^4$$

$$f_{xy} = 0$$

$$\text{using } (f_{xy})^2 - f_{xx}f_{yy}$$

$$= (0)^2 - (-12a^4)(6a^4)$$

$$= 72a^8 > 0$$

Hence  $(-a, 0)$  is a NODE.

⑪

$$y^2 = ax^2 + ax^3$$

$$y^2 - ax^2 - ax^3 = 0$$

To find the nature of singular point  $(0, 0)$

Equate the lowest degree term to zero.

We get tangents at orig.

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$$y^2 - ax^2 = 0$$

$$y^2 = ax^2$$

$$y = \pm a\sqrt{x} \quad \text{--- (1)}$$

if  $a > 0$ , then eq (1) represents two real and distinct

tangent lines at O.

So origin is Node.

ii) if  $a = 0$ , then eq (1) represents two real and coincident tangent lines at O.

So origin is Cusp.

iii) if  $a < 0$ , then eq (1) represents two imaginary tangent lines at O.

So origin is CONJUGATE point.

$$(12) \quad f(x, y) = x^4 - 4ax^3 - 2ay^3 + 4a^2x^2 + 3a^2y^2 - a^4$$

$$f_x = 4x^3 - 12ax^2 + 8a^2x$$

$$f_y = -6ay^2 + 6a^2y$$

$$f_x = 0$$

$$4x^3 - 12ax^2 + 8a^2x = 0$$

$$4x(x^2 - 3ax + 2a^2) = 0$$

$$4x = 0 \quad \text{or} \quad x^2 - 3ax + 2a^2 = 0$$

$$x = 0$$

$$(x - 2a)(x - a) = 0$$

$$x = a, 2a$$

$$f_y = 0$$

$$-6ay^2 + 6a^2y = 0$$

$$-6ay(y - a) = 0$$

$$-6ay = 0 \quad \text{or} \quad y - a = 0$$

$$y = 0$$

$$y = a$$

So the possible multiple points are

$$(0, 0), (0, a), (a, 0), (a, a), (2a, 0), (2a, a)$$

out of these only  $(0, a), (a, 0)$

$(2a, a)$  satisfy the equation

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Table 10, a)

Shifting the origin, by putting

$$x = X + a$$

$y = Y + a$  in given equation.

$$X^4 - 4aX^3 - 2a(Y + a)^3 + 4a^2X^2 + 3a^2(Y + a)^2 - a^4 = 0$$

$$\Rightarrow X^4 - 4aX^3 - 2a(Y^3 + a^3 + 3aY^2 + 3a^2Y) + 4a^2X^2 + 3a^2(Y^2 + a^2 + 2aY) - a^4 = 0$$

$$\Rightarrow X^4 - 4aX^3 - 2aY^3 - 2a^4 - 6a^2Y^2 - 6a^3Y + 4a^2X^2 + 3a^2Y^2 + 3a^4 + 6a^3Y - a^4 = 0$$

$$\Rightarrow X^4 - 4aX^3 - 2aY^3 - 3a^2Y^2 + 4a^2X^2 = 0$$

$$X^4 - 4aX^3 - 2aY^3 + a^2(4X^2 - 3Y^2) = 0$$

The eq of tangent at new origin  $(0, a)$ .

Equate the lowest degree term to zero.

$$a^2(4X^2 - 3Y^2) = 0$$

$$4X^2 - 3Y^2 = 0$$

$$3Y^2 = 4X^2$$

$$Y^2 = \frac{4}{3}X^2$$

$$Y = \pm \frac{2}{\sqrt{3}}X$$

$$\text{Put } Y = y - a, \quad X = x$$

$$y - a = \pm \frac{2}{\sqrt{3}}x$$

$$y = \pm \frac{2}{\sqrt{3}}x + a$$

Table (a, 0)

Shifting the origin, by

putting  $x = X + a$

$y = Y$  in given equation

$$(X + a)^4 - 4a(X + a)^3 - 2aY^3 + 4a^2(X + a)^2 + 3a^2Y^2 - a^4 = 0$$

$$\Rightarrow X^4 + 4aX^3 + 6a^2X^2 + 4a^3X + a^4 - 4a(X^3 + 3aX^2 + 3a^2X + a^3) - 2aY^3 + 4a^2(X^2 + 2aX + a^2) + 3a^2Y^2 - a^4 = 0$$

$$\Rightarrow X^4 - 4aX^3 + 6a^2X^2 + 4a^3X + a^4 - 4aX^3 - 12a^2X^2 - 12a^3X - 4a^4 - 2aY^3 + 4a^2X^2 + 8a^3X + 4a^4 + 3a^2Y^2 - a^4 = 0$$



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$$2a^2x^2 - 2ay^3 + 3a^2y^2 = 0$$

$$x^4 - 2ay^3 - 2a^2x^2 + 3a^2y^2 = 0$$

$$x^4 - 2ay^3 + a^2(3y^2 - 2x^2) = 0$$

The eq of tangents at new origin (a, 0)

Equate the lowest degree term to zero.

$$a^2(3y^2 - 2x^2) = 0$$

$$3y^2 - 2x^2 = 0$$

$$3y^2 = 2x^2$$

$$y^2 = \frac{2}{3}x^2$$

$$y = \pm \sqrt{\frac{2}{3}}x$$

Put  $y = y$

$$x = x - a$$

$$y = \pm \sqrt{\frac{2}{3}}(x - a)$$

are eqs of tangents.

Take (2a, a)

shifting the origin, by

putting  $x = X + 2a$

$y = Y + a$  in given equation.

$$(x+2a)^4 - 4a(x+2a)^3 - 2a(y+a)^3$$

$$+ 4a^2(x+2a)^2 + 3a^2(y+a)^2 - a^4 = 0$$

$$x^4 + 4x^3(2a) + 6x^2(4a^2) + 4x(8a^3)$$

$$+ 16a^4 - 4a(x^3 + 3x^2(2a) + 3x(4a^2) + 8a^3)$$

$$+ 2a(y^3 + 3y^2a + 3ya^2 + a^3) + 4a^2(x^2 + 4ax + 4a^2)$$

$$+ 3a^2(y^2 + a^2 + 2ay) - a^4 = 0$$

$$x^4 + 8x^3a + 24x^2a^2 + 32xa^3 + 16a^4$$

$$- 4ax^3 - 24x^2a^2 - 48xa^3 - 32a^4$$

$$+ 2ay^3 - 6y^2a^2 - 6ya^3 - 3a^4 + 4x^2a^2$$

$$+ 16xa^3 + 16a^4 + 3a^2y^2 + 6a^3y$$

$$+ 3a^4 - a^4 = 0$$

$$x^4 + 4ax^3 - 2ay^3 - 3a^2y^2 + 4a^2x^2 = 0$$

$$x^4 + 4ax^3 - 2ay^3 + a^2(-3y^2 + 4x^2) = 0$$

The eq of tangents at new origin.

Equate the lowest degree term to zero.

$$a^2(-3y^2 + 4x^2) = 0$$

$$-3y^2 + 4x^2 = 0$$

$$3y^2 = 4x^2$$

$$y^2 = \frac{4}{3}x^2$$

$$y = \pm \frac{2}{\sqrt{3}}x$$

Put  $X = x - 2a$

$Y = y - a$

$$y - a = \pm \frac{2}{\sqrt{3}}(x - 2a)$$

are eqs of tangents.

$$(13) (y-2)^2 = x(x-1)^2$$

$$f(x, y) = (y-2)^2 - x(x-1)^2 = 0$$

$$f_x = -(x \cdot 2(x-1) + (x-1)^2 \cdot 1)$$

$$= -(2x^2 - 2x + x^2 + 1 - 2x)$$

$$= -(3x^2 - 4x + 1)$$

$$f_y = 2(y-2)$$

$$f_x = 0$$

$$-(3x^2 - 4x + 1) = 0$$

$$3x^2 - 4x + 1 = 0$$

$$(x-1)(3x-1) = 0$$

$$x-1=0 \text{ or } 3x-1=0$$

$$x=1$$

$$3x=1$$

$$x = \frac{1}{3}$$

$$f_y = 0$$

$$2(y-2) = 0$$

$$y-2=0$$

$$y=2$$

So possible multiple pts are

$$(1, 2), \left(\frac{1}{3}, 2\right)$$



Out of these only (1,2)

Satisfies the equation.

So (1,2) is multiple point.

Take (1,2)

Shifting the origin, by

putting  $x = X+1$

$y = Y+2$  in

given equation.

$$(Y+2-2)^2 = (X+1)(X+1-1)^2$$

$$Y^2 = (X+1)X^2$$

$$Y^2 = X^3 + X^2$$

The eqs of tangents at new origin (1,2).

Equate the lowest degree term to zero.

$$Y^2 = X^2$$

$$Y = \pm X$$

$$\text{Put } x = X-1$$

$$y = Y-2$$

$$Y-2 = \pm(X-1)$$

are eqs of tangents.

$$(14) \quad f(x,y) = x^2(x-y) + y^2$$

$$f(x,y) = x^3 - x^2y + y^2$$

$$f_x = 3x^2 - 2xy, \quad f_y = -x^2 + 2y$$

$$f_x = 0$$

$$f_y = 0$$

$$3x^2 - 2xy = 0 \quad \text{or} \quad -x^2 + 2y = 0$$

$$2y = x^2$$

$$y = x^2/2$$

①

put in (2)

$$3x^2 - 2x(x^2/2) = 0$$

$$3x^2 - x^3 = 0$$

$$x^2(3-x) = 0$$

$$x = 0$$

$$3-x = 0$$

$$x = 3$$

put in (1)

put in (1)

$$y = 0$$

$$y = 9/2$$

So (0,0), (3, 9/2) are possible multiple points.

out of these only (0,0) satisfies the equation.

So (0,0) is a multiple point.

Now Equation of tangent at (0,0)

Equate the lowest degree term to zero.

$$Y^2 = 0$$

are coincident tangents at origin.

Hence (0,0) is a CUSP.

with x-axis as the common tangent.

As given eq. is

$$x^3 - x^2y + y^2 = 0$$

$$y^2 - x^2y + x^3 = 0$$

$$a=1 \quad b=-x^2 \quad c=x^3$$

$$y = \frac{x^2 \pm \sqrt{x^4 - 4x^3}}{2}$$

$$y = \frac{x^2 \pm x\sqrt{x^2 - 4x}}{2}$$

Species Put  $x > 0$  in above eq.

Then  $y$  is imag.

which shows that no part of the curve lies to the right of  $x=0$

Also when  $x < 0$ ,  $y$  has two real values with different signs showing that the origin is a cusp of first specie.

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(15)  $f(x, y) = x^3 + y^3 - 2axy^2$

$$f_x = 3x^2$$

$$f_y = 3y^2 - 4ay$$

$$f_x = 0$$

$$f_y = 0$$

$$3x^2 = 0$$

$$x = 0$$

$$3y^2 - 4ay = 0$$

$$y(3y - 4a) = 0$$

$$y = 0, 3y - 4a = 0$$

$$3y = 4a$$

$$y = \frac{4a}{3}$$

possible

So multiple pts are  $(0, 0)$  and  $(0, \frac{4a}{3})$

only  $(0, 0)$  is a multiple pt.

The eq of tangent at  $(0, 0)$  is

$$-2ay^2 = 0$$

$y^2 = 0$  are coincident tangents at origin

Hence  $(0, 0)$  is a cusp with

$x$ -axis.

(16)  $f(x, y) = x^6 - axy^4 - a^3x^2y + a^4y^2$

$$f_x = 6x^5 - 4ax^3y - 2a^3xy$$

$$f_y = -ax^4 - a^3x^2 + 2a^4y$$

$$f_x = 0$$

$$f_y = 0$$

$$6x^5 - 4ax^3y - 2a^3xy = 0$$

$$\text{--- (1) } -ax^4 - a^3x^2 + 2a^4y = 0$$

$$\text{--- (2)}$$

We see that  $(0, 0)$  satisfies

both eqs (1) and (2).

So  $(0, 0)$  is a multiple point.

Eq of tangent at  $(0, 0)$

$$a^4y^2 = 0$$

$y^2 = 0$  are coincident tangents at origin

Hence  $(0, 0)$  is a cusp.

As given eq can be written as

$$a^4y^2 - a(x^4 + a^2x^2)y + x^6 = 0$$

$$y = \frac{a(x^4 + a^2x^2) \pm \sqrt{a^2(x^4 + a^2x^2)^2 - 4a^4x^6}}{2a^4}$$

$$y = \frac{a(x^4 + a^2x^2) \pm \sqrt{a^2(x^4 + a^2x^2 + 2a^2x^6 - 4a^4x^6)}}{2a^4}$$

$$y = \frac{a(x^4 + a^2x^2) \pm a\sqrt{(x^4 + a^2x^2)^2 - 4a^4x^6}}{2a^4}$$

$$y = \frac{x^4 + a^2x^2 \pm (x^4 + a^2x^2 - 2a^2x^6)}{2a^3}$$

$$y = \frac{x^4 + a^2x^2 + x^4 - a^2x^2}{2a^3}, y = \frac{x^4 + a^2x^2 - x^4 + a^2x^2}{2a^3}$$

$$y = \frac{2x^4}{2a^3}, y = \frac{2a^2x^2}{2a^3}$$

$$y = \frac{x^4}{a^3}, y = \frac{x^2}{a}$$

Species put  $x > 0$ , we get two real values of  $y$  with same sign.

Put  $x < 0$ , we get two real values of  $y$  with same sign.

Hence  $(0, 0)$  is a double cusp of 2nd species.

(17)  $f(x, y) = (x-a)^2(2x-a) - y^3$

$$f_x = 2(3x^2 - 5ax + 2a^2)$$

$$f_y = -3y^2$$

$$f_x = 0$$

$$f_y = 0$$

$$2(3x^2 - 5ax + 2a^2) = 0$$

$$-3y^2 = 0$$

$$3x^2 - 5ax + 2a^2 = 0$$

$$y = 0$$

$$x = a, \frac{2a}{3}$$

So possible multiple pts are  $(a, 0)$  and  $(\frac{2a}{3}, 0)$ . out of these only  $(a, 0)$  is a multiple pt.

Shifting the origin, by

$$\text{putting } x = X + a$$

$$y = Y$$

$$Y^3 = (X + a - a)^2(2(X + a) - a)$$

$$Y^3 = 2X^3 + aX^2$$

Equate the lowest degree term to zero

$$aX^2 = 0$$

$$X^2 = 0$$

$$(X - a)^2 = 0$$

$$X - a = 0$$

$$X = a$$

which are coincident tangents

So  $(a, 0)$  is a cusp.

Given eq can be written as

$$y^3 = (x-a)^2(2x-a)$$

$$y = \pm(x-a)^{2/3}(2x-a)^{1/3}$$

Species when  $x > a$ ,

$y$  has two real values with different signs.

which show cusp is of 1st species.

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## Exercise 7.5

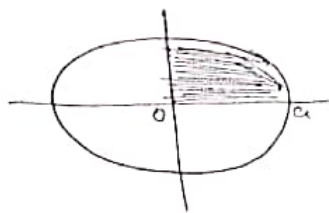
Q. No. 1  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$\frac{y^2}{b^2} = \frac{a^2 - x^2}{a^2}$$

$$y^2 = \frac{b^2}{a^2} (a^2 - x^2)$$

$$y = \frac{b}{a} \sqrt{a^2 - x^2}$$



Area of 1st region bounded by the curve = 4 (Shaded area)

$$= 4 \int_0^a y \, dx$$

$$= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx$$

$$= \frac{4b}{a} \left[ \frac{x \sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a$$

$$= \frac{4b}{a} \left[ 0 + \frac{a^2}{2} \sin^{-1}(1) \right]$$

$$= \frac{4b}{a} \cdot \frac{a^2}{2} \cdot \frac{\pi}{2}$$

$$A = \pi ab$$

(2)

$$y = \ln x$$

$$A = \int_a^b y \, dx$$

$$= \int_a^b \ln x \, dx$$

$$= [x \ln x - x]_a^b$$

$$= (b \ln b - b) - (a \ln a - a)$$

(3)

$$xy = c^2$$

$$y = \frac{c^2}{x}$$

$$A = \int_a^b y \, dx$$

$$= \int_a^b c^2 \cdot \frac{1}{x} \, dx$$

$$= c^2 \int_a^b \frac{1}{x} \, dx$$

$$= c^2 [\ln x]_a^b$$

$$= c^2 (\ln b - \ln a)$$

(4)

$$y^2 = 4ax$$

$$y = \sqrt{4ax}$$

Area of 1st region bounded by parabola and its L.R. = 2 (Shaded region)

$$= 2 \int_0^a y \, dx$$

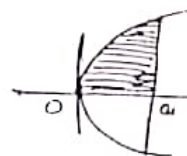
$$= 2 \int_0^a \sqrt{4ax} \, dx$$

$$= 2 \sqrt{4a} \int_0^a x^{1/2} \, dx$$

$$= 4\sqrt{a} \cdot \left[ \frac{x^{3/2}}{3/2} \right]_0^a$$

$$= \frac{8\sqrt{a}}{3} [a^{3/2}]$$

$$A = \frac{8}{3} a^2$$



(5)

$$y = c \cosh \frac{x}{c}$$

$$A = \int_a^b y \, dx$$

$$= \int_a^b c \cosh \frac{x}{c} \, dx$$

$$= c \int_a^b \cosh \frac{x}{c} \, dx$$

$$= c \cdot \left[ \frac{\sinh \frac{x}{c}}{1/c} \right]_a^b$$

$$= c^2 (\sinh \frac{b}{c} - \sinh \frac{a}{c})$$

(6) Let circular disc be

$$x^2 + y^2 = a^2$$

$$y^2 = a^2 - x^2$$

$$y = \sqrt{a^2 - x^2}$$

Req. Area = 2 (Shaded region)

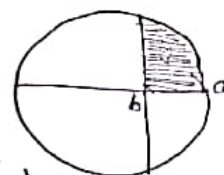
$$= 2 \int_0^a y \, dx$$

$$= 2 \int_0^a \sqrt{a^2 - x^2} \, dx$$

$$= 2 \left[ \frac{x \sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a$$

$$= 2 \left[ \frac{a^2}{2} \sin^{-1}(1) - \frac{b \sqrt{a^2 - b^2}}{2} - \frac{a^2}{2} \sin^{-1} \left( \frac{b}{a} \right) \right]$$

$$= 2 \left( \frac{\pi a^2}{4} - \frac{b \sqrt{a^2 - b^2}}{2} - \frac{a^2}{2} \sin^{-1} \left( \frac{b}{a} \right) \right)$$





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$$3ay^2 = x(x-a)^2$$

as powers of  $y$  even and  
no constant term.

So the curve is symmetrical  
about  $x$ -axis and passes  
through the origin.

The curve  $x$ -axis at  $(a, 0)$

$$3ay^2 = x(x-a)^2$$

$$y^2 = \frac{x(x-a)^2}{3a}$$

$$y = \frac{1}{\sqrt{3a}} \sqrt{x} (x-a)$$

$$= \frac{1}{\sqrt{3a}} (x^{3/2} - ax^{1/2})$$

2 area of the  
region bounded  
the loop of  
the curve



$$= 2 \text{ (Shaded area)}$$

$$= 2 \int_0^a y \, dx$$

$$= 2 \int_0^a \frac{1}{\sqrt{3a}} (x^{3/2} - ax^{1/2}) \, dx$$

$$= \frac{2}{\sqrt{3a}} \int_0^a (x^{3/2} - ax^{1/2}) \, dx$$

$$= \frac{2}{\sqrt{3a}} \left[ \frac{x^{5/2}}{5/2} - a \frac{x^{3/2}}{3/2} \right]_0^a$$

$$= \frac{2}{\sqrt{3a}} \left[ \frac{2}{5} x^{5/2} - \frac{2a}{3} x^{3/2} \right]_0^a$$

$$= \frac{2}{\sqrt{3a}} \left[ \frac{2a^{5/2}}{5} - \frac{2a}{3} a^{3/2} \right]$$

$$= \frac{2}{\sqrt{3a}} \left[ \frac{2a^{5/2}}{5} - \frac{2a^{5/2}}{3} \right]$$

$$= \frac{2 \cdot a^{5/2}}{\sqrt{3a}} \left( \frac{2}{5} - \frac{2}{3} \right)$$

$$= \frac{2}{\sqrt{3}} \frac{a^{5/2}}{a^{1/2}} \left( -\frac{4}{15} \right)$$

$$= \frac{2}{\sqrt{3}} a^{5/2} \left( -\frac{4}{15} \right)$$

$$= \frac{8a^2}{15\sqrt{3}} \text{ (Area always +ve)}$$

(8)

$$x^2(x^2+y^2) = a^2(y^2-x^2)$$

$$x^4+x^2y^2 = a^2y^2-a^2x^2$$

$$x^4+a^2x^2 = a^2y^2-x^2y^2$$

$$x^2(x^2+a^2) = y^2(a^2-x^2)$$

$$y^2 = \frac{x^2(x^2+a^2)}{(a^2-x^2)}$$

$$y = x \sqrt{\frac{a^2+x^2}{a^2-x^2}}$$

Powers of  $y$  and powers of  $x$   
are even, so symmetrical  
about  $x$ -axis and  $y$ -axis  
No constant term, it passes  
through the origin.

For vertical asymptotes, equate  
the coefficient of highest  
power of  $y$  to zero.

$$\text{So } x^2 - a^2 = 0$$

$$x^2 = a^2$$

$$x = \pm a$$



$$\text{Req. Area} = 4 \text{ (shaded region)}$$

$$= 4 \int_0^a y \, dx$$

$$= 4 \int_0^a x \sqrt{\frac{a^2+x^2}{a^2-x^2}} \, dx$$

$$\text{put } x^2 = a^2 \cos \theta$$

$$2x \, dx = a^2 (-\sin \theta) \, d\theta$$

$$x \, dx = -\frac{a^2 \sin \theta}{2} \, d\theta$$

we also change the

limit when  $x=0$ ,  $\theta = \pi/2$

$x=a$ ,  $\theta = 0$

$$= 4 \int_{\pi/2}^0 \sqrt{\frac{a^2+a^2 \cos \theta}{a^2-a^2 \cos \theta}} \left( -\frac{a^2}{2} \sin \theta \right) \, d\theta$$

$$= 2a^2 \int_0^{\pi/2} \sqrt{\frac{1+\cos \theta}{1-\cos \theta}} \cdot \sin \theta \, d\theta$$

$$= 2a^2 \int_0^{\pi/2} \sqrt{\frac{2 \cos^2 \theta/2}{2 \sin^2 \theta/2}} \cdot \sin \theta \, d\theta$$

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$$\begin{aligned}
 &= 2a^2 \int_0^{\pi/2} \frac{\cos \theta/2}{\sin \theta/2} \cdot 2 \sin \theta/2 \cos \theta/2 d\theta \\
 &= 2a^2 \int_0^{\pi/2} 2 \cos^2 \theta/2 d\theta \\
 &= 2a^2 \int_0^{\pi/2} (1 + \cos \theta) d\theta \\
 &= 2a^2 \left[ \theta + \sin \theta \right]_0^{\pi/2} \\
 &= 2a^2 \left[ \frac{\pi}{2} + \sin \frac{\pi}{2} - (0 + \sin 0) \right] \\
 &= 2a^2 \left( \frac{\pi}{2} + 1 \right) \\
 &= a^2 (\pi + 2)
 \end{aligned}$$

⑨

$$\begin{aligned}
 xy^2 &= 4(2-x) \\
 xy^2 &= 8-4x \\
 xy^2 + 4x &= 8 \\
 x(y^2 + 4) &= 8 \\
 x &= \frac{8}{y^2 + 4}
 \end{aligned}$$

Symmetry  
Power of  $y$  is even, Symmetrical about  $x$ -axis, Constant term shows that it does not pass through the origin.

$x$ -Intercept Put  $y=0$   
 $x=2$

Vertical Asymptotes  $x=0$

Req. Area  $= 2$  (Shaded region)

$$\begin{aligned}
 &= 2 \int_0^{\infty} x dy \\
 &= 2 \int_0^{\infty} \frac{8}{y^2 + 4} dy \\
 &= 2 \cdot 8 \int_0^{\infty} \frac{1}{y^2 + 2^2} dy \\
 &= 16 \cdot \left[ \frac{1}{2} \tan^{-1} \frac{y}{2} \right]_0^{\infty} \\
 &= 8 (\tan^{-1} \infty - \tan^{-1}(0)) \\
 &= 8 \left( \frac{\pi}{2} - 0 \right) = 4\pi
 \end{aligned}$$

⑩

$$\begin{aligned}
 x^2 y^2 &= a^2 (y^2 - x^2) \\
 x^2 y^2 &= a^2 y^2 - a^2 x^2 \\
 x^2 y^2 - a^2 y^2 &= -a^2 x^2 \\
 y^2 (x^2 - a^2) &= -a^2 x^2 \\
 y^2 &= \frac{-a^2 x^2}{x^2 - a^2} \\
 y^2 &= \frac{a^2 x^2}{a^2 - x^2} \\
 y &= \frac{ax}{\sqrt{a^2 - x^2}}
 \end{aligned}$$

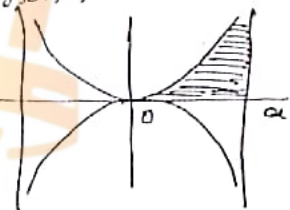
Symmetry powers of  $y$  and powers of  $x$  are even, so it is symmetrical about both the axes. No constant term, it passes through the origin.

Vertical Asymptotes

$$\begin{aligned}
 x^2 - a^2 &= 0 \\
 x^2 &= a^2 \\
 x &= \pm a
 \end{aligned}$$

Req. Area  $= 4$  (Shaded region)

$$\begin{aligned}
 &= 4 \int_0^a y dx \\
 &= 4 \int_0^a \frac{ax}{\sqrt{a^2 - x^2}} dx
 \end{aligned}$$



$$\begin{aligned}
 &= \frac{4a}{-2} \int_0^a (a^2 - x^2)^{-1/2} (-2x) dx \\
 &= -2a \cdot \left[ \frac{(a^2 - x^2)^{1/2}}{1/2} \right]_0^a \\
 &= -4a [0 - a] \\
 &= 4a^2
 \end{aligned}$$

⑪

$$\begin{aligned}
 ay^2 &= x^2(a-x) \\
 y^2 &= \frac{x^2(a-x)}{a} \\
 y &= \frac{1}{\sqrt{a}} x \sqrt{a-x}
 \end{aligned}$$

Symmetry :- power of  $y$  is even, so it is symmetrical about  $x$ -axis. No constant term, it passes through the origin.

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x-intercept put  $y=0$

$$x^2(a-x)=0$$

$$x^2=0 \quad \text{or} \quad a-x=0$$

$$x=0 \quad \quad \quad x=a$$

Req. Area = 2 (Shaded area)

$$2 \int_0^a y \, dx$$

$$2 \int_0^a \frac{1}{\sqrt{a}} \cdot x \sqrt{a-x} \, dx$$

$$\frac{2}{\sqrt{a}} \int_0^a (-x) \sqrt{a-x} \, dx$$

$$-\frac{2}{\sqrt{a}} \int_0^a (a-x) \sqrt{a-x} \, dx$$

$$-\frac{2}{\sqrt{a}} \int_0^a (a-x) \sqrt{a-x} - a \sqrt{a-x} \, dx$$

$$-\frac{2}{\sqrt{a}} \int_0^a (a-x)^{3/2} - a (a-x)^{1/2} \, dx$$

$$= \frac{2}{\sqrt{a}} \int_0^a (a-x)(-1) \, dx + a \int_0^a (a-x)(-1) \, dx$$

$$= \frac{2}{\sqrt{a}} \left[ \frac{(a-x)^{5/2}}{5/2} \right]_0^a + a \left[ \frac{(a-x)^{3/2}}{3/2} \right]_0^a$$

$$= \frac{2}{\sqrt{a}} \left[ \frac{2}{5} (a-x)^{5/2} \right]_0^a + \frac{2a}{3} (a-x)^{3/2} \Big|_0^a$$

$$= \frac{2}{\sqrt{a}} \left[ -\frac{2}{5} a^{5/2} + \frac{2a}{3} \cdot a^{3/2} \right]$$

$$= + \frac{4}{\sqrt{a}} \left[ a^{5/2} \left( -\frac{1}{5} + \frac{1}{3} \right) \right]$$

$$= 4a^2 \left( \frac{2}{15} \right)$$

$$= \frac{8a^2}{15}$$

(2)  $a^2 x^2 = y^3 (2a-y)$

$$a^2 x^2 = y^3 (2a-y)$$

$$x^2 = \frac{y^3 (2a-y)}{a^2}$$

$$x = \frac{y^{3/2} (2a-y)^{1/2}}{a}$$

Symmetry power of  $x$  is even.  
so it is symmetrical  
about  $y$ -axis  
No constant term, it  
passes through the origin.

y-intercept put  $x=0$

$$y^3 (2a-y) = 0$$

$$y^3 = 0 \quad \text{or} \quad 2a-y=0$$

$$y=0 \quad \quad \quad y=2a$$

Req. Area = 2 (Shaded region)

$$= 2 \int_0^{2a} x \, dy$$

$$= 2 \int_0^{2a} \frac{y^{3/2}}{a} (2a-y)^{1/2} \, dy$$

put  $y = 2a \sin^2 \theta$

$$dy = 4a \sin \theta \cos \theta \, d\theta$$

as  $y \rightarrow 0$ ,  $\theta \rightarrow 0$

$y \rightarrow 2a$ ,  $\theta \rightarrow \pi/2$

$$= \frac{2}{a} \int_0^{\pi/2} (2a \sin^2 \theta)^{3/2} (2a - 2a \sin^2 \theta)^{1/2} \cdot 4a \sin \theta \cos \theta \, d\theta$$

$$= \frac{2}{a} \int_0^{\pi/2} (2a)^{3/2} \cdot \sin^3 \theta \cdot (2a)^{1/2} (1 - \sin^2 \theta)^{1/2} \cdot 4a \sin \theta \cos \theta \, d\theta$$

$$= \frac{2}{a} \cdot (2a)^{3/2} \cdot (2a)^{1/2} \cdot 4a \cdot \int_0^{\pi/2} \sin^4 \theta \cdot (\cos^2 \theta)^{1/2} \cdot \sin \theta \cos \theta \, d\theta$$

$$= 4 \cdot 8 a^2 \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta \, d\theta$$

$$= 32a^2 \cdot \frac{3 \cdot 1 \cdot 1}{6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2}$$

$$= \pi a^2$$

(13)  $y^2(a+x) = x^2(a-x)$

$$y^2 = \frac{x^2(a-x)}{(a+x)}$$

$$y = \pm x \sqrt{\frac{a-x}{a+x}}$$

Symmetry:- power of  $y$  is even  
so it is symmetrical about  
 $x$ -axis.

No constant term, it passes  
through the origin.

x-intercept put  $y=0$

$$x^2(a-x)=0$$

$$x^2=0 \quad \text{or} \quad a-x=0$$

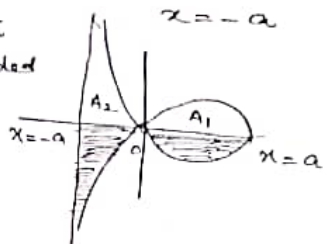
$$x=0 \quad \quad \quad x=a$$



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Vertical asymptotes

$A_1$  = Area of the region bounded by the loop of the curve.



$A_1$  = Req. Area = 2 (Shaded region)

$$= 2 \int_0^a y \, dx \quad (\text{Take +ve } y)$$

$$= 2 \int_0^a x \sqrt{\frac{a-x}{a+x}} \, dx$$

Put  $x = a \cos \theta$   
 $dx = -a \sin \theta \, d\theta$

When  $x = 0$ ,  $\theta = \pi/2$   
 $x = a$ ,  $\theta = 0$

$$= 2 \int_{\pi/2}^0 a \cos \theta \sqrt{\frac{a-a \cos \theta}{a+a \cos \theta}} (-a \sin \theta \, d\theta)$$

$$= 2 \int_0^{\pi/2} a^2 \cos \theta \sin \theta \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \, d\theta$$

$$= 2 a^2 \int_0^{\pi/2} \cos \theta \sin \theta \sqrt{\frac{2 \sin^2 \theta/2}{2 \cos^2 \theta/2}} \, d\theta$$

$$= 2 a^2 \int_0^{\pi/2} \cos \theta \cdot 2 \sin \theta/2 \cos \theta/2 \cdot \frac{\sin \theta/2}{\cos \theta/2} \, d\theta$$

$$= 2 a^2 \int_0^{\pi/2} \cos \theta \cdot 2 \sin^2 \theta/2 \, d\theta$$

$$= 2 a^2 \int_0^{\pi/2} \cos \theta (1 - \cos \theta) \, d\theta$$

$$= 2 a^2 \int_0^{\pi/2} (\cos \theta - \cos^2 \theta) \, d\theta$$

using wallis's formula

$$= 2 a^2 (1 - \frac{1}{2} \cdot \frac{\pi}{2})$$

$$= 2 a^2 (1 - \frac{\pi}{4})$$

$$A_1 = a^2 (2 - \frac{\pi}{2})$$

2ndly

$A_2$  = Area of the region bounded by the curve and its asymptotes.  
 = 2 (Shaded region)

$$A_2 = 2 \int_{-a}^0 y \, dx$$

$$= 2 \int_{-a}^0 -x \sqrt{\frac{a-x}{a+x}} \, dx \quad (\text{Take -ve})$$

Put  $x = a \cos \theta$   
 $dx = -a \sin \theta \, d\theta$

When  $x = 0$ ,  $\theta = \pi/2$

$x = -a$ ,  $\theta = \pi$

$$= 2 \int_{\pi}^{\pi/2} -(a \cos \theta) \sqrt{\frac{a-a \cos \theta}{a+a \cos \theta}} (-a \sin \theta \, d\theta)$$

$$= 2 a^2 \int_{\pi}^{\pi/2} \cos \theta \sin \theta \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \, d\theta$$

$$= 2 a^2 \int_{\pi}^{\pi/2} \cos \theta \sin \theta \sqrt{\frac{2 \sin^2 \theta/2}{2 \cos^2 \theta/2}} \, d\theta$$

$$= 2 a^2 \int_{\pi}^{\pi/2} \cos \theta \sin \theta \cdot \frac{\sin \theta/2}{\cos \theta/2} \, d\theta$$

$$= 2 a^2 \int_{\pi}^{\pi/2} \cos \theta \cdot 2 \sin \theta/2 \cos \theta/2 \cdot \frac{\sin \theta/2}{\cos \theta/2} \, d\theta$$

$$= 2 a^2 \int_{\pi}^{\pi/2} \cos \theta \cdot 2 \sin^2 \theta/2 \, d\theta$$

$$= 2 a^2 \int_{\pi}^{\pi/2} \cos \theta (1 - \cos \theta) \, d\theta$$

$$= 2 a^2 \int_{\pi}^{\pi/2} (\cos \theta - \cos^2 \theta) \, d\theta$$

$$= 2 a^2 \int_{\pi}^{\pi/2} (\cos \theta - \frac{1+\cos 2\theta}{2}) \, d\theta$$

$$= 2 a^2 \left[ \sin \theta - \frac{1}{2} (\theta - \frac{\sin 2\theta}{2}) \right]_{\pi}^{\pi/2}$$

$$= 2 a^2 \left[ \left( \sin \frac{\pi}{2} - \frac{1}{2} (\frac{\pi}{2} - \frac{\sin \pi}{2}) \right) - \left( \sin \pi - \frac{1}{2} (\pi - \frac{\sin 2\pi}{2}) \right) \right]$$

$$= 2 a^2 \left[ 1 - \frac{1}{2} (\frac{\pi}{2} - 0) - (0 - \frac{1}{2} \pi) \right]$$

$$= 2 a^2 (1 - \frac{\pi}{4} + \frac{\pi}{2})$$

$$= 2 a^2 (1 - \frac{\pi}{4})$$

$$A_2 = a^2 (2 - \frac{\pi}{2})$$

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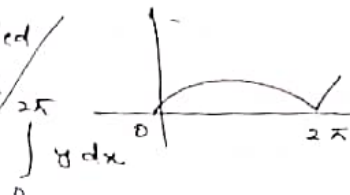
(14)

$$y = a(1 - \cos \theta)$$

$$x = a(\theta - \sin \theta)$$

$$dx = a(1 - \cos \theta) d\theta$$

Area bounded  
by cycloid and  
its base



$$= \int_0^{2\pi} y dx$$

$$= \int_0^{2\pi} a(1 - \cos \theta) a(1 - \cos \theta) d\theta$$

$$= a^2 \int_0^{2\pi} (1 - \cos \theta)^2 d\theta$$

$$= a^2 \int_0^{2\pi} (2 \sin^2 \theta/2)^2 d\theta$$

$$= 4a^2 \int_0^{2\pi} \sin^4 \theta/2 d\theta$$

$$\theta \rightarrow 0, z \rightarrow 0$$

$$\theta \rightarrow 2\pi, z \rightarrow \pi$$

$$\left. \begin{array}{l} \text{Put } \theta/2 = z \\ \theta = 2z \\ d\theta = 2dz \end{array} \right\}$$

$$= 4a^2 \int_0^{\pi} \sin^4 z \cdot 2 dz$$

$$= 8a^2 \int_0^{\pi} \sin^4 z dz$$

$$= 8a^2 \cdot 2 \int_0^{\pi/2} \sin^4 z dz$$

using wallis sine formula

$$= 16a^2 \cdot \frac{3 \cdot 1}{4 \cdot 2} \cdot \frac{\pi}{2}$$

$$= 3\pi a^2$$

(15)

$$x^4 + y^4 = 2a^2 xy$$

First, we convert it into  
polar form

$$\text{Put } x = r \cos \theta, \quad y = r \sin \theta$$

$$r^4 \cos^4 \theta + r^4 \sin^4 \theta = 2a^2 \cdot r \cos \theta \cdot r \sin \theta$$

$$r^4 (\cos^4 \theta + \sin^4 \theta) = 2a^2 r^2 \cos \theta \sin \theta$$

$$r^2 (\cos^4 \theta + \sin^4 \theta) = 2a^2 \cos \theta \sin \theta$$

$$r^2 = \frac{2a^2 \cos \theta \sin \theta}{\cos^4 \theta + \sin^4 \theta}$$

For limits of  $\theta$   
put  $r=0$

$$\frac{2a^2 \cos \theta \sin \theta}{\cos^4 \theta + \sin^4 \theta} = 0$$

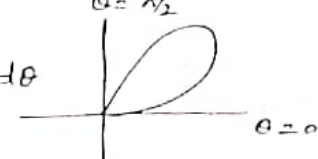
$$2a^2 \cos \theta \sin \theta = 0$$

$$\cos \theta \sin \theta = 0$$

$$\cos \theta = 0, \quad \sin \theta = 0$$

$$\theta = \pi/2, \quad \theta = 0$$

Req. Area =  $\int_0^{\pi/2} \frac{1}{2} r^2 d\theta$



$$= \frac{1}{2} \int_0^{\pi/2} \frac{2a^2 \cos \theta \sin \theta}{\cos^4 \theta + \sin^4 \theta} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{2a^2 \cos \theta \sin \theta / \cos^4 \theta}{\frac{\cos^4 \theta}{\cos^4 \theta} + \frac{\sin^4 \theta}{\cos^4 \theta}} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{2a^2 \tan \theta \sec^2 \theta}{1 + \tan^4 \theta} d\theta$$

when  $\left. \begin{array}{l} \theta = 0, z = 0 \\ \theta = \pi/2, z = \infty \end{array} \right\} \text{ Put } \tan^2 \theta = z$

$$2 \tan \theta \sec^2 \theta d\theta = dz$$

$$= \frac{1}{2} \int_0^{\infty} \frac{a^2 \cdot dz}{1 + z^2}$$

$$= \frac{a^2}{2} \int_0^{\infty} \frac{1}{1 + z^2} dz$$

$$= \frac{a^2}{2} \cdot \left[ \tan^{-1} z \right]_0^{\infty}$$

$$= \frac{a^2}{2} \left[ \tan^{-1} \infty - \tan^{-1}(0) \right]$$

$$= \frac{a^2}{2} \left[ \frac{\pi}{2} - 0 \right] = \frac{\pi a^2}{4}$$

(16)

$$(x+y)^2(x^2+y^2) = 2a^2 xy$$

First, we convert it into  
polar form.

$$\text{Put } x = r \cos \theta, \quad y = r \sin \theta$$

$$(r \cos \theta + r \sin \theta)^2 (r^2 \cos^2 \theta + r^2 \sin^2 \theta) =$$

$$2a^2 \cdot r \cos \theta \cdot r \sin \theta$$

$$\Rightarrow r^2 (\cos \theta + \sin \theta)^2 \cdot r^2 (\cos^2 \theta + \sin^2 \theta) =$$

$$2a^2 r^2 \cos \theta \sin \theta$$

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$$r^2 (\cos \theta + \sin \theta)^2 = 2a \cos \theta \sin \theta$$

$$r^2 = \frac{2a \cos \theta \sin \theta}{(\cos \theta + \sin \theta)^2}$$

For limit of  $\theta$

Put  $r = 0$

$$\frac{2a \cos \theta \sin \theta}{(\cos \theta + \sin \theta)^2} = 0$$

$$2a \cos \theta \sin \theta = 0$$

$$\cos \theta = 0, \sin \theta = 0$$

$$\theta = \pi/2, \theta = 0$$

So area

$$A = \int_0^{\pi/2} \frac{1}{2} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{2a \cos \theta \sin \theta}{(\cos \theta + \sin \theta)^2} d\theta$$

$$= \frac{a}{2} \int_0^{\pi/2} \frac{\sin 2\theta}{\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta} d\theta$$

$$= \frac{a}{2} \int_0^{\pi/2} \frac{\sin 2\theta}{1 + \sin 2\theta} d\theta$$

$$= \frac{a}{2} \int_0^{\pi/2} \frac{1 + \sin 2\theta - 1}{1 + \sin 2\theta} d\theta$$

$$= \frac{a}{2} \int_0^{\pi/2} \left( \frac{1 + \sin 2\theta}{1 + \sin 2\theta} - \frac{1}{1 + \sin 2\theta} \right) d\theta$$

$$= \frac{a}{2} \int_0^{\pi/2} \left( 1 - \frac{1}{1 + \sin 2\theta} \right) d\theta$$

$$= \frac{a}{2} \left[ \int_0^{\pi/2} 1 d\theta - \int_0^{\pi/2} \frac{1}{1 + \sin 2\theta} \cdot \frac{1 - \sin 2\theta}{1 - \sin 2\theta} d\theta \right]$$

$$= \frac{a}{2} \left[ \theta \Big|_0^{\pi/2} - \int_0^{\pi/2} \frac{1 - \sin 2\theta}{1 - \sin^2 \theta} d\theta \right]$$

$$= \frac{a}{2} \left[ \frac{\pi}{2} - \int_0^{\pi/2} \frac{1 - \sin 2\theta}{\cos^2 \theta} d\theta \right]$$

$$= \frac{a}{2} \left[ \frac{\pi}{2} - \int_0^{\pi/2} \left( \frac{1}{\cos^2 \theta} - \frac{\sin 2\theta}{\cos^2 \theta} \right) d\theta \right]$$

$$= \frac{a}{2} \left[ \frac{\pi}{2} - \left[ \sec^2 \theta - \sec \theta \tan \theta \right]_0^{\pi/2} \right]$$

$$= \frac{a}{2} \left[ \frac{\pi}{2} - \left[ \frac{\tan 2\theta}{2} - \frac{\sec 2\theta}{2} \right]_0^{\pi/2} \right]$$

$$= \frac{a}{2} \left[ \frac{\pi}{2} - 0 + \left( \frac{\sec \pi}{2} - \frac{\sec 0}{2} \right) \right]$$

$$= \frac{a}{2} \left[ \frac{\pi}{2} + (-1/2 - 1/2) \right]$$

$$= \frac{a}{2} (\pi/2 - 1)$$

(17) Circle:  $r = a$

Five leaved Rose:  $r = a \cos 5\theta$

For limit of  $\theta$

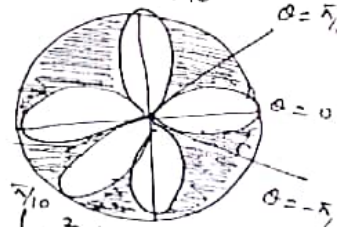
Put  $r = 0$

$$a \cos 5\theta = 0$$

$$\cos 5\theta = 0$$

$$5\theta = \pm \pi/2$$

$$\theta = \pm \pi/10$$



It has ten portions.  
limit of one portion is  $\pi/10$ .

Area of one portion =  $\frac{1}{2} \int_0^{\pi/10} r^2 d\theta$

$$= \frac{1}{2} \int_0^{\pi/10} a^2 \cos^2 5\theta d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/10} \cos^2 5\theta d\theta$$

Put  $5\theta = z$

$$5d\theta = dz$$

when  $\theta = 0, z = 0$

$$= \frac{a^2}{5 \cdot 2} \int_0^{\pi/10} \cos^2 z dz$$

$$= \frac{a^2}{10} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi a^2}{40}$$

Req. Area of 10 portions =  $10 \left( \frac{\pi a^2}{40} \right) = \frac{\pi a^2}{4}$

Req. Area :- Area of circle - Area of five leaves rose

$$= \pi a^2 - \frac{\pi a^2}{4}$$

$$= \frac{3}{4} \pi a^2$$

$$= \frac{3}{4} (\text{Area of circle})$$



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(18)

$$r = a(1 + \cos \theta)$$

For the limit  
of  $\theta$

$$\text{Put } r = 0$$

$$\cos \theta = -1$$

$$A = 2 \int_0^\pi \frac{1}{2} r^2 d\theta$$

$$= \int_0^\pi a^2 (1 + \cos \theta)^2 d\theta$$

$$= a^2 \int_0^\pi (2 \cos^2 \frac{\theta}{2})^2 d\theta$$

$$= 4a^2 \int_0^\pi \cos^4 \frac{\theta}{2} d\theta$$

Let

$$\theta = 0, z = 0$$

$$\theta = \pi, z = \pi/2$$

$$\text{Put } \frac{\theta}{2} = z$$

$$\theta = 2z$$

$$d\theta = 2dz$$

$$= 4a^2 \int_0^{\pi/2} \cos^4 z \cdot 2dz$$

$$= 8a^2 \cdot \frac{3 \cdot 1}{4 \cdot 2} \cdot \frac{\pi}{2} = \frac{3\pi a^2}{2}$$



(20)

$$r = \frac{3a \sin \theta \cos \theta}{\cos^3 \theta + \sin^3 \theta}$$

For the limit of  $\theta$ :

$$\text{Put } r = 0$$

$$\frac{3a \sin \theta \cos \theta}{\cos^3 \theta + \sin^3 \theta} = 0$$

$$3a \sin \theta \cos \theta = 0$$

$$\sin \theta \cos \theta = 0$$

$$\sin \theta = 0 \text{ or } \cos \theta = 0$$

$$\theta = 0$$

$$\theta = \pi/2$$

$$\text{Area of the loop} = \int_0^{\pi/2} \frac{1}{2} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{9a^2 \sin^2 \theta \cos^2 \theta}{(\cos^3 \theta + \sin^3 \theta)^2} d\theta$$

$$= \frac{9a^2}{2} \int_0^{\pi/2} \frac{\sin^2 \theta \cos^2 \theta}{(\cos^3 \theta + \sin^3 \theta)^2} d\theta$$

$$= \frac{9a^2}{2} \int_0^{\pi/2} \frac{\sin^2 \theta \cos^2 \theta}{\cos^4 \theta (1 + \tan^3 \theta)^2} d\theta$$

$$= \frac{9a^2}{2} \int_0^{\pi/2} \frac{\tan^2 \theta \sec^2 \theta}{(1 + \tan^3 \theta)^2} d\theta$$

$$\text{For } \theta = 0, z = 1$$

$$\theta = \pi/2, z = \infty$$

$$\text{Put } 1 + \tan^3 \theta = z$$

$$3 \tan^2 \theta \sec^2 \theta d\theta = dz$$

$$\tan^2 \theta \sec^2 \theta d\theta = \frac{1}{3} dz$$

$$= \frac{9a^2}{2} \int_1^\infty \frac{1}{z^2} \cdot \frac{1}{3} dz$$

$$= \frac{3a^2}{2} \left[ \left( -\frac{1}{z} \right) \right]_1^\infty$$

$$= \frac{3a^2}{2} \left( -\frac{1}{\infty} + 1 \right)$$

$$= \frac{3a^2}{2} (0 + 1)$$

$$= \frac{3a^2}{2}$$



For the limit  
of  $\theta$ :

$$\text{Put } r = 0$$

$$\cos 2\theta = 0$$

$$2\theta = \pi/2$$

$$\theta = \pi/4$$

$$\text{Req. Area} = 4 \left( \text{shaded region} \right)$$

$$= 4 \int_0^{\pi/4} \frac{1}{2} r^2 d\theta$$

$$= 2 \int_0^{\pi/4} a^2 \cos 2\theta d\theta$$

$$= 2a^2 \left[ \frac{\sin 2\theta}{2} \right]_0^{\pi/4}$$

$$= a^2 (\sin \pi/2 - \sin 0)$$

$$= a^2 (1 - 0)$$

$$= a^2$$

(21)

$$r = \frac{6}{2 - \cos \theta}$$

transform into rectangular coordinates.

$$2r - r \cos \theta = 6$$

$$\text{Put } r \cos \theta = x$$

$$r = \sqrt{x^2 + y^2}$$

$$2\sqrt{x^2 + y^2} - x = 6$$

$$2\sqrt{x^2 + y^2} = 6 + x$$

Squaring both sides

$$4(x^2 + y^2) = 36 + x^2 + 12x$$

$$4x^2 + 4y^2 = 36 + x^2 + 12x$$

$$3x^2 - 12x + 4y^2 = 36$$

$$3(x^2 - 4x) + 4y^2 = 36$$

$$3(x^2 - 4x + 4 - 4) + 4y^2 = 36$$

$$3(x^2 - 4x + 4) - 12 + 4y^2 = 36$$

$$3(x-2)^2 + 4y^2 = 36 + 12$$

$$3(x-2)^2 + 4y^2 = 48$$

$$\frac{3(x-2)^2}{48} + \frac{4y^2}{48} = 1$$

$$\frac{(x-2)^2}{16} + \frac{y^2}{12} = 1$$

$$\frac{(x-2)^2}{4^2} + \frac{y^2}{(2\sqrt{3})^2} = 1$$

is an ellipse.

$$a = 4, \quad b = 2\sqrt{3}$$

$$\text{Area of ellipse} = \pi ab$$

$$= \pi(4)(2\sqrt{3})$$

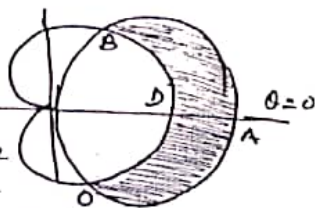
$$= 8\sqrt{3}\pi$$

(22)

$$\text{Cardioid: } r = 1 + \cos \theta$$

$$\text{Circle: } r = 3 \cos \theta$$

we are to find the area of shaded region which is difference of area of circle and area of cardioid.



$$\text{As } r = 1 + \cos \theta, \quad r = 3 \cos \theta$$

$$1 + \cos \theta = 3 \cos \theta$$

$$2 \cos \theta = 1$$

$$\cos \theta = 1/2$$

$$\theta = \pm \pi/3$$

$$\text{Req. Area} = 2 \left[ \text{Area of } OABO - \text{Area of } ODBO \right]$$

$$= 2 \left[ \int_0^{\pi/3} \frac{1}{2} r_1^2 d\theta - \int_0^{\pi/3} \frac{1}{2} r_2^2 d\theta \right]$$

$$= \int_0^{\pi/3} (r_1^2 - r_2^2) d\theta$$

$$= \int_0^{\pi/3} (3 \cos \theta)^2 - (1 + \cos \theta)^2 d\theta$$

$$= \int_0^{\pi/3} (9 \cos^2 \theta - (1 + \cos^2 \theta + 2 \cos \theta)) d\theta$$

$$= \int_0^{\pi/3} (8 \cos^2 \theta - 2 \cos \theta - 1) d\theta$$

$$= \int_0^{\pi/3} (4(1 + \cos 2\theta) - 2 \cos \theta - 1) d\theta$$

$$= \int_0^{\pi/3} (4 + 4 \cos 2\theta - 2 \cos \theta - 1) d\theta$$

$$= \int_0^{\pi/3} (3 + 4 \cos 2\theta - 2 \cos \theta) d\theta$$

$$= \left[ 3\theta + 2 \sin 2\theta - 2 \sin \theta \right]_0^{\pi/3}$$

$$= 2 \sin 2\pi/3 - 2 \sin \pi/3 + 3\pi/3$$

$$= 2 \cdot \frac{\sqrt{3}}{2} - 2 \cdot \frac{1}{2} + \pi$$

$$= \pi$$

(23)

$$r^2 = 2a^2 \cos 2\theta$$

$$r = a$$

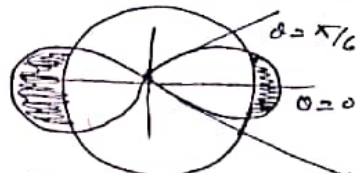
$$\text{As } a^2 = 2a^2 \cos 2\theta$$

$$2 \cos 2\theta = 1$$

$$\cos 2\theta = 1/2$$

$$2\theta = \pm \pi/3$$

$$\theta = \pm \pi/6$$



we are to find the area of shaded region.

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$$\begin{aligned}
 \therefore \text{Area} &= 4 \left[ \int_0^{\pi/6} \frac{1}{2} r_1^2 d\theta - \int_0^{\pi/6} \frac{1}{2} r_2^2 d\theta \right] \\
 &= 4 \cdot \frac{1}{2} \left[ \int_0^{\pi/6} (r_1^2 - r_2^2) d\theta \right] \\
 &= 2 \left[ \int_0^{\pi/6} (2a^2 \cos 2\theta - a^2) d\theta \right] \\
 &= 2a^2 \left[ \int_0^{\pi/6} (2\cos 2\theta - 1) d\theta \right] \\
 &= 2a^2 \left[ \frac{2 \sin 2\theta}{2} - \theta \right]_0^{\pi/6} \\
 &= 2a^2 \left[ \sin \frac{\pi}{3} - \frac{\pi}{6} \right] \\
 &= 2a^2 \left( \frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) \\
 &= a^2 \left( \sqrt{3} - \frac{\pi}{3} \right)
 \end{aligned}$$

$$r = 2 - 3 \sin \theta$$

$$\text{Put } r = 0$$

$$2 - 3 \sin \theta = 0$$

$$3 \sin \theta = 2$$

$$\sin \theta = \frac{2}{3}$$

$$\theta = \sin^{-1}(\frac{2}{3})$$

The left half of smaller loop is determined by

$$\theta = \sin^{-1}(\frac{2}{3}) \text{ to } \theta = \frac{\pi}{2}$$

$$\text{Req. Area} = 2 \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} \frac{1}{2} r^2 d\theta$$

$$= \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} (2 - 3 \sin \theta)^2 d\theta$$

$$= \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} (4 + 9 \sin^2 \theta - 12 \sin \theta) d\theta$$

$$= \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} (4 + \frac{9}{2} (1 - \cos 2\theta) - 12 \sin \theta) d\theta$$

$$\begin{aligned}
 &= \left[ 4\theta + \frac{9}{2}\theta - \frac{9}{2} \cdot \frac{\sin 2\theta}{2} - 12 \cos \theta \right]_{\sin^{-1}(\frac{2}{3})}^{\pi/2} \\
 &= \left[ \frac{17}{2}\theta - \frac{9}{4} \sin 2\theta - 12 \cos \theta \right]_{\sin^{-1}(\frac{2}{3})}^{\pi/2}
 \end{aligned}$$

$$\begin{aligned}
 \text{as } \theta = \sin^{-1}(\frac{2}{3}) \quad \left| \begin{aligned} \cos \theta &= \sqrt{1 - \sin^2 \theta} \\ &= \sqrt{1 - 4/9} \\ \cos \theta &= \frac{\sqrt{5}}{3} \end{aligned} \right.
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{17}{2} \cdot \frac{\pi}{2} + 12 \cos \frac{\pi}{2} - \frac{9}{4} \sin \pi - \frac{17}{2} \sin^{-1} \frac{2}{3} \\
 &- 12 \cos (\sin^{-1} \frac{2}{3}) + \frac{9}{4} \cdot \sin 2 (\sin^{-1} \frac{2}{3}) \\
 &= \frac{17\pi}{4} + 0 - 0 - \frac{17}{2} \sin^{-1} \frac{2}{3} - 12 \cos \theta \\
 &\quad + \frac{9}{4} \sin 2\theta \\
 &= \frac{17\pi}{4} - \frac{17}{2} \sin^{-1} \frac{2}{3} - 12 \cdot \frac{\sqrt{5}}{3} + \frac{9}{4} \cdot 2 \sin \theta \cos \theta \\
 &= \frac{17\pi}{4} - \frac{17}{2} \sin^{-1} \frac{2}{3} - 4\sqrt{5} + \frac{9}{4} \cdot 2 \cdot \frac{2}{3} \cdot \frac{\sqrt{5}}{3} \\
 &= \frac{17\pi}{4} - \frac{17}{2} \sin^{-1} \frac{2}{3} - 4\sqrt{5} + \sqrt{5} \\
 &= \frac{17\pi}{4} - \frac{17}{2} \sin^{-1} \frac{2}{3} - 3\sqrt{5}
 \end{aligned}$$

(25)

$$r = 2 + 3 \cos \theta$$

$$\text{Put } r = 0$$

$$2 + 3 \cos \theta = 0$$

$$3 \cos \theta = -2$$

$$\cos \theta = -\frac{2}{3}$$

$$\theta = \cos^{-1}(-\frac{2}{3})$$

The upper half of the outer loop has limits  $\theta = 0$

$$\text{and } \theta = \cos^{-1}(-\frac{2}{3}) = 4 \text{ (say)}$$

The lower half of the smaller loop has limits  $\theta = \cos^{-1}(\frac{2}{3}) = 4$

$$\text{Req. Area} = 2 \int_0^4 \frac{1}{2} r^2 d\theta - 2 \int_4^{\pi} \frac{1}{2} r^2 d\theta$$

$$= \int_0^4 (2 + 3 \cos \theta)^2 d\theta - \int_4^{\pi} (2 + 3 \cos \theta)^2 d\theta$$

$$= \int_0^4 (4 + 9 \cos^2 \theta + 12 \cos \theta) d\theta - \int_4^{\pi} (4 + 9 \cos^2 \theta + 12 \cos \theta) d\theta$$

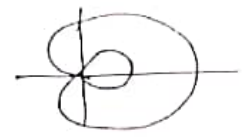
$$= \int_0^4 (4 + \frac{9}{2} (1 + \cos 2\theta) + 12 \cos \theta) d\theta - \int_4^{\pi} (4 + \frac{9}{2} (1 + \cos 2\theta) + 12 \cos \theta) d\theta$$

$$= \int_0^4 (4 + \frac{9}{2} + \frac{9}{2} \cos 2\theta + 12 \cos \theta) d\theta$$

$$- \int_4^{\pi} (4 + \frac{9}{2} + \frac{9}{2} \cos 2\theta + 12 \cos \theta) d\theta$$

$$= \int_0^4 (\frac{17}{2} + \frac{9}{2} \cos 2\theta + 12 \cos \theta) d\theta$$

$$- \int_4^{\pi} (\frac{17}{2} + \frac{9}{2} \cos 2\theta + 12 \cos \theta) d\theta$$





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$$\left( \frac{17}{2} \theta + \frac{9}{2} \frac{\sin 2\theta}{2} + 12 \sin \theta \right) - \left( \frac{17\pi}{2} + \frac{9}{2} \frac{\sin 2\pi}{2} + 12 \sin \pi \right)$$

$$\left( \frac{17}{2} 4 + \frac{9}{4} \sin 24 + 12 \sin 4 \right) - \left( \frac{17\pi}{2} + \frac{9}{4} \sin 2\pi + 12 \sin \pi - \frac{17}{2} 4 - \frac{9}{4} \sin 24 - 12 \sin 4 \right)$$

$$\frac{17}{2} 4 + \frac{9}{4} \sin 24 + 12 \sin 4 - \frac{17\pi}{2} - \frac{9}{4} \sin 2\pi - 12 \sin \pi + \frac{17}{2} 4 + \frac{9}{4} \sin 24 + 12 \sin 4$$

$$= \frac{17}{2} 4 + \frac{9}{4} \sin 24 + 12 \sin 4 - \frac{17\pi}{2} + \frac{17}{2} 4 + \frac{9}{4} \sin 24 + 12 \sin 4$$

$$= 174 + \frac{9}{2} \sin 24 + 24 \sin 4 - \frac{17\pi}{2}$$

$$= 174 + 9 \sin 4 \cos 4 + 24 \sin 4 - \frac{17\pi}{2}$$

$$\text{put } u = \cos^{-1}(-\frac{2}{3})$$

$$\cos u = -\frac{2}{3}$$

$$\sin u = \sqrt{1 - \cos^2 u}$$

$$= \sqrt{1 - 4/9} = \frac{\sqrt{5}}{3}$$

$$= 17 \cos^{-1}(-\frac{2}{3}) + 9 \sin(\cos^{-1}(-\frac{2}{3})) \cos(\cos^{-1}(-\frac{2}{3}))$$

$$+ 24 \sin(\cos^{-1}(-\frac{2}{3})) - \frac{17\pi}{2}$$

$$= 17 \cos^{-1}(-\frac{2}{3}) + 9 \sin u(-\frac{2}{3})$$

$$+ 24 \sin u - \frac{17\pi}{2}$$

$$= 17 \cos^{-1}(-\frac{2}{3}) + 9 \cdot \frac{\sqrt{5}}{3} (-\frac{2}{3})$$

$$+ 24 \cdot \frac{\sqrt{5}}{3} - \frac{17\pi}{2}$$

$$= 17 \cos^{-1}(-\frac{2}{3}) - 2\sqrt{5} + 8\sqrt{5} - \frac{17\pi}{2}$$

$$= 17 \cos^{-1}(-\frac{2}{3}) + 6\sqrt{5} - \frac{17\pi}{2}$$

~~~~~ x ~~~~~

Exercise: 7.6

① parabola :- $y^2 = 4ax$

st. line :- $3y = 8x$

point of intersection:-

$$y^2 = 4ax \quad \text{--- ①}$$

$$3y = 8x \quad \text{--- ②}$$

$$y = \frac{8x}{3} \quad \text{--- ③}$$

put (3) in ①

$$\left(\frac{8x}{3} \right)^2 = 4ax$$

$$\frac{64x^2}{9} = 4ax$$

$$64x^2 = 36ax$$

$$64x^2 - 36ax = 0$$

$$4x(16x - 9a) = 0$$

$$4x = 0 \quad \text{or} \quad 16x - 9a = 0$$

$$x = 0$$

$$16x = 9a$$

$$x = \frac{9a}{16}$$

now $y^2 = 4ax$

$$2y \frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{2a}{y}$$

$$\text{so } S = \int_0^{\frac{9a}{16}} \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

$$= \int_0^{\frac{9a}{16}} \sqrt{1 + \left(\frac{2a}{y} \right)^2} dx$$

$$= \int_0^{\frac{9a}{16}} \sqrt{1 + \frac{4a^2}{y^2}} dx$$

$$= \int_0^{\frac{9a}{16}} \sqrt{1 + \frac{4a^2}{4ax}} dx$$

$$= \int_0^{\frac{9a}{16}} \sqrt{1 + \frac{a}{x}} dx$$

$$= \int_0^{\frac{9a}{16}} \sqrt{\frac{x+a}{x}} dx$$

$$\text{put } \sqrt{x} = z$$

$$x = z^2$$

$$dx = 2z dz$$

$$\text{as } x = 0, z = 0$$

$$x = \frac{9a}{16}$$

$$x = \frac{3}{4} \sqrt{a}$$

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$$\int_0^{\frac{3\sqrt{a}}{4}} \frac{\sqrt{z^2+a}}{z} \cdot 2z \cdot dz$$

$$\int_0^{\frac{3\sqrt{a}}{4}} \sqrt{z^2+(\sqrt{a})^2} dz$$

$$\left[\frac{z \sqrt{z^2+(\sqrt{a})^2}}{2} + \frac{(\sqrt{a})^2}{2} \ln \left| \frac{z + \sqrt{z^2+(\sqrt{a})^2}}{\sqrt{a}} \right| \right]_0^{\frac{3\sqrt{a}}{4}}$$

$$\frac{\frac{3\sqrt{a}}{4} \sqrt{\frac{9a}{16}+a}}{2} + \frac{a}{2} \ln \left| \frac{\frac{3\sqrt{a}}{4} + \sqrt{\frac{9a}{16}+a}}{\sqrt{a}} \right|$$

$$\frac{15a}{16} + a \ln 2$$

~~~~~ x ~~~~~

$$y = c \cosh(x/c)$$

$$\frac{dy}{dx} = c \cdot \sinh(x/c) \cdot \frac{1}{c}$$

$$\frac{dy}{dx} = \sinh(x/c)$$

$$S = \int_0^x \sqrt{1 + (dy/dx)^2} dx$$

$$= \int_0^x \sqrt{1 + \sinh^2 x/c} dx$$

$$= \int_0^x \sqrt{\cosh^2 x/c} dx$$

$$= \int_0^x \cosh x/c dx$$

$$= \frac{\sinh x/c}{1/c} \Big|_0^x$$

$$= c \sinh x/c$$

$$3ay^2 = x(a-x)^2$$

$$\sqrt{3a} y = \sqrt{x} (a-x)$$

$$y = \frac{\sqrt{x} (a-x)}{\sqrt{3a}}$$

$$y = \frac{1}{\sqrt{3a}} (a\sqrt{x} - x\sqrt{x})$$

$$y = \frac{1}{\sqrt{3a}} (ax^{1/2} - x^{3/2})$$

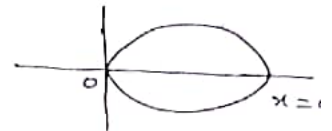
$$\frac{dy}{dx} = \frac{1}{\sqrt{3a}} (a \cdot \frac{1}{2} x^{-1/2} - \frac{3}{2} x^{1/2})$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{3a}} \left( \frac{a}{2\sqrt{x}} - \frac{3\sqrt{x}}{2} \right)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{3a}} \left( \frac{a-3x}{2\sqrt{x}} \right)$$

Symmetry

power of y  
is even, so



Symmetrical about  
x-axis.

no constant term  
passes through the  
origin.

x-intercept put y=0

$$x(a-x)^2=0$$

$$x=0 \text{ or } (a-x)^2=0$$

$$a-x=0$$

$$x=a$$

So S = 2 (length of half curve  
of loop)

$$= 2 \int_0^a \sqrt{1 + (dy/dx)^2} dx$$

$$= 2 \int_0^a \sqrt{1 + \left( \frac{a-3x}{\sqrt{3a} \cdot 2\sqrt{x}} \right)^2} dx$$

$$= 2 \int_0^a \sqrt{1 + \frac{a^2+9x^2-6ax}{12ax}} dx$$

$$= 2 \int_0^a \sqrt{\frac{12ax+a^2+9x^2-6ax}{12ax}} dx$$

$$= 2 \int_0^a \sqrt{\frac{9x^2+6ax+a^2}{12ax}} dx$$

$$= 2 \int_0^a \sqrt{\frac{(3x+a)^2}{12ax}} dx$$

$$= \frac{2}{\sqrt{12a}} \int_0^a \frac{3x+a}{\sqrt{x}} dx$$

$$= \frac{2}{\sqrt{12a}} \int_0^a \left( \frac{3x}{\sqrt{x}} + \frac{a}{\sqrt{x}} \right) dx$$

$$= \frac{2}{\sqrt{12a}} \int_0^a \left( 3\sqrt{x} + \frac{a}{\sqrt{x}} \right) dx$$

$$= \frac{2}{\sqrt{12a}} \int_0^a \left( 3x^{1/2} + ax^{-1/2} \right) dx$$

$$\begin{aligned}
 &= \frac{2}{\sqrt{12a}} \left( 3 \frac{x^{3/2}}{3/2} + a \frac{x^{1/2}}{1/2} \right) \Big|_0^a \\
 &= \frac{2}{\sqrt{12a}} \left[ 2 x^{3/2} + 2a x^{1/2} \right]_0^a \\
 &= \frac{2}{\sqrt{12a}} (2a^{3/2} + 2a a^{1/2}) \\
 &= \frac{2}{\sqrt{12a}} (2a^{3/2} + 2a^{3/2}) \\
 &= \frac{2}{2\sqrt{3a}} (4a^{3/2}) = \frac{4a}{\sqrt{3}}
 \end{aligned}$$

④  $x = e^a \sin \theta$  ,  $y = e^a \cos \theta$ .

$$\frac{dx}{d\theta} = e^a (\sin \theta + \cos \theta) , \quad \frac{dy}{d\theta} = e^a (\cos \theta - \sin \theta)$$

$$\begin{aligned}
 S &= \int_0^{\pi/2} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\
 &= \int_0^{\pi/2} \sqrt{e^{2a} (\sin \theta + \cos \theta)^2 + e^{2a} (\cos \theta - \sin \theta)^2} d\theta \\
 &= \int_0^{\pi/2} \sqrt{e^{2a} [\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta - 2 \sin \theta \cos \theta]} d\theta \\
 &= \int_0^{\pi/2} \sqrt{2e^{2a}} d\theta \\
 &= \sqrt{2} \int_0^{\pi/2} e^a d\theta \\
 &= \sqrt{2} (e^{\pi/2} - e^0) \\
 &= \sqrt{2} (e^{\pi/2} - 1)
 \end{aligned}$$

⑤  $x^2 + y^2 = a^2$   
parametric eqs are

$$x = a \cos \alpha , \quad y = a \sin \alpha$$

$$\frac{dx}{d\alpha} = -a \sin \alpha , \quad \frac{dy}{d\alpha} = a \cos \alpha$$

$$\text{when } x = a \cos \alpha \quad \left| \quad \begin{array}{l} x = a \cos \alpha \\ x = a \cos \beta \end{array} \right.$$

$$a \cos \alpha = a \cos \beta , \quad a \sin \alpha = a \sin \beta$$

$$\alpha = \alpha , \quad \alpha = \beta$$

$$\text{so } S = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{d\alpha}\right)^2 + \left(\frac{dy}{d\alpha}\right)^2} d\alpha$$

$$\begin{aligned}
 S &= \int_{\alpha}^{\beta} \sqrt{a^2 \sin^2 \theta + a^2 \cos^2 \theta} d\theta \\
 &= \int_{\alpha}^{\beta} \sqrt{a^2 (\sin^2 \theta + \cos^2 \theta)} d\theta \\
 &= \int_{\alpha}^{\beta} \sqrt{a^2 (1)} d\theta \\
 &= \int_{\alpha}^{\beta} a d\theta \\
 &= a [\theta]_{\alpha}^{\beta}
 \end{aligned}$$

$$S = a(\beta - \alpha)$$

⑥  $x = a(\theta + \sin \theta)$  ,  $y = a(1 - \cos \theta)$

$$\frac{dx}{d\theta} = a(1 + \cos \theta) , \quad \frac{dy}{d\theta} = a \sin \theta$$

$$\begin{aligned}
 S &= \int_0^{2\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\
 &= \int_0^{2\pi} \sqrt{a^2 (1 + \cos \theta)^2 + a^2 \sin^2 \theta} d\theta \\
 &= \int_0^{2\pi} \sqrt{a^2 (1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta)} d\theta \\
 &= \int_0^{2\pi} \sqrt{a^2 (1 + 1 + 2 \cos \theta)} d\theta \\
 &= \int_0^{2\pi} \sqrt{a^2 (2 + 2 \cos \theta)} d\theta \\
 &= \int_0^{2\pi} \sqrt{a^2 \cdot 2(1 + \cos \theta)} d\theta \\
 &= \int_0^{2\pi} \sqrt{a^2 \cdot 2 \cdot 2 \cos^2 \theta/2} d\theta \\
 &= \int_0^{2\pi} \sqrt{4a^2 \cos^2 \theta/2} d\theta \\
 &= \int_0^{2\pi} 2a \cos \theta/2 d\theta \\
 &= 2a \cdot \left[ \frac{\sin \theta/2}{1/2} \right]_0^{2\pi} \\
 &= 4a \sin \pi
 \end{aligned}$$



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$$x = a(\theta - \sin\theta) ; y = a(1 - \cos\theta)$$

$$= a(1 - \cos\theta) ; \frac{dy}{d\theta} = a \sin\theta$$

length of arch of cycloid is

$$\int_0^{2\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$\int_0^{2\pi} \sqrt{a^2(1 - \cos\theta)^2 + a^2 \sin^2\theta} d\theta$$

$$= \int_0^{2\pi} \sqrt{a^2(1 + \cos^2\theta - 2\cos\theta + \sin^2\theta)} d\theta$$

$$= \int_0^{2\pi} \sqrt{a^2(1 + 1 - 2\cos\theta)} d\theta$$

$$= \int_0^{2\pi} \sqrt{a^2(2 - 2\cos\theta)} d\theta$$

$$= \int_0^{2\pi} \sqrt{a^2 \cdot 2(1 - \cos\theta)} d\theta$$

$$= \int_0^{2\pi} \sqrt{a^2 \cdot 2 \cdot 2\sin^2\frac{\theta}{2}} d\theta$$

$$= \int_0^{2\pi} \sqrt{4a^2 \sin^2\frac{\theta}{2}} d\theta$$

$$= \int_0^{2\pi} 2a \sin\frac{\theta}{2} d\theta$$

$$= 2a \left[ -\frac{\cos\frac{\theta}{2}}{1/2} \right]_0^{2\pi}$$

$$= 4a(-\cos\pi + \cos 0)$$

$$= 4a(1 + 1) = 8a$$

8)

$$x^{2/3} + y^{2/3} = a^{2/3}$$

parametric eqs  
of curve are

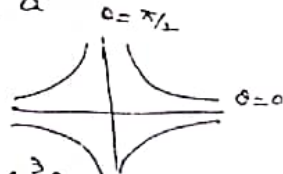
$$x = a \cos^3\theta, y = a \sin^3\theta$$

$$\frac{dx}{d\theta} = -3a \cos^2\theta \sin\theta, \frac{dy}{d\theta} = 3a \sin^2\theta \cos\theta$$

So length in one quadrant

$$s = \int_0^{\pi/2} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$= \int_0^{\pi/2} \sqrt{(-3a \cos^2\theta \sin\theta)^2 + (3a \sin^2\theta \cos\theta)^2} d\theta$$



$$= \int_0^{\pi/2} \sqrt{9a^2 \cos^4\theta \sin^2\theta + 9a^2 \sin^4\theta \cos^2\theta} d\theta$$

$$= \int_0^{\pi/2} \sqrt{9a^2 \cos^2\theta \sin^2\theta (\cos^2\theta + \sin^2\theta)} d\theta$$

$$= \int_0^{\pi/2} \sqrt{9a^2 \cos^2\theta \sin^2\theta (1)} d\theta$$

$$= \int_0^{\pi/2} 3a \cos\theta \sin\theta d\theta$$

$$= 3a \cdot \frac{\sin^2\theta}{2} \Big|_0^{\pi/2}$$

$$= \frac{3a}{2} (1 - 0) = \frac{3a}{2}$$

$$\text{2nd part } \left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$$

the parametric eqs are

$$x = a \cos^3\theta, y = b \sin^3\theta$$

$$\frac{dx}{d\theta} = -3a \cos^2\theta \sin\theta, \frac{dy}{d\theta} = 3b \sin^2\theta \cos\theta$$

So length in 1st quadrant

$$s = \int_0^{\pi/2} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$= \int_0^{\pi/2} \sqrt{(-3a \cos^2\theta \sin\theta)^2 + (3b \sin^2\theta \cos\theta)^2} d\theta$$

$$= \int_0^{\pi/2} \sqrt{9a^2 \cos^4\theta \sin^2\theta + 9b^2 \sin^4\theta \cos^2\theta} d\theta$$

$$= \int_0^{\pi/2} \sqrt{9 \cos^2\theta \sin^2\theta (a^2 \cos^2\theta + b^2 \sin^2\theta)} d\theta$$

$$= \int_0^{\pi/2} 3 \cos\theta \sin\theta \cdot \sqrt{a^2 \cos^2\theta + b^2 \sin^2\theta} d\theta$$

$$\text{Put } \sqrt{a^2 \cos^2\theta + b^2 \sin^2\theta} = z$$

$$a^2 \cos^2\theta + b^2 \sin^2\theta = z^2$$

$$(-2a^2 \cos\theta \sin\theta + 2b^2 \sin\theta \cos\theta) d\theta =$$

$$2z dz$$

$$\cos\theta \sin\theta (b^2 - a^2) d\theta = z dz$$

$$d\theta = \frac{z dz}{\cos\theta \sin\theta (b^2 - a^2)}$$

we also change limits

$$\text{when } \theta = 0, z = a$$

$$\theta = \pi/2, z = b$$

$$= 3 \int_a^b \cos\theta \sin\theta \cdot z \cdot \frac{z dz}{\cos\theta \sin\theta (b^2 - a^2)}$$

$$\begin{aligned}
 &= \frac{3}{b^2 - a^2} \int_a^b x^2 dx \\
 &= \frac{3}{b^2 - a^2} \cdot \left[ \frac{x^3}{3} \right]_a^b \\
 &= \frac{1}{b^2 - a^2} (b^3 - a^3) \\
 &= \frac{(b-a)(b^2 + a^2 + ab)}{(b-a)(b+a)} \\
 &= \frac{b^2 + a^2 + ab}{b+a}
 \end{aligned}$$

⑨  $r = a(1 - \cos \theta)$

$$\frac{dr}{d\theta} = a \sin \theta$$



Limit of upper half is 0 to  $\pi$ .

Entire length is its twice.

$$\begin{aligned}
 S &= 2 \int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\
 &= 2 \int_0^\pi \sqrt{a^2(1 - \cos \theta)^2 + a^2 \sin^2 \theta} d\theta \\
 &= 2 \int_0^\pi \sqrt{a^2(1 + \cos^2 \theta - 2\cos \theta + \sin^2 \theta)} d\theta \\
 &= 2 \int_0^\pi \sqrt{a^2(1 + 1 - 2\cos \theta)} d\theta \\
 &= 2 \int_0^\pi \sqrt{a^2(2 - 2\cos \theta)} d\theta \\
 &= 2 \int_0^\pi \sqrt{a^2 \cdot 2(1 - \cos \theta)} d\theta \\
 &= 2 \int_0^\pi \sqrt{a^2 \cdot 2 \cdot 2\sin^2 \frac{\theta}{2}} d\theta \\
 &= 2 \int_0^\pi \sqrt{4a^2 \sin^2 \frac{\theta}{2}} d\theta \\
 &= 2 \int_0^\pi 2a \cdot \sin \frac{\theta}{2} d\theta \\
 &= 4a \cdot \left[ -\frac{\cos \frac{\theta}{2}}{1/2} \right]_0^\pi \\
 &= 8a(-\cos \frac{\pi}{2} + \cos 0) \\
 &= 8a(0 + 1) \\
 &= 8a
 \end{aligned}$$

2nd part:-

Now length of the arc from

$$\theta = 0 \text{ to } \theta = 2\pi/3$$

$$S = \int_0^{2\pi/3} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

After solving

we get

$$= 2a \left[ -\frac{\cos \frac{\theta}{2}}{1/2} \right]_0^{2\pi/3}$$

$$= 4a[-\cos \pi/3 + \cos 0]$$

$$= 4a[-1/2 + 1]$$

$$= 4a(1/2) = 2a$$

which is equal to half of upper portion which is 4a.

⑩

$$r = \sin^2 \theta/2$$

from (0,0) to (1,  $\pi$ )

$\theta$  varies from 0 to  $\pi$ .

$$\frac{dr}{d\theta} = \sin \theta/2 \cos \theta/2 \cdot \frac{1}{2}$$

$$\frac{dr}{d\theta} = \sin \theta/2 \cos \theta/2$$

$$\begin{aligned}
 S &= \int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\
 &= \int_0^\pi \sqrt{(\sin^2 \theta/2)^2 + (\sin \theta/2 \cos \theta/2)^2} d\theta \\
 &= \int_0^\pi \sqrt{\sin^4 \theta/2 + \sin^2 \theta/2 \cos^2 \theta/2} d\theta \\
 &= \int_0^\pi \sqrt{\sin^2 \theta/2 (\sin^2 \theta/2 + \cos^2 \theta/2)} d\theta \\
 &= \int_0^\pi \sqrt{\sin^2 \theta/2 (1)} d\theta \\
 &= \int_0^\pi \sin \theta/2 d\theta \\
 &= -\frac{\cos \theta/2}{1/2} \Big|_0^\pi \\
 &= 2(-\cos \pi/2 + \cos 0) \\
 &= 2(0 + 1) \\
 &= 2
 \end{aligned}$$

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$$r = a(1 + \cos \theta) \quad \text{--- (1)}$$

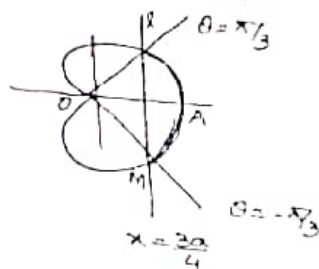
$$4r \cos \theta = 3a \quad \text{--- (2)}$$

$$\text{As } x = r \cos \theta$$

$$4x = 3a$$

$$x = \frac{3a}{4}$$

which is the line parallel to y-axis



For pt of intersection  
Solve (1) & (2)

$$4r \cos \theta = 3a$$

$$r = \frac{3a}{4 \cos \theta}$$

$$r = a(1 + \cos \theta)$$

$$\frac{3a}{4 \cos \theta} = a(1 + \cos \theta)$$

$$\frac{3}{4 \cos \theta} = 1 + \cos \theta$$

$$\cos \theta + 4 \cos^2 \theta = 3$$

$$4 \cos^2 \theta + \cos \theta - 3 = 0$$

$$4 \cos^2 \theta + 6 \cos \theta - 2 \cos \theta - 3 = 0$$

$$4 \cos^2 \theta + 4 \cos \theta - 3 = 0$$

$$(2 \cos \theta + 3)(2 \cos \theta - 1) = 0$$

$$2 \cos \theta + 3 = 0 \quad \text{or}$$

$$2 \cos \theta - 1 = 0$$

$$\cos \theta = -\frac{3}{2}$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = \pm \frac{\pi}{3}$$

$$S_1 = \text{M.A.L} = \int_{-\pi/3}^{\pi/3} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$S_1 = 2 \int_0^{\pi/3} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$2 \int_0^{\pi/3} \sqrt{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta} d\theta$$

$$2 \int_0^{\pi/3} \sqrt{a^2(1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta)} d\theta$$

$$= 2 \int_0^{\pi/3} \sqrt{a^2(2 + 2 \cos \theta)} d\theta$$

$$= 2 \int_0^{\pi/3} \sqrt{2a^2(1 + \cos \theta)} d\theta$$

$$= 2 \int_0^{\pi/3} \sqrt{2a^2 \cdot 2 \cos^2 \frac{\theta}{2}} d\theta$$

$$= 2 \int_0^{\pi/3} \sqrt{4a^2 \cos^2 \frac{\theta}{2}} d\theta$$

$$= 2 \int_0^{\pi/3} 2a \cos \frac{\theta}{2} d\theta$$

$$= 4a \int_0^{\pi/3} \cos \frac{\theta}{2} d\theta$$

$$= 4a \left[ \frac{\sin \frac{\theta}{2}}{\frac{1}{2}} \right]_0^{\pi/3}$$

$$= 8a \left[ \sin \frac{\pi}{6} - \sin 0 \right]$$

$$= 8a \left( \frac{1}{2} - 0 \right)$$

$$S_1 = 4a$$

$$\text{Also } S_2 = \text{L.O.M} = 2 \int_{\pi/3}^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

after simplification

$$= 2 \cdot 4a \left[ \sin \frac{\theta}{2} \right]_{\pi/3}^{\pi}$$

$$= 8a \left( \sin \frac{\pi}{2} - \sin \frac{\pi}{6} \right)$$

$$= 8a \left( 1 - \frac{1}{2} \right)$$

$$S_2 = 8a \left( \frac{1}{2} \right) = 4a$$

$$\text{Hence ratio} = \frac{S_1}{S_2} = \frac{4a}{4a} = 1$$

Thus the line bisects the whole arc of the curve i.e. ratio of the arc on the two sides of the line is 1:1.

$$(12) \quad x = (a+b) \cos \theta - b \cos \left( \frac{a+b}{b} \theta \right)$$

$$\frac{dx}{d\theta} = -(a+b) \sin \theta + b \sin \left( \frac{a+b}{b} \theta \right) \left( \frac{a+b}{b} \right)$$

$$= (a+b) \left[ -\sin \theta + \sin \left( \frac{a+b}{b} \theta \right) \right]$$

$$\text{and } y = (a+b) \sin \theta - b \sin \left( \frac{a+b}{b} \theta \right)$$

$$\frac{dy}{d\theta} = (a+b) \cos \theta - b \cos \left( \frac{a+b}{b} \theta \right) \left( \frac{a+b}{b} \right)$$

$$= (a+b) \cos \theta - \cos \left( \frac{a+b}{b} \theta \right) (a+b)$$

$$\frac{dy}{d\theta} = (a+b) \left[ \cos \theta - \cos \left( \frac{a+b}{b} \theta \right) \right]$$

Then, req. length is

$$S = \int_{\frac{\pi b}{a}}^{\frac{\pi b}{a}} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$



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$$\int \frac{(a+b)^2 \left[ -\sin \theta + \sin\left(\frac{a+b}{b}\theta\right) \right]^2 + (a+b)^2 \left[ \cos \theta - \cos\left(\frac{a+b}{b}\theta\right) \right]^2}{(a+b)^2} d\theta$$

$$\int \frac{(a+b)^2 \left[ \sin^2 \theta + \sin^2\left(\frac{a+b}{b}\theta\right) - 2\sin \theta \sin\left(\frac{a+b}{b}\theta\right) + \cos^2 \theta + \cos^2\left(\frac{a+b}{b}\theta\right) - 2\cos \theta \cos\left(\frac{a+b}{b}\theta\right) \right]}{(a+b)^2} d\theta$$

$$(a+b) \int \sqrt{1 + 1 - 2 \left\{ \sin \theta \sin\left(\frac{a+b}{b}\theta\right) + \cos \theta \cos\left(\frac{a+b}{b}\theta\right) \right\}} d\theta$$

$$(a+b) \int \sqrt{2 - 2 \left( \cos\left(\frac{a+b}{b}\theta\right) - \cos \theta \right)} d\theta$$

$$\sqrt{2}(a+b) \int \sqrt{1 - \cos \frac{a\theta}{b}} d\theta$$

$$\sqrt{2}(a+b) \int \sqrt{2 \sin^2 \frac{a\theta}{2b}} d\theta$$

$$= \sqrt{2}(a+b) \sqrt{2} \int \sin \frac{a\theta}{2b} d\theta$$

$$= 2(a+b) \left[ -\frac{\cos \frac{a\theta}{2b}}{\frac{a}{2b}} \right]_0^{\frac{\pi b}{a}}$$

$$= \frac{4b}{a}(a+b) \left[ -\cos \frac{a\theta}{2b} - 0 \right]$$

$$= \frac{4b(a+b)}{a} \cdot \cos \frac{a\theta}{2b}$$

(a)  $r = a + b \cos \theta$

$$\frac{dr}{d\theta} = -b \sin \theta$$

$$S_0 = 2 \int_0^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$= 2 \int_0^{\pi} \sqrt{(a + b \cos \theta)^2 + (-b \sin \theta)^2} d\theta$$

$$= 2 \int_0^{\pi} \sqrt{a^2 + b^2 \cos^2 \theta + 2ab \cos \theta + b^2 \sin^2 \theta} d\theta$$

$$= 2 \int_0^{\pi} \sqrt{a^2 + b^2 + 2ab \cos \theta} d\theta$$

$$= 2a \int_0^{\pi} \sqrt{1 + \frac{b^2}{a^2} + 2\frac{b}{a} \cos \theta} d\theta \quad \text{let } \frac{b}{a} = k$$

$$= 2a \int_0^{\pi} (1 + k^2 + 2k \cos \theta)^{1/2} d\theta$$

$$= 2a \int_0^{\pi} 1 + \frac{1}{2}(k^2 + 2k \cos \theta) + \frac{1}{2!} \frac{(k^2 + 2k \cos \theta)^2}{2!} d\theta$$

neglecting  $k^3$  & higher powers as  $k$  is so small.

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$$\begin{aligned} &= 2a \int_0^{\pi} \left( 1 + k \cos \theta + \frac{k^2}{2} - \frac{1}{8}(4k^2 \cos^2 \theta) \right) d\theta \\ &= 2a \int_0^{\pi} \left( 1 + k \cos \theta + \frac{k^2}{2} - \frac{k^2}{2} \cos^2 \theta \right) d\theta \\ &= 2a \int_0^{\pi} \left( 1 + k \cos \theta + \frac{k^2}{2}(1 - \cos^2 \theta) \right) d\theta \\ &= 2a \int_0^{\pi} \left( 1 + k \cos \theta + \frac{k^2}{2} \sin^2 \theta \right) d\theta \\ &= 2a \int_0^{\pi} \left( 1 + k \cos \theta + \frac{k^2}{2} \cdot \frac{(1 - \cos 2\theta)}{2} \right) d\theta \\ &= 2a \int_0^{\pi} \left( 1 + k \cos \theta + \frac{k^2}{4} - \frac{k^2}{4} \cos 2\theta \right) d\theta \end{aligned}$$

$$= 2a \left[ \theta + k \sin \theta + \frac{k^2}{4} \theta - \frac{k^2}{4} \frac{\sin 2\theta}{2} \right]_0^{\pi}$$

$$= 2a \left[ \pi + k \sin \pi + \frac{k^2}{4} \pi - \frac{k^2}{4} \frac{\sin 2\pi}{2} - \left( 0 + k \sin 0 + \frac{k^2}{4}(0) - \frac{k^2}{8} \sin 0 \right) \right]$$

$$= 2a \left( \pi + \frac{k^2 \pi}{4} \right)$$

$$= 2\pi a \left( 1 + \frac{k^2}{4} \right) \quad \text{as } k = \frac{b}{a}$$

$$= 2\pi a \left( 1 + \frac{b^2}{4a^2} \right)$$

(14)  $r = a + b \cos \theta$  ( $\frac{a}{b}$  is small)

$$\frac{dr}{d\theta} = -b \sin \theta$$

For the limits of loop.

$$\text{put } r = 0$$

$$a + b \cos \theta = 0$$

$$b \cos \theta = -a$$

$$\cos \theta = -\frac{a}{b}$$

$$\theta = \cos^{-1} \left( -\frac{a}{b} \right)$$

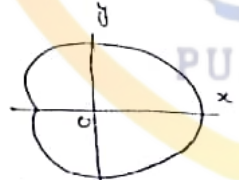
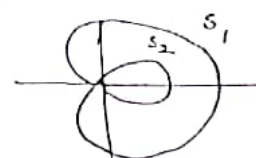
For this value of  $\theta$ , curve enters the pole.

Now for the upper half of the outer loop,  $\theta$  varies from 0 to  $\cos^{-1} \left( -\frac{a}{b} \right)$ .

For the lower half of the inner loop,  $\theta$  varies from  $\cos^{-1} \left( -\frac{a}{b} \right)$  to  $\pi$ .

So the difference of the lengths of these loops are  $\pi$

$$S = S_1 - S_2 = 2 \int_0^{\cos^{-1}(-\frac{a}{b})} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta - 2 \int_{\cos^{-1}(-\frac{a}{b})}^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$



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$$\int_{\cos^{-1}(-a/b)}^{\pi} \sqrt{(a+b\cos\theta)^2 + (-b\sin\theta)^2} - 2 \int_{\cos^{-1}(-a/b)}^{\pi} \sqrt{(a+b\cos\theta)^2 + (-b\sin\theta)^2} d\theta$$

(15)

$$z\theta = a$$

$$z = \frac{a}{\theta}$$

$$\frac{dz}{d\theta} = -\frac{a}{\theta^2}$$

when

$$z = a, \quad \theta = 1$$

$$z = 2a, \quad \theta = 1/2$$

So req. length is

$$S = \int_{1/2}^1 \sqrt{z^2 + \left(\frac{dz}{d\theta}\right)^2} d\theta$$

$$= \int \sqrt{\frac{a^2}{\theta^2} + \frac{a^2}{\theta^4}} d\theta$$

$$= \int \sqrt{\frac{a^2\theta^2 + a^2}{\theta^4}} d\theta$$

$$= a \int \frac{\sqrt{\theta^2 + 1}}{\theta^2} d\theta$$

$$= a \int \frac{\sqrt{\theta^2 + 1}}{\theta^2} \cdot \frac{\sqrt{\theta^2 + 1}}{\sqrt{\theta^2 + 1}} d\theta$$

$$= a \int \frac{\theta^2 + 1}{\theta^2 \sqrt{\theta^2 + 1}} d\theta$$

$$= a \int \left( \frac{\theta^2}{\theta^2 \sqrt{\theta^2 + 1}} + \frac{1}{\theta^2 \sqrt{\theta^2 + 1}} \right) d\theta$$

$$= a \left( \int \frac{1}{\sqrt{\theta^2 + 1}} d\theta + \int \frac{1}{\theta^2 \sqrt{\theta^2 + 1}} d\theta \right)$$

$$= a (I_1 + I_2) \quad \text{--- (1)}$$

$$I_1 = \int_{1/2}^1 \frac{1}{\sqrt{\theta^2 + 1}} d\theta$$

$$= \ln(\theta + \sqrt{\theta^2 + 1}) \Big|_{1/2}^1$$

$$= \ln(1 + \sqrt{2}) - \ln\left(\frac{1}{2} + \sqrt{\frac{1}{4} + 1}\right)$$

$$= \ln(1 + \sqrt{2}) - \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$$

$$= \ln(1 + \sqrt{2}) - \ln\left(\frac{1 + \sqrt{5}}{2}\right)$$

$$= \ln\left(\frac{1 + \sqrt{2}}{\frac{1 + \sqrt{5}}{2}}\right) = \ln \frac{2(1 + \sqrt{2})}{(1 + \sqrt{5})}$$

$$I_1 = \ln\left(\frac{2 + \sqrt{8}}{1 + \sqrt{5}}\right)$$

$$\text{Also } I_2 = \int_{1/2}^1 \frac{1}{\theta^2 \sqrt{\theta^2 + 1}} d\theta$$

$$a = 1/2, \quad z = 2$$

$$a = 1, \quad z = 1$$

$$\text{put } \theta = \frac{1}{z}$$

$$d\theta = -\frac{1}{z^2} dz$$

$$= \int \frac{-1/z^2 dz}{\frac{1}{z^2} \sqrt{\frac{1}{z^2} + 1}}$$

$$a^2 + b^2 \cos^2 \theta + 2ab \cos \theta + b^2 \sin^2 \theta d\theta$$

$$\sqrt{a^2 + b^2 \cos^2 \theta + 2ab \cos \theta + b^2 \sin^2 \theta} d\theta$$

$$\left[ \sqrt{a^2 + b^2 + 2ab \cos \theta} d\theta - 2 \int \sqrt{a^2 + b^2 + 2ab \cos \theta} d\theta \right]$$

$$\left[ \sqrt{\frac{a^2}{b^2} + 1 + 2\frac{a}{b} \cos \theta} d\theta - 2b \int \sqrt{\frac{a^2}{b^2} + 1 + 2\frac{a}{b} \cos \theta} d\theta \right]$$

$$\text{we neglect } \frac{a^2}{b^2}, \quad \text{as } b > a$$

$$a < b$$

$$\frac{a}{b} < 1$$

$$\int \left(1 + \frac{2a}{b} \cos \theta\right)^{1/2} d\theta - 2b \int \left(1 + \frac{2a}{b} \cos \theta\right)^{1/2} d\theta$$

$$\int \left(1 + \frac{1}{2} \cdot \frac{2a}{b} \cos \theta\right) d\theta - 2b \int \left(1 + \frac{1}{2} \cdot \frac{2a}{b} \cos \theta\right) d\theta$$

neglecting  $\frac{a^2}{b^2}$  and higher powers

$$\left[ \theta + \frac{a}{b} \sin \theta \right]_{\cos^{-1}(-a/b)}^{\pi} - 2b \left[ \theta + \frac{a}{b} \sin \theta \right]_{\cos^{-1}(-a/b)}^{\pi}$$

$$b \left[ \cos^{-1}\left(-\frac{a}{b}\right) + \frac{a}{b} \sin \cos^{-1}\left(-\frac{a}{b}\right) - \theta - \frac{a}{b} \sin \theta \right]$$

$$b \left[ \pi + \frac{a}{b} \sin \pi - \cos^{-1}\left(-\frac{a}{b}\right) - \frac{a}{b} \sin \cos^{-1}\left(-\frac{a}{b}\right) \right]$$

$$b \left[ 2 \cos^{-1}\left(-\frac{a}{b}\right) + \frac{2a}{b} \sin \cos^{-1}\left(-\frac{a}{b}\right) - \pi \right]$$

$$\text{As } \cos^{-1}\left(-\frac{a}{b}\right) = \theta$$

$$-\frac{a}{b} = \cos \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \frac{a^2}{b^2}} = \frac{\sqrt{b^2 - a^2}}{b}$$

using Maclaurin's series

$$f(x) = \cos^{-1} x, \quad f(0) = \pi/2$$

$$f'(x) = -\frac{1}{\sqrt{1-x^2}}, \quad f'(0) = -1$$

$$f(x) = f(0) + x f'(0) + \dots$$

$$\cos^{-1} x = \pi/2 - x$$

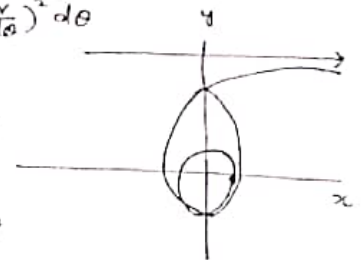
$$\cos^{-1}(-a/b) = \pi/2 + a/b$$

$$2b \left[ 2\left(\pi/2 + \frac{a}{b}\right) + \frac{2a}{b} \cdot \frac{\sqrt{b^2 - a^2}}{b} - \pi \right]$$

$$2b \left[ \pi + \frac{2a}{b} + \frac{2a}{b} (1 - \frac{a^2}{b^2})^{1/2} - \pi \right] \text{ (neglect)}$$

$$2b \left( \pi + \frac{4a}{b} - \pi \right)$$

$$\Rightarrow 2b \left( \frac{4a}{b} \right) = 8a$$



$$\begin{aligned}
 &= \int_1^2 \frac{dz}{\frac{\sqrt{1+z^2}}{z}} \\
 &= \int_1^2 \frac{z dz}{\sqrt{1+z^2}} \\
 &= \frac{1}{2} \int_1^2 (1+z^2)^{-1/2} (2z) dz \\
 &= \frac{1}{2} \left[ \frac{(1+z^2)^{1/2}}{1/2} \right]_1^2 \\
 &= (1+z^2)^{1/2} - (1+1^2)^{1/2} \\
 I_2 &= \sqrt{5} - \sqrt{2}
 \end{aligned}$$

① becomes  $a(I_1 + I_2)$   
 $= a \left( \ln \frac{2+\sqrt{5}}{1+\sqrt{2}} + \sqrt{5} - \sqrt{2} \right)$

⑫  $y = c \cosh\left(\frac{x}{c}\right)$   
 $\frac{dy}{dx} = c \sinh\left(\frac{x}{c}\right) \cdot \frac{1}{c}$   
 $\frac{dy}{dx} = \sinh\left(\frac{x}{c}\right)$   
 As  $\frac{dy}{dx} = \tanh \alpha$   
 $\tanh \alpha = \sinh\left(\frac{x}{c}\right)$   
 Also  $S = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$   
 $= \int_0^x \sqrt{1 + \sinh^2 \frac{x}{c}} dx$   
 $= \int_0^x \sqrt{\cosh^2 \frac{x}{c}} dx$   
 $= \int_0^x \cosh \frac{x}{c} dx$   
 $= \left[ \frac{\sinh \frac{x}{c}}{1/c} \right]_0^x$   
 $= c \sinh \frac{x}{c}$   
 from (i)  
 $S = c \tanh \frac{x}{c}$   
 is req. intrinsic equation.

⑬  $x^2 = 4ay$   
 Diff. w.r.t.  $x$   
 $2x = 4a \frac{dy}{dx}$   
 $\frac{x}{2a} = \frac{dy}{dx}$   
 as  $\frac{dy}{dx} = \tanh \alpha$

$$\tanh \alpha = \frac{x}{2a}$$

$$x = 2a \tanh \alpha$$

$$\begin{aligned}
 \therefore S &= \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\
 &= \int_0^x \sqrt{1 + \left(\frac{x}{2a}\right)^2} dx \\
 &= \int_0^x \sqrt{1 + \frac{x^2}{4a^2}} dx \\
 &= \int_0^x \sqrt{\frac{4a^2 + x^2}{4a^2}} dx \\
 &= \frac{1}{2a} \left[ \frac{x \sqrt{x^2 + 4a^2}}{2} + \frac{4a^2}{2} \sinh^{-1} \frac{x}{2a} \right]_0^x \\
 &= \frac{1}{2a} \left[ \frac{x \sqrt{x^2 + 4a^2}}{2} + 2a^2 \ln \left( \frac{x + \sqrt{x^2 + 4a^2}}{2a} \right) \right]_0^x \\
 &= \frac{1}{2a} \left[ \frac{x \sqrt{x^2 + 4a^2}}{2} + 2a^2 \ln \left| \frac{x + \sqrt{x^2 + 4a^2}}{2a} \right| \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2a} \left[ \frac{2a \tanh \alpha \sqrt{4a^2 \tanh^2 \alpha + 4a^2}}{2} + \right. \\
 &\quad \left. 2a^2 \ln \left| \frac{2a \tanh \alpha + \sqrt{4a^2 \tanh^2 \alpha + 4a^2}}{2a} \right| \right] \\
 &= \frac{1}{2a} \left[ a \tanh \alpha \sqrt{4a^2 (\tanh^2 \alpha + 1)} + 2a^2 \ln \left| \frac{2a \tanh \alpha + \sqrt{4a^2 (\tanh^2 \alpha + 1)}}{2a} \right| \right] \\
 &= \frac{1}{2a} \left[ a \tanh \alpha \cdot 2a \operatorname{sech} \alpha + 2a^2 \ln (\tanh \alpha + \operatorname{sech} \alpha) \right] \\
 S &= a \tanh \alpha \operatorname{sech} \alpha + a \ln (\tanh \alpha + \operatorname{sech} \alpha)
 \end{aligned}$$

⑭  $x^{2/3} + y^{2/3} = a^{2/3}$

parametric eqs are

$$x = a \cos^3 \theta, \quad y = a \sin^3 \theta$$

$$\frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta, \quad \frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$= 3a \sin^2 \theta \cos \theta \cdot \frac{1}{-3a \cos^2 \theta \sin \theta}$$

$$= -\frac{\sin \theta}{\cos \theta}$$

$$\frac{dy}{dx} = -\tanh \alpha$$

$$\text{as } \frac{dy}{dx} = \tanh \alpha$$

$$\tanh \alpha = \tanh(-\alpha)$$

$$\alpha = -\theta$$

$$\text{or } \theta = -\alpha$$



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$$\int_0^{\theta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$\int_0^{\theta} \sqrt{9a^2 \cos^4 \theta + 9a^2 \sin^4 \theta} d\theta$$

$$\int_0^{\theta} \sqrt{9a^2 \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta)} d\theta$$

$$= \int_0^{\theta} \sqrt{9a^2 \cos^2 \theta \sin^2 \theta (1)} d\theta$$

$$= \int_0^{\theta} 3a \cos \theta \sin \theta d\theta$$

$$S = 3a \left[ \frac{\sin^2 \theta}{2} \right]_0^{\theta}$$

$$S = \frac{3a}{2} \sin^2 \theta$$

$$S = \frac{3a}{2} \sin^2(-\alpha) \quad \text{As } \theta = -\alpha$$

$$S = \frac{3a}{2} \sin^2 \alpha$$

$$x = a(\theta + \sin \theta) ; y = a(1 - \cos \theta)$$

$$\frac{x}{a} = \theta + \sin \theta ; \frac{dy}{d\theta} = a \sin \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = a \sin \theta \cdot \frac{1}{a(1 + \cos \theta)} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\frac{dy}{dx} = \tan \frac{\theta}{2}$$

$$\text{As } \frac{dy}{dx} = \tan \alpha$$

$$\tan \alpha = \tan \frac{\theta}{2}$$

$$\alpha = \frac{\theta}{2} \dots \dots \dots i)$$

$$= \int_0^{\theta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$= \int_0^{\theta} \sqrt{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta} d\theta$$

$$= a \int_0^{\theta} \sqrt{1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta} d\theta$$

$$= a \int_0^{\theta} \sqrt{2 + 2 \cos \theta} d\theta$$

$$= a \int_0^{\theta} \sqrt{2(1 + \cos \theta)} d\theta$$

$$= a \int_0^{\theta} \sqrt{2 \cdot 2 \cos^2 \frac{\theta}{2}} d\theta$$

$$= a \int_0^{\theta} \sqrt{4 \cos^2 \frac{\theta}{2}} d\theta$$

$$= a \cdot \int_0^{\theta} \sqrt{2^2 \cos^2 \frac{\theta}{2}} d\theta$$

$$= a \cdot \int_0^{\theta} 2 \cos \frac{\theta}{2} d\theta$$

$$= 2a \cdot \left[ \frac{\sin \frac{\theta}{2}}{\frac{1}{2}} \right]_0^{\theta}$$

$$S = 4a \sin \frac{\theta}{2} \quad \text{from i)}$$

$$S = 4a \sin \alpha$$

$$(20) \quad r = a(1 + \cos \theta)$$

$$\frac{dr}{d\theta} = -a \sin \theta$$

In  $\Delta OPM$ ,

$$0 + \phi + \frac{\pi}{2} - \alpha = \pi$$

$$0 + \phi = \frac{\pi}{2} + \alpha \dots \dots \dots ii)$$

we know that

$$r \frac{dr}{du} = \frac{a(1 + \cos \theta)}{-a \sin \theta}$$

$$\tan \phi = \frac{r \cos^2 \frac{\theta}{2}}{-r \sin \theta \cos \frac{\theta}{2}}$$

$$\tan \phi = -\cot \frac{\theta}{2}$$

$$\tan \phi = \tan(\frac{\pi}{2} + \frac{\theta}{2})$$

$$\phi = \frac{\pi}{2} + \frac{\theta}{2}$$

Put in ii)

$$\theta + \frac{\pi}{2} + \frac{\theta}{2} = \frac{\pi}{2} + \alpha$$

$$\frac{3\theta}{2} = \alpha$$

$$\theta = \frac{2\alpha}{3} \dots \dots \dots iii)$$

$$\text{now } S = \int_0^{\theta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$= \int_0^{\theta} \sqrt{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta} d\theta$$

$$= a \int_0^{\theta} \sqrt{2(1 + \cos \theta)} d\theta$$

$$= a \int_0^{\theta} \sqrt{2 \cdot 2 \cos^2 \frac{\theta}{2}} d\theta$$

$$= 2a \int_0^{\theta} \cos \frac{\theta}{2} d\theta$$

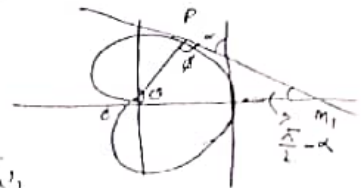
$$= 2a \cdot \left[ \frac{\sin \frac{\theta}{2}}{\frac{1}{2}} \right]_0^{\theta}$$

$$S = 4a \sin \frac{\theta}{2}$$

from iii)

$$S = 4a \sin \frac{1}{2} \cdot \frac{2\alpha}{3}$$

$$S = 4a \sin \frac{\alpha}{3}$$



①

$$\begin{aligned} y &= c \cosh \frac{x}{c} \\ y_1 &= \sinh \frac{x}{c} \\ y_2 &= \cosh \frac{x}{c} \cdot \frac{1}{c} \\ y_2 &= \frac{1}{c} \cosh \frac{x}{c} \\ \rho &= \frac{(1 + y_1^2)^{3/2}}{y_2} \\ &= \frac{(1 + \sinh^2 \frac{x}{c})^{3/2}}{\frac{1}{c} \cosh \frac{x}{c}} \\ &= c \frac{(\cosh^2 \frac{x}{c})^{3/2}}{\cosh \frac{x}{c}} \\ &= c \frac{\cosh^3 \frac{x}{c}}{\cosh \frac{x}{c}} \\ \rho &= c \cosh^2 \frac{x}{c} \end{aligned}$$

$$\begin{aligned} y_2 &= - \csc^2 t/2 \cdot \frac{1}{2} \cdot \frac{dt}{dx} \\ &= - \frac{1}{2} \cdot \frac{1}{\sin^2 t/2} \cdot \frac{1}{a(1 - \cot t)} \\ &= - \frac{1}{2a} \cdot \frac{1}{\sin^2 t/2} \cdot \frac{1}{2 \sin^2 t/2} \end{aligned}$$

$$\begin{aligned} y_2 &= - \frac{1}{4a \sin^4 t/2} \\ \rho &= \frac{(1 + y_1^2)^{3/2}}{y_2} \\ &= \frac{(1 + \cot^2 t/2)^{3/2}}{- \frac{1}{4a \sin^4 t/2}} \\ &= - 4a \sin^4 t/2 (\csc^2 t/2)^{3/2} \\ &= - 4a \sin^4 t/2 \cdot \frac{1}{\sin^3 t/2} \end{aligned}$$

$$\begin{aligned} \rho &= - 4a \sin t/2 \\ \rho &= 4a \sin t/2 \end{aligned}$$

②  $x = a(\cos t + t \sin t); y = a(\sin t - t \cos t)$

$$\frac{dx}{dt} = a(-\sin t + \cos t + t \cos t); \frac{dy}{dt} = a(\cos t + t \sin t - \sin t)$$

$$\frac{dx}{dt} = a t \cos t \quad \frac{dy}{dt} = a t \sin t$$

$$\begin{aligned} y_1 &= \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} \\ &= a t \sin t \cdot \frac{1}{a t \cos t} \\ y_1 &= \tan t \end{aligned}$$

$$\begin{aligned} y_2 &= \sec^2 t \cdot \frac{dt}{dx} \\ y_2 &= \sec^2 t \cdot \frac{1}{a t \cos t} = \frac{\sec^3 t}{a t} \end{aligned}$$

$$\begin{aligned} \rho &= \frac{(1 + y_1^2)^{3/2}}{y_2} \\ &= \frac{(1 + \tan^2 t)^{3/2}}{\frac{\sec^3 t}{a t}} \\ &= \frac{(\sec^2 t)^{3/2} \cdot \frac{1}{\sec^2 t}}{\frac{\sec^3 t}{a t}} \\ \rho &= a t \end{aligned}$$

③  $x = a(t - \sin t); y = a(1 - \cos t)$

$$\frac{dx}{dt} = a(1 - \cos t); \frac{dy}{dt} = a \sin t$$

$$\begin{aligned} y_1 &= \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} \\ &= a \sin t \cdot \frac{1}{a(1 - \cos t)} = \frac{\sin t}{1 - \cos t} \\ y_1 &= \cot t/2 \end{aligned}$$

④  $x = a \cos^3 \theta; y = a \sin^3 \theta$

$$\frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta; \frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

$$\begin{aligned} y_1 &= \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} \\ &= 3a \sin^2 \theta \cos \theta \cdot \frac{1}{-3a \cos^2 \theta \sin \theta} \\ y_1 &= - \tan \theta \\ y_2 &= - \sec^2 \theta \cdot \frac{d\theta}{dx} \\ &= - \sec^2 \theta \cdot \frac{1}{-3a \cos^2 \theta \sin \theta} \end{aligned}$$

$$\begin{aligned} y_2 &= \frac{1}{3a \cos^4 \theta \sin \theta} \\ \rho &= \frac{(1 + y_1^2)^{3/2}}{y_2} \\ &= \frac{(1 + \tan^2 \theta)^{3/2}}{\frac{1}{3a \cos^4 \theta \sin \theta}} \\ &= 3a \cos^4 \theta \sin \theta (\sec^2 \theta)^{3/2} \\ &= 3a \cos^4 \theta \sin \theta \cdot \sec^3 \theta \\ &= 3a \cos \theta \cdot \sin \theta \cdot \frac{1}{\cos^3 \theta} \\ \rho &= 3a \sec^2 \theta \sin \theta \end{aligned}$$

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$$r = 2 \cos 2\theta \quad \text{at } \theta = \frac{\pi}{12}$$

$$r_1 = -4 \sin 2\theta$$

$$r_2 = -8 \cos 2\theta$$

$$\text{at } \theta = \pi/12$$

$$r = 2 \cos 2(\pi/12) = 2 \cos \pi/6 = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$$

$$r_1 = -4 \sin 2(\pi/12) = -4 \sin \frac{\pi}{6} = -4 \cdot \frac{1}{2} = -2$$

$$r_2 = -8 \cos 2(\pi/12) = -8 \cos \frac{\pi}{6} = -8 \cdot \frac{\sqrt{3}}{2} = -4\sqrt{3}$$

$$S = \frac{(r^2 + r_1^2)^{3/2}}{(r^2 + 2r_1^2 - r_2^2)}$$

$$= \frac{(3 + 4)^{3/2}}{3 + 2(-2)^2 - (-4\sqrt{3})^2}$$

$$= \frac{7^{3/2}}{3 + 8 + 12} = \frac{7\sqrt{7}}{23}$$

$$(6) \quad r \cos \theta = a$$

$$r = \frac{a}{\cos \theta}$$

$$r_1 = -\frac{a}{\sin \theta}, \quad r_2 = \frac{2a}{\cos \theta}$$

$$S = \frac{(r^2 + r_1^2)^{3/2}}{(r^2 + 2r_1^2 - r_2^2)}$$

$$S = \frac{\left(\frac{a^2}{\cos^2 \theta} + \frac{a^2}{\sin^2 \theta}\right)^{3/2}}{\frac{a^2}{\cos^2 \theta} + \frac{2a^2}{\sin^2 \theta} - \frac{4a^2}{\cos^2 \theta}}$$

$$= \frac{(a^2 \cos^2 \theta + a^2 \sin^2 \theta)^{3/2}}{(a^2 \cos^2 \theta + 2a^2 \sin^2 \theta - 4a^2 \cos^2 \theta)}$$

$$= \frac{a^3 (\cos^2 \theta + \sin^2 \theta)^{3/2}}{a^2 \cos^2 \theta}$$

$$= \frac{a(1 + \cos^2 \theta)^{3/2}}{\cos^4 \theta}$$

$$1) \quad r^n = a^n \sin \theta$$

Diff. w.r.t  $\theta$

$$n r^{n-1} \cdot r_1 = a^n \cos \theta$$

$$r_1 = \frac{a^n \cos \theta}{r^{n-1}}$$

$$r_1 = \frac{a^n \cos \theta}{r^{n-1}} \cdot \frac{r}{r}$$

$$r_1 = \frac{r a^n \cos \theta}{r^n} \quad \text{from (1)}$$

$$r_1 = \frac{r a^n \cos \theta}{a^n \sin^n \theta}$$

$$r_1 = r \cot^n \theta$$

$$r_2 = r(-n \csc^n \theta) + \cot^n \theta \cdot r_1$$

$$= -n r \csc^n \theta + \cot^n \theta \cdot r \cot^n \theta$$

$$r_2 = -n r \csc^n \theta + r \cot^{2n} \theta$$

$$\text{Num: } (r^2 + r_1^2)^{3/2}$$

$$= (r^2 + r^2 \cot^{2n} \theta)^{3/2}$$

$$= (r^2)^{3/2} (1 + \cot^{2n} \theta)^{3/2}$$

$$= r^3 (\csc^n \theta)^{3/2}$$

$$= r^3 \csc^{3n} \theta$$

$$\text{Den: } (r^2 + 2r_1^2 - r_2^2)$$

$$= r^2 + 2r^2 \cot^{2n} \theta + n^2 r^2 \csc^{2n} \theta - r^2 \cot^{2n} \theta$$

$$= r^2 + r^2 \cot^{2n} \theta + n^2 r^2 \csc^{2n} \theta$$

$$= r^2 (1 + \cot^{2n} \theta) + n^2 r^2 \csc^{2n} \theta$$

$$= r^2 \csc^{2n} \theta (1 + n)$$

$$= r^2 \csc^{2n} \theta (1 + n)$$

$$S = \frac{(r^2 + r_1^2)^{3/2}}{(r^2 + 2r_1^2 - r_2^2)}$$

$$= \frac{r^3 \csc^{3n} \theta}{r^2 \csc^{2n} \theta (1 + n)}$$

$$= \frac{r \csc^n \theta}{(1 + n)}$$

$$= \frac{r}{\sin^n \theta (1 + n)}$$

$$= \frac{r}{\sin^n \theta (1 + n)}$$

$$S = \frac{r}{\frac{r^n}{a^n} (1 + n)} = \frac{a^n}{r^{n-1} (1 + n)} \quad \text{from given (1)}$$

$$(8) \quad r(1 + \cos \theta) = a$$

$$r = \frac{a}{1 + \cos \theta}$$

$$r = \frac{a}{2 \cos^2 \theta/2}$$

$$r = \frac{a}{2} \sec^2 \theta/2$$

Diff. w.r.t  $\theta$

$$r_1 = \frac{a}{2} (2 \sec \theta/2 \tan \theta/2 \cdot \frac{1}{2})$$

$$= \frac{a}{2} \sec^2 \theta/2 \cdot \tan \theta/2$$

$$r_1 = r \tan \theta/2 \quad \text{from (1)}$$



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$$r_2 = r \cdot \sec^2 \theta_1 \cdot \frac{1}{2} + \tan \theta_1 \cdot r_1$$

$$r_2 = \frac{r}{2} \sec^2 \theta_1 + r \tan^2 \theta_1 \quad \text{from ii}$$

$$\begin{aligned} \text{Num: } & (r^2 + r_1^2)^{3/2} \\ &= (r^2 + r^2 \tan^2 \theta_1)^{3/2} \\ &= (r^2 (1 + \tan^2 \theta_1))^{3/2} \\ &= (r^2 \sec^2 \theta_1)^{3/2} \\ &= r^3 \sec^3 \theta_1 \end{aligned}$$

$$\begin{aligned} \text{Den: } & (r^2 + 2r_1^2 - r_1 r_2) \\ &= r^2 + 2r^2 \tan^2 \theta_1 - \frac{r^2}{2} \sec^2 \theta_1 - r^2 \tan^2 \theta_1 \\ &= r^2 + r^2 \tan^2 \theta_1 - \frac{r^2}{2} \sec^2 \theta_1 \\ &= r^2 (1 + \tan^2 \theta_1) - \frac{r^2}{2} \sec^2 \theta_1 \\ &= r^2 \sec^2 \theta_1 - \frac{1}{2} r^2 \sec^2 \theta_1 \\ &= \frac{1}{2} r^2 \sec^2 \theta_1 \end{aligned}$$

$$\begin{aligned} \rho &= \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r_1 r_2} \\ &= \frac{r^3 \sec^3 \theta_1}{\frac{1}{2} r^2 \sec^2 \theta_1} \\ \rho &= 2r \sec \theta_1 \quad \text{from i} \\ &= 2r \cdot \sqrt{\frac{2r}{a}} = \sqrt{\frac{8r^3}{a}} \end{aligned}$$

$$(9) \quad x^2 y = a(x^2 + y^2)$$

$$f(x, y) = ax^2 + ay^2 - x^2 y = 0$$

$$f_x = 2ax - 2xy, \quad f_y = 2ay - x^2$$

$$f_{xx} = 2a - 2y, \quad f_{yy} = 2a$$

$$f_{xy} = -2x$$

$$\text{At } (2a, 2a)$$

$$f_x = 2a(2a) - 2(2a)(2a)$$

$$f_x = 4a^2 - 8a^2 = -4a^2$$

$$f_{xx} = 2a - 2(2a) = 2a - 4a = -2a$$

$$f_y = (2a)(2a) - 4a^2 = 0$$

$$f_{yy} = 2a$$

$$f_{xy} = -2(2a) = -4a$$

$$\rho = \frac{(f_x^2 + f_y^2)^{3/2}}{f_y \cdot f_{xx} - 2f_x f_y f_{xy} + f_x^2 \cdot f_{yy}}$$

$$\rho = \frac{(6a^4)^{3/2}}{0 - 0 + 16a^4(-2a)}$$

$$= -1 - 2a$$

$$\rho = 2a$$

(10) Given Equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

its parametric equations are

$$x = a \cos \theta, \quad y = b \sin \theta$$

$$\frac{dx}{d\theta} = -a \sin \theta, \quad \frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$= b \cos \theta \cdot \frac{1}{-a \sin \theta} = -\frac{b \cos \theta}{a \sin \theta}$$

$$y_1 = -\frac{b}{a} \cot \theta$$

Diff. w.r.t x

$$y_2 = -\frac{b}{a} \cdot -\csc^2 \theta \cdot \frac{d\theta}{dx}$$

$$= \frac{b}{a} \cdot \frac{1}{\sin^2 \theta} \cdot \frac{1}{-a \sin \theta}$$

$$y_2 = -\frac{b}{a^2 \sin^3 \theta}$$

The radius of curvature at any point is

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$\rho = \frac{(1 + \frac{b^2 \cos^2 \theta}{a^2 \sin^2 \theta})^{3/2}}{-\frac{b}{a^2 \sin^3 \theta}}$$

$$\rho = -\frac{a^2 \sin^3 \theta}{b} \left( \frac{a^2 \sin^2 \theta + b^2 \cos^2 \theta}{a^2 \sin^2 \theta} \right)^{3/2}$$

$$\rho = -\frac{a^2 \sin^3 \theta}{b} \left( \frac{a^2 \sin^2 \theta + b^2 \cos^2 \theta}{a^2 \sin^2 \theta} \right)^{3/2}$$

neglect (-ve sign)

$$\rho = \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2}}{ab}$$

Now Eq of tangent at point  $(a \cos \theta, b \sin \theta)$  is

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$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = 1$$

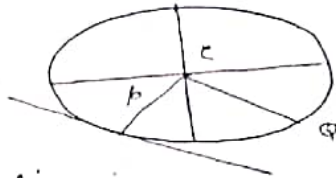
$$\frac{x \cdot a \cos \theta}{a^2} + \frac{y \cdot b \sin \theta}{b^2} = 1$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$b \cos \theta x + a \sin \theta y = ab$$

$$b \cos \theta x + a \sin \theta y - ab = 0$$

Let  $p$  be the length of the perp from the origin (centre) to the tangent line is



$$p = \frac{|b \cos \theta(0) + a \sin \theta(0) - ab|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$p = \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$p^2 = \frac{a^2 b^2}{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$a^2 \sin^2 \theta + b^2 \cos^2 \theta = \frac{a^2 b^2}{p^2}$$

Put in (i)

$$s = \left( \frac{a^2 b^2 / p^2}{ab} \right)^{3/2}$$

$$s = \frac{a^3 b^3}{p^3 \cdot ab}$$

$$s = \frac{a^2 b^2}{p^3}$$

The coordinates of pt Q are  $(-a \sin \theta, b \cos \theta)$ .

$$\text{then } |CQ| = \sqrt{(0 + a \sin \theta)^2 + (0 - b \cos \theta)^2}$$

$$|CQ| = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$|CQ|^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta$$

put in (i)

$$s = \frac{(|CQ|^2)^{3/2}}{ab}$$

$$s = \frac{|CQ|^3}{ab}$$

$$(ii) \quad y^2 = 4ax \quad \dots \dots (i)$$

$$2xy_1 = 4a$$

$$xy_1 = 2a$$

$$y_1 = \frac{2a}{x}$$

$$y_1 = 2a \bar{y}^{-1}$$

$$y_2 = 2ac(-1) \bar{y}^{-1}$$

$$= -2a \bar{y}^{-2} (2a \bar{y}^{-1})$$

$$y_2 = -4a^2 \bar{y}^{-3}$$

$$y_2 = \frac{-4a^2}{y^3}$$

$$s = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + (2a/y)^2)^{3/2}}{-4a^2/y^3}$$

$$s = \frac{(1 + \frac{4a^2}{y^2})^{3/2}}{-4a^2/y^3}$$

$$s = \frac{(y^2 + 4a^2)^{3/2} / y^3}{-4a^2 / y^3}$$

Squaring both sides

$$s^2 = \frac{(y^2 + 4a^2)^3}{16a^4}$$

from (i)  $y^2 = 4ax$

$$s^2 = \frac{(4ax + 4a^2)^3}{16a^4}$$

$$s^2 = \frac{(4a)^3 (x+a)^3}{16a^4}$$

$$s^2 = \frac{4}{a} (x+a)^3 \quad \dots \dots (ii)$$

$$|FP| = \sqrt{(x-a)^2 + (y-0)^2}$$

$$= \sqrt{x^2 + a^2 - 2ax + y^2}$$

from (i)  $y^2 = 4ax$

$$= \sqrt{x^2 + a^2 - 2ax + 4ax}$$

$$= \sqrt{x^2 + a^2 + 2ax}$$

$$|FP| = \sqrt{(x+a)^2}$$

$$|FP| = (x+a) \quad \dots \dots (iii)$$

Put (iii) in (ii)

$$s^2 = \frac{4}{a} (FP)^3$$

$$s^2 = k (FP)^3$$

$$s^2 \propto (FP)^3$$

$$(12) \quad r = a(1 + \cos \theta) \quad \dots \dots (i)$$

$$r_1 = -a \sin \theta$$

$$r_2 = -a \cos \theta$$

$$\text{Num:- } (r^2 + r_1^2)^{3/2}$$

$$= (a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta)^{3/2}$$

$$= (a^2(1 + \cos^2 \theta + 2\cos \theta + \sin^2 \theta))^{3/2}$$

$$= (a^2(1 + 1 + 2\cos \theta))^{3/2}$$

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$$\begin{aligned}
 &= (a^2(2+2\cos\theta))^{3/2} \\
 &= (2a^2(1+\cos\theta))^{3/2} \\
 &= (2a^2 \cdot 2\cos^2\theta/2)^{3/2} \\
 &= (4a^2\cos^2\theta/2)^{3/2} \\
 &= (2^2 a^2 \cos^2\theta/2)^{3/2}
 \end{aligned}$$

$$\text{Num} = 8a^3 \cos^3\theta/2$$

$$\text{Den: } r^2 + 2r_1^2 - r_2^2$$

$$a^2(1+\cos\theta)^2 + 2a^2\sin^2\theta + a^2\cos\theta(1+\cos\theta)$$

$$= a^2[1+\cos^2\theta + 2\cos\theta + 2\sin^2\theta + \cos\theta + \cos^2\theta]$$

$$= a^2[1+2\cos^2\theta + 2\sin^2\theta + 3\cos\theta]$$

$$= a^2[1+2(\cos^2\theta + \sin^2\theta) + 3\cos\theta]$$

$$= a^2[1+2(1) + 3\cos\theta]$$

$$= a^2(3+3\cos\theta)$$

$$= 3a^2(1+\cos\theta)$$

$$= 3a^2 \cdot 2\cos^2\theta/2$$

$$= 6a^2 \cos^2\theta/2$$

$$\text{now } \rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r_2^2}$$

$$\rho = \frac{8a^3 \cos^3\theta/2}{6a^2 \cos^2\theta/2}$$

$$\rho = \frac{4a}{3} \cos\theta/2$$

Squaring both sides

$$\rho^2 = \frac{16a^2}{9} \cos^2\theta/2$$

$$\rho^2 = \frac{8a^2}{9} \cdot 2\cos^2\theta/2$$

$$\rho^2 = \frac{8a^2}{9} (1+\cos\theta)$$

$$\rho^2 = \frac{8a^2}{9} \cdot \frac{r}{a} \quad \text{from (1)}$$

$$\frac{\rho^2}{r} = \frac{8a}{9}$$

$$= \text{Const.}$$

(13) Show that for a pedal curve  $\rho = r \frac{dr}{dp}$

Solution:-

Consider a polar curve

$$r = f(\theta)$$

Let T be the tangent line

at any pt  $P(r, \theta)$  on curve.

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Let  $p$  be the length of the perp. from origin to the tangent line.

From Fig.

$$\frac{p}{r} = \sin\phi$$

$$p = r \sin\phi$$

Diff. w.r.t  $r$

$$\frac{dp}{dr} = r \cos\phi \frac{d\phi}{dr} + \sin\phi$$

$$\text{As } \tan\phi = r \frac{d\theta}{dr}$$

$$\frac{\sin\phi}{\cos\phi} = \frac{r \frac{d\theta}{dr}}{\frac{dr}{ds}}$$

$$\therefore \sin\phi = r \frac{d\theta}{ds}$$

$$\cos\phi = \frac{dr}{ds}$$

$$\text{now } \frac{dp}{dr} = r \frac{dr}{ds} \frac{d\phi}{dr} + r \frac{d\theta}{ds}$$

$$= r \frac{d\phi}{ds} + r \frac{d\theta}{ds}$$

$$= r \frac{d(\phi + \theta)}{ds}$$

$$\text{as } \theta + \phi = \psi$$

$$\frac{d\psi}{ds} = k = \frac{1}{\rho}$$

$$\frac{dp}{dr} = r \cdot \frac{1}{\rho}$$

$$\rho = r \frac{dr}{dp}$$

$$(14) \quad p^2 = ar \Rightarrow r = \frac{p^2}{a}$$

Diff.

$$2p \frac{dp}{dr} = a$$

$$\frac{dp}{dr} = \frac{a}{2p}$$

$$\frac{dr}{dp} = \frac{2p}{a}$$

$$\rho = r \frac{dr}{dp}$$

$$= \frac{p^2}{a} \cdot \frac{2p}{a}$$

$$\rho = \frac{2p^3}{a^2}$$





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$$\frac{1}{p^2} = \frac{A}{r^2} + B$$

$$\bar{p}^2 = A\bar{r}^2 + B$$

Diff.

$$-2\bar{p}^{-3} \frac{d\bar{p}}{d\bar{r}} = -2A\bar{r}^{-3}$$

$$\bar{p}^{-3} \frac{d\bar{p}}{d\bar{r}} = A\bar{r}^{-3}$$

$$\frac{d\bar{p}}{d\bar{r}} = \frac{A\bar{r}^3}{\bar{p}^3}$$

$$\frac{d\bar{p}}{d\bar{r}} = \frac{A\bar{p}^3}{\bar{r}^3}$$

$$\frac{d\bar{r}}{d\bar{p}} = \frac{\bar{r}^3}{A\bar{p}^3}$$

$$\therefore S = \bar{r} \frac{d\bar{r}}{d\bar{p}}$$

$$= \bar{r} \cdot \frac{\bar{r}^3}{A\bar{p}^3}$$

$$S = \frac{\bar{r}^4}{A\bar{p}^3}$$

$$p^2(a^2 + b^2 - r^2) = a^2b^2 \quad \text{--- i}$$

Diff.

$$p^2(-2r) + (a^2 + b^2 - r^2)2p \frac{dp}{dr} = 0$$

$$(a^2 + b^2 - r^2)2p \frac{dp}{dr} = 2p^2r$$

$$\frac{dp}{dr} = \frac{rp}{a^2 + b^2 - r^2}$$

$$\frac{dr}{dp} = \frac{a^2 + b^2 - r^2}{rp}$$

$$\therefore S = r \frac{dr}{dp}$$

$$= r \cdot \frac{(a^2 + b^2 - r^2)}{rp}$$

$$= \frac{a^2 + b^2 - r^2}{p}$$

$$= \frac{a^2b^2/p^2}{p} \quad \text{from i}$$

$$S = \frac{a^2b^2}{p^3}$$

$$(17) \quad r = a(1 + \cos \theta)$$

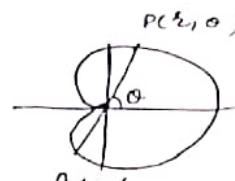
$$r_1 = -a \sin \theta$$

$$r_2 = -a \cos \theta$$

$$S = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r_2^2} = \frac{8a^3 \cos^3 \theta/2}{6a^2 \cos^2 \theta/2}$$

$$S = \frac{4a}{3} \cos \theta/2 \quad \text{--- ii}$$

Let PQ be the chord which passes through the pole.



$$P(r', \pi + \theta)$$

Let  $P(r, \theta)$  and  $Q(r', \pi + \theta)$

Let  $S_1$  and  $S_2$  be the radii of curvature at P and Q, then

$$S_1 = \frac{4a}{3} \cos \theta/2 \quad \text{--- ii}$$

$$S_2 = \frac{4a}{3} \cos(\frac{\pi + \theta}{2})$$

$$= \frac{4a}{3} \cos(\frac{\pi}{2} + \frac{\theta}{2})$$

$$S_2 = \frac{4a}{3} \sin \theta/2 \quad \text{--- iii}$$

$$\text{Now } S_1^2 + S_2^2 = \frac{16a^2}{9} \cos^2 \theta/2 + \frac{16a^2}{9} \sin^2 \theta/2$$

$$= \frac{16a^2}{9} (\cos^2 \theta/2 + \sin^2 \theta/2)$$

$$= \frac{16a^2}{9} (1)$$

$$= \frac{16a^2}{9}$$

$$(18) \quad r = a(1 + \cos \theta)$$

$$r_1 = -a \sin \theta$$

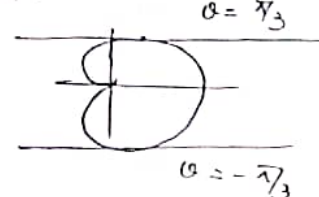
$$r_2 = -a \cos \theta$$

$$S = \frac{(r^2 + r_1^2)^{3/2}}{(r^2 + 2r_1^2 - r_2^2)} = \frac{8a^3 \cos^3 \theta/2}{6a^2 \cos^2 \theta/2}$$

$$S = \frac{4a}{3} \cos \theta/2 \quad \text{--- ii}$$

$$\text{At } \theta = \pm \pi/3$$

tangents are parallel to the initial line



$$\text{At } \theta = \pi/3$$

$$S = \frac{4a}{3} \cos \pi/6$$

$$= \frac{4a}{3} \cdot \frac{\sqrt{3}}{2} = \frac{2a\sqrt{3}}{3} = \frac{2a}{\sqrt{3}}$$

$$\text{At } \theta = -\pi/3$$

$$S = \frac{4a}{3} \cos(-\pi/6) = \frac{4a}{3} \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{2a\sqrt{3}}{3}$$

$$= \frac{2a}{\sqrt{3}}$$

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(19)

$$y = ax^2 + bx + c$$

$$ax^2 + bx = y - c$$

$$x^2 + \frac{b}{a}x = \frac{1}{a}(y - c)$$

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \frac{1}{a}\left(y + \frac{b^2}{4a} - c\right)$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{1}{a}\left(y + \frac{b^2 - 4ac}{4a}\right)$$

$$\text{vertex} \left(-\frac{b}{2a}, -\frac{b^2 - 4ac}{4a}\right)$$

$$\text{Again } y = ax^2 + bx + c$$

$$y_1 = 2ax + b$$

$$y_2 = 2a$$

$$S = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$= \frac{(1 + (2ax + b)^2)^{3/2}}{2a}$$

$$S = \frac{1}{2a}(1 + b^2 + 4abx + 4a^2x^2)^{3/2}$$

we see that  $S$  will be minimum

if  $4abx + 4a^2x^2$  is minimum

$$\text{let } f(x) = 4abx + 4a^2x^2$$

$$f'(x) = 4ab + 8a^2x$$

$$f''(x) = 8a^2$$

$$\text{at } \left(-\frac{b}{a}, -\frac{(b^2 - 4ac)}{4a}\right)$$

$$f''(x) = 8a^2 > 0$$

so  $S$  is min at vertex.

(20)

$$y = \ln x$$

$$y_1 = \frac{1}{x}$$

$$y_2 = -\frac{1}{x^2}$$

The curvature of the curve at any point is

$$K = \frac{y_2}{(1 + y_1^2)^{3/2}}$$

$$K = \frac{1 - \frac{1}{x^2}}{(1 + \frac{1}{x^2})^{3/2}}$$

$$K = \frac{1}{x^2(x^2 + 1)^{3/2}}$$

$$K = \frac{1}{x^2(x^2 + 1)^{3/2}}$$

$$K = \frac{x}{(x^2 + 1)^{3/2}}$$

$$\frac{dK}{dx} = \frac{\text{Diff. w.r.t. } x}{(x^2 + 1)^3} = \frac{(x^2 + 1)^{3/2} \cdot 1 - 3/2(x^2 + 1)^{1/2} \cdot (2x) \cdot x}{(x^2 + 1)^3}$$

$$= \frac{(x^2 + 1)^{1/2}(x^2 + 1 - 3x^2)}{(x^2 + 1)^3}$$

$$= \frac{(x^2 + 1)^{1/2}(1 - 2x^2)}{(x^2 + 1)^3}$$

$$\frac{dK}{dx} = \frac{(1 - 2x^2)}{(x^2 + 1)^{5/2}}$$

$$\text{put } \frac{dK}{dx} = 0$$

$$\frac{1 - 2x^2}{(x^2 + 1)^{5/2}} = 0$$

$$1 - 2x^2 = 0$$

$$-2x^2 = -1$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

$$\text{Take } x = \frac{1}{\sqrt{2}}$$

$$x = -\frac{1}{\sqrt{2}} \quad (\text{neglect})$$

$$\frac{d^2K}{dx^2} = \frac{(x^2 + 1)^{5/2}(-4x) - (1 - 2x^2)^{5/2}(x^2 + 1)^{3/2} \cdot 2x}{(x^2 + 1)^5}$$

$$= \frac{-4x(x^2 + 1)^{5/2} - 5x(1 - 2x^2)(x^2 + 1)^{3/2}}{(x^2 + 1)^5}$$

$$\text{at } x = \frac{1}{\sqrt{2}}$$

$$\frac{d^2K}{dx^2} < 0$$

$$K \text{ is max at } x = \frac{1}{\sqrt{2}}$$

$$\text{Max value of } y$$

$$\text{Put } x = \frac{1}{\sqrt{2}} \text{ in } y$$

$$y = \ln \frac{1}{\sqrt{2}}$$

$$y = -\ln \sqrt{2}$$

$$\text{So } \left(\frac{1}{\sqrt{2}}, -\ln \sqrt{2}\right) \text{ is req. pt.}$$

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## Exercise No. 7-8

①

$$y = \ln x$$

$$y_1 = \frac{1}{x} \text{ at } (1, 0) \Rightarrow y_1 = 1$$

$$y_2 = -\frac{1}{x^2} \text{ at } (1, 0) \Rightarrow y_2 = -1$$

let  $(\alpha, \beta)$  be the coordinates of center of curvature, then

$$\alpha = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$\begin{aligned} &\text{at } (1, 0) \\ &= 1 - \frac{1(1+1)}{-1} \\ &= 1+2 \\ &\alpha = 3 \end{aligned}$$

$$\text{and } \beta = y + \frac{(1+y_1^2)}{y_2}$$

$$\begin{aligned} &\text{at } (1, 0) \\ &= 0 + \frac{(1+1)}{-1} \\ &\beta = -2 \end{aligned}$$

$$\therefore (\alpha, \beta) = (3, -2)$$

The radius of curvature is

$$\begin{aligned} S &= \frac{(1+y_1^2)^{3/2}}{y_2} \\ &= \frac{(1+1)^{3/2}}{-1} \\ S &= 2^{3/2} \end{aligned}$$

Eq of osculating circle is

$$(x-\alpha)^2 + (y-\beta)^2 = S^2$$

$$(x-3)^2 + (y+2)^2 = (2^{3/2})^2$$

$$(x-3)^2 + (y+2)^2 = 8$$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

$$\frac{2x}{4} - \frac{2yy_1}{9} = 1$$

$$-\frac{2yy_1}{9} = -\frac{2x}{4}$$

$$\frac{yy_1}{9} = \frac{x}{4}$$

$$y_1 = \frac{9}{4} \cdot \frac{x}{y}$$

$$\begin{aligned} y_2 &= \frac{9}{4} \cdot \frac{y \cdot 1 - x \cdot y_1}{y^2} \\ &= \frac{9}{4} \cdot \frac{y - x \cdot (9x/4y)}{y^2} \end{aligned}$$

$$y_2 = \frac{9}{4} \cdot \frac{4y^2 - 9x^2}{4y \cdot y^2}$$

$$y_2 = \frac{36y^2 - 81x^2}{16y^3}$$

let  $(\alpha, \beta)$  be the coordinates of center of curvature, then

$$\alpha = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$\begin{aligned} &= x - \frac{\frac{9x}{4y} \left(1 + \frac{81x^2}{16y^2}\right)}{\frac{36y^2 - 81x^2}{16y^3}} \\ &= x - \frac{\frac{9x}{4y} \left(\frac{16y^2 + 81x^2}{16y^2}\right) \cdot \frac{16y^3}{36y^2 - 81x^2}} \end{aligned}$$

$$= x - \frac{9x}{4} \cdot \frac{16y^2 + 81x^2}{36y^2 - 81x^2}$$

$$= x - \frac{9x}{4} \cdot \frac{16y^2 + 81x^2}{36y^2 - 81x^2}$$

$$\begin{aligned} &\text{at } (-2, 0) \\ \alpha &= -2 - \frac{18}{4} \cdot \frac{81(-2)^2}{-81(-2)^2} \end{aligned}$$

$$\alpha = -2 - \frac{9}{2} = -\frac{13}{2}$$

$$\beta = y + \frac{(1+y_1^2)}{y_2}$$

$$\beta = y + \frac{\left(1 + \frac{81x^2}{16y^2}\right)}{\frac{36y^2 - 81x^2}{16y^3}}$$

$$= y + \left(\frac{16y^2 + 81x^2}{16y^2}\right) \cdot \frac{16y^3}{36y^2 - 81x^2}$$

$$\text{at } (-2, 0)$$

$$\beta = 0$$

$$\therefore (\alpha, \beta) = \left(-\frac{13}{2}, 0\right)$$

$$S = \frac{(1+y_1^2)^{3/2}}{y_2}$$

$$= \frac{\left(1 + \frac{81x^2}{16y^2}\right)^{3/2}}{\frac{36y^2 - 81x^2}{16y^3}}$$

$$= \frac{(16y^2 + 81x^2)^{3/2}}{(16y^2)^{3/2}} \cdot \frac{16y^3}{(36y^2 - 81x^2)}$$

$$= \frac{(16y^2 + 81x^2)^{3/2}}{4 \cdot 64y^3} \cdot \frac{16y^3}{(36y^2 - 81x^2)}$$

$$\begin{aligned} &\text{at } (-2, 0) \\ S &= \frac{81(4)^{3/2}}{|4(-81)(4)|} = \frac{(9 \cdot 2)^{3/2}}{4 \cdot 81 \cdot 4} = \frac{9}{2} \end{aligned}$$



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Eq of osculating circle is

$$(x - \alpha)^2 + (y - \beta)^2 = r^2$$

$$(x + \frac{13}{2})^2 + (y - 0)^2 = (\frac{9}{2})^2$$

$$(x + \frac{13}{2})^2 + y^2 = \frac{81}{4}$$

③  $x = a \cos \theta$   $y = b \sin \theta$

$$\frac{dx}{d\theta} = -a \sin \theta$$

$$\frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$= b \cos \theta \cdot \frac{1}{-a \sin \theta} = -\frac{b \cos \theta}{a \sin \theta}$$

$$y_1 = -\frac{b}{a} \cot \theta$$

$$y_2 = -\frac{b}{a} \cdot -\cos^2 \theta \cdot \frac{d\theta}{dx}$$

$$= \frac{b}{a} \cdot \frac{1}{\sin^2 \theta} \cdot \frac{1}{-a \sin \theta}$$

$$y_2 = -\frac{b}{a^2 \sin^3 \theta}$$

Let  $(\alpha, \beta)$  be coordinates of center of curvature

$$\alpha = x - \frac{y_1(1 + y_1^2)}{y_2}$$

$$= a \cos \theta - \frac{-\frac{b \cos \theta}{a \sin \theta} \left(1 + \frac{b^2 \cos^2 \theta}{a^2 \sin^2 \theta}\right)}{-\frac{b}{a^2 \sin^3 \theta}}$$

$$= a \cos \theta - \frac{b \cos \theta}{a \sin \theta} \cdot \frac{a^2 \sin^3 \theta}{b} \left(\frac{a^2 \sin^2 \theta + b^2 \cos^2 \theta}{a^2 \sin^2 \theta}\right)$$

$$\alpha = a \cos \theta - \frac{\cos \theta}{a} (a^2 \sin^2 \theta + b^2 \cos^2 \theta)$$

$$= \frac{\cos \theta}{a} (a^2 - a^2 \sin^2 \theta - b^2 \cos^2 \theta)$$

$$= \frac{\cos \theta}{a} (a^2 (1 - \sin^2 \theta) - b^2 \cos^2 \theta)$$

$$= \frac{\cos \theta}{a} (a^2 \cos^2 \theta - b^2 \cos^2 \theta)$$

$$= \frac{\cos \theta}{a} \cdot \cos^2 \theta (a^2 - b^2)$$

$$\alpha = \frac{\cos^3 \theta}{a} (a^2 - b^2)$$

$$a \alpha = \cos^3 \theta (a^2 - b^2)$$

$$(a \alpha)^{2/3} = \cos^2 \theta (a^2 - b^2)^{2/3} \quad \text{--- (i)}$$

$$\beta = y + \frac{(1 + y_1^2)}{y_2}$$

$$\begin{aligned} \beta &= b \sin \theta + \frac{(1 + \frac{b^2 \cos^2 \theta}{a^2 \sin^2 \theta})}{-\frac{b}{a^2 \sin^3 \theta}} \\ &= b \sin \theta - \frac{a^2 \sin^3 \theta}{b} \left( \frac{a^2 \sin^2 \theta + b^2 \cos^2 \theta}{a^2 \sin^2 \theta} \right) \\ &= b \sin \theta - \frac{\sin \theta}{b} (a^2 \sin^2 \theta + b^2 \cos^2 \theta) \\ &= \frac{\sin \theta}{b} (b^2 - a^2 \sin^2 \theta - b^2 \cos^2 \theta) \\ &= \frac{\sin \theta}{b} [b^2 (1 - \cos^2 \theta) - a^2 \sin^2 \theta] \\ &= \frac{\sin \theta}{b} (b^2 \sin^2 \theta - a^2 \sin^2 \theta) \\ \beta &= \frac{\sin \theta}{b} \cdot -\sin^2 \theta (a^2 - b^2) \\ b \beta &= -\sin^3 \theta (a^2 - b^2) \\ (b \beta)^{2/3} &= \sin^2 \theta (a^2 - b^2)^{2/3} \quad \text{--- (ii)} \end{aligned}$$

Adding (i) & (ii)

$$\begin{aligned} (a \alpha)^{2/3} + (b \beta)^{2/3} &= (a^2 - b^2)^{2/3} \cos^2 \theta + (a^2 - b^2)^{2/3} \sin^2 \theta \\ &= (a^2 - b^2)^{2/3} (\cos^2 \theta + \sin^2 \theta) \\ &= (a^2 - b^2)^{2/3} (1) \\ (a \alpha)^{2/3} + (b \beta)^{2/3} &= (a^2 - b^2)^{2/3} \end{aligned}$$

change  $(\alpha, \beta)$  into current coordinates  $(x, y)$

$$(a x)^{2/3} + (b y)^{2/3} = (a^2 - b^2)^{2/3}$$

④  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

parametric eqs of hyperbola are

$$x = a \sec \theta \quad ; \quad y = b \tan \theta$$

$$\frac{dx}{d\theta} = a \sec \theta \tan \theta, \quad \frac{dy}{d\theta} = b \sec^2 \theta$$

$$y_1 = \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$y_1 = b \sec^2 \theta \cdot \frac{1}{a \sec \theta \tan \theta}$$

$$y_1 = \frac{b \sec \theta}{a \tan \theta}$$

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$$y_1 = \frac{b}{a \cos \theta} \frac{\sin \theta}{\cos \theta}$$

$$y_1 = \frac{b}{a \sin \theta}$$

$$y_1 = \frac{b}{a} \operatorname{cosec} \theta$$

$$y_2 = -\frac{b}{a} \operatorname{cosec} \theta \cot \theta \cdot \frac{d\theta}{dx}$$

$$= -\frac{b}{a} \operatorname{cosec} \theta \cot \theta \cdot \frac{1}{a \sec \theta \tan \theta}$$

$$= -\frac{b}{a} \frac{1}{\sin \theta} \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{a \cdot \frac{1}{\cos \theta} \cdot \frac{\sin \theta}{\cos \theta}}$$

$$y_2 = -\frac{b \cos^3 \theta}{a^2 \sin^3 \theta}$$

Let  $(\alpha, \beta)$  be the coordinates of center of curvature.

$$\alpha = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$= a \sec \theta - \frac{b}{a \sin \theta} \left(1 + \frac{b^2}{a^2 \sin^2 \theta}\right)$$

$$= a \sec \theta + \frac{b}{a \sin \theta} \cdot \frac{a^2 \sin^2 \theta}{b \cos^3 \theta} \left(\frac{a^2 \sin^2 \theta + b^2}{a^2 \sin^2 \theta}\right)$$

$$= a \sec \theta + \frac{1}{a \cos^3 \theta} (a^2 \sin^2 \theta + b^2)$$

$$= a \sec \theta + \frac{1}{a} \left(\frac{a^2 \sin^2 \theta}{\cos^3 \theta} + \frac{b^2}{\cos^3 \theta}\right)$$

$$= a \sec \theta + \frac{1}{a} (a^2 \tan^2 \theta \sec \theta + b^2 \sec^3 \theta)$$

$$\frac{1}{a} (a^2 \sec \theta + a^2 \tan^2 \theta \sec \theta + b^2 \sec^3 \theta)$$

$$= \frac{1}{a} [a^2 \sec \theta (1 + \tan^2 \theta) + b^2 \sec^3 \theta]$$

$$= \frac{1}{a} [a^2 \sec \theta \cdot \sec^2 \theta + b^2 \sec^3 \theta]$$

$$= \frac{1}{a} (a^2 \sec^3 \theta + b^2 \sec^3 \theta)$$

$$x = \frac{1}{a} \sec^3 \theta (a^2 + b^2)$$

$$a x = \sec^3 \theta (a^2 + b^2)$$

$$(ax)^{2/3} = \sec^2 \theta (a^2 + b^2) \quad \text{--- (i)}$$

Now  $\beta = y + \frac{(1+y_1^2)}{y_2}$

$$\beta = b \tan \theta + \frac{(1 + \frac{b^2}{a^2 \sin^2 \theta})}{-\frac{b \cos^3 \theta}{a^2 \sin^3 \theta}}$$

$$\beta = b \tan \theta - \frac{a^2 \sin^2 \theta}{b \cos^3 \theta} \left(\frac{a^2 \sin^2 \theta + b^2}{a^2 \sin^2 \theta}\right)$$

$$= b \tan \theta - \frac{\sin \theta}{b \cos^3 \theta} (a^2 \sin^2 \theta + b^2)$$

$$= b \tan \theta - \frac{1}{b} \left(\frac{a^2 \sin^2 \theta}{\cos^3 \theta} + \frac{b^2 \sin \theta}{\cos^3 \theta}\right)$$

$$\beta = b \tan \theta - \frac{1}{b} (a^2 \tan^3 \theta + b^2 \tan \theta \sec^2 \theta)$$

$$\beta = \frac{1}{b} [b^2 \tan \theta - a^2 \tan^3 \theta - b^2 \tan \theta \sec^2 \theta]$$

$$= \frac{1}{b} [-b^2 \tan \theta (-1 + \sec^2 \theta) - a^2 \tan^3 \theta]$$

$$= \frac{1}{b} [-b^2 \tan \theta \cdot \tan^2 \theta - a^2 \tan^3 \theta]$$

$$= \frac{1}{b} (-b^2 \tan^3 \theta - a^2 \tan^3 \theta)$$

$$\beta = -\frac{\tan^3 \theta}{b} (a^2 + b^2)$$

$$b\beta = -\tan^3 \theta (a^2 + b^2)$$

$$(b\beta)^{2/3} = -\tan^2 \theta (a^2 + b^2)^{2/3} \quad \text{--- (ii)}$$

Subtracting (ii) from (i)

$$(ax)^{2/3} - (b\beta)^{2/3} = (a^2 + b^2)^{2/3} \sec^2 \theta -$$

$$(a^2 + b^2)^{2/3} \tan^2 \theta$$

$$= (a^2 + b^2)^{2/3} (\sec^2 \theta - \tan^2 \theta)$$

$$= (a^2 + b^2)^{2/3} (1)$$

$$(ax)^{2/3} - (b\beta)^{2/3} = (a^2 + b^2)^{2/3}$$

change  $(\alpha, \beta)$  into current coordinates  $(x, y)$

$$(ax)^{2/3} - (by)^{2/3} = (a^2 + b^2)^{2/3}$$

⑤  $2xy = a^2$

$$y = \frac{a^2}{2x}$$

$$y_1 = -\frac{a^2}{2x^2}$$

$$y_2 = \frac{a^3}{x^3}$$

Let  $(\alpha, \beta)$  be the coordinates of center of curvature.

$$\alpha = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$\alpha = x - \left(-\frac{a^2}{2x^2}\right) \left(1 + \frac{a^4}{4x^4}\right) \frac{1}{\frac{a^3}{x^3}}$$

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$$= x + \frac{a^2}{2x^2} \cdot \frac{x^3}{a^2} \left( \frac{4x^4 + a^4}{4x^4} \right)$$

$$\alpha = x + \frac{1}{8x^3} (4x^4 + a^4)$$

$$= x + \frac{4x^4}{8x^3} + \frac{a^4}{8x^3}$$

$$\alpha = x + \frac{x}{2} + \frac{a^4}{8x^3}$$

$$\alpha = \frac{3x}{2} + \frac{a^4}{8x^3} \quad \text{--- (1)}$$

now  $\beta = y + \frac{(1+y_1^2)}{y_2}$

$$= y + \frac{(1 + \frac{a^4}{4x^4})}{\frac{a^2}{x^3}}$$

$$= y + \frac{x^3}{a^2} \left( \frac{4x^4 + a^4}{4x^4} \right)$$

put  $y = \frac{a^2}{2x}$

$$= \frac{a^2}{2x} + \frac{1}{a^2} \left( \frac{4x^4}{4x} + \frac{a^4}{4x} \right)$$

$$= \frac{a^2}{2x} + \frac{1}{a^2} \left( x^3 + \frac{a^4}{4x} \right)$$

$$\beta = \frac{a^2}{2x} + \frac{x^3}{a^2} + \frac{a^2}{4x}$$

$$\beta = \frac{3a^2}{4x} + \frac{x^3}{a^2} \quad \text{--- (2)}$$

now, adding (1) & (2)

$$\alpha + \beta = \frac{3x}{2} + \frac{a^4}{8x^3} + \frac{3a^2}{4x} + \frac{x^3}{a^2}$$

$$= \frac{12a^2x^4 + a^6 + 6a^4x^2 + 8x^6}{8a^2x^3}$$

$$\alpha + \beta = \frac{(a^2 + 2x^2)^3}{8a^2x^3}$$

$$(\alpha + \beta)^{2/3} = \frac{(a^2 + 2x^2)^2}{8^{2/3} a^{4/3} x^2}$$

$$(\alpha + \beta) = \frac{(a^2 + 2x^2)^2}{4a^{4/3}x^2} \quad \text{--- (3)}$$

Now, subtracting (1) & (2)

$$\alpha - \beta = \frac{3x}{2} + \frac{a^4}{8x^3} - \frac{3a^2}{4x} - \frac{x^3}{a^2}$$

$$= \frac{12a^2x^4 + a^6 - 6a^4x^2 - 8x^6}{8a^2x^3}$$

$$\alpha - \beta = \frac{(a^2 - 2x^2)^3}{8a^2x^3}$$

$$(\alpha - \beta)^{2/3} = \frac{(a^2 - 2x^2)^2}{(8a^2x^3)^{2/3}}$$

$$(\alpha - \beta)^{2/3} = \frac{(a^2 - 2x^2)^2}{4a^{4/3}x^2} \quad \text{--- (4)}$$

Subtracting (3) & (4)

$$(\alpha + \beta)^{2/3} - (\alpha - \beta)^{2/3} = \frac{(a^2 + 2x^2)^2}{4a^{4/3}x^2} - \frac{(a^2 - 2x^2)^2}{4a^{4/3}x^2}$$

$$= \frac{a^4 + 4x^4 + 4a^2x^2 - a^4 - 4x^4 + 4a^2x^2}{4a^{4/3}x^2}$$

$$= \frac{8a^2x^2}{4a^{4/3}x^2}$$

$$(\alpha + \beta)^{2/3} - (\alpha - \beta)^{2/3} = 2a^{2/3}$$

change  $(\alpha, \beta)$  into current coordinates  $(x, y)$

$$(x + y)^{2/3} - (x - y)^{2/3} = 2a^{2/3}$$

(6)

$$x = a(\theta - \sin\theta) ; \quad y = a(1 - \cos\theta)$$

$$\frac{dx}{d\theta} = a(1 - \cos\theta) ; \quad \frac{dy}{d\theta} = a \sin\theta$$

$$y_1 = \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$= \frac{a \sin\theta}{a(1 - \cos\theta)} \cdot \frac{1}{a(1 - \cos\theta)}$$

$$= \frac{\sin\theta/2 \cos\theta/2}{2 \sin^2\theta/2}$$

$$y_1 = \cot\theta/2$$

$$y_2 = -\cos^2\theta/2 \cdot \frac{1}{2} \cdot \frac{d\theta}{dx}$$

$$= -\frac{1}{\sin^2\theta/2} \cdot \frac{1}{2} \cdot \frac{1}{a(1 - \cos\theta)}$$

$$= -\frac{1}{2 \sin^2\theta/2} \cdot \frac{1}{a \cdot 2 \sin^2\theta/2}$$

$$y_2 = -\frac{1}{4a \sin^4\theta/2}$$

let  $(\alpha, \beta)$  be the coordinates of center of curvature.

$$\alpha = x - \frac{y_1(1 + y_1^2)}{y_2}$$

$$\alpha = a(\theta - \sin\theta) - \frac{\cot\theta/2}{\sin^4\theta/2} \left( \frac{1 + \cot^2\theta/2}{-\frac{1}{4a \sin^4\theta/2}} \right)$$



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$$= a(a - \sin \alpha) + \frac{\cos \alpha/2}{\sin \alpha/2} \cdot 4a \sin^4 \alpha/2 \cdot \sec^2 \alpha/2$$

$$a(a - \sin \alpha) + \frac{\cos \alpha/2}{\sin \alpha/2} \cdot 4a \sin^4 \alpha/2 \cdot \frac{1}{\sin^2 \alpha/2}$$

$$+ 4a \sin \alpha/2 \cos \alpha/2$$

$$+ 2a \cdot 2 \sin \alpha/2 \cos \alpha/2$$

$$= a(a - \sin \alpha) + 2a \sin \alpha$$

$$= a(a - \sin \alpha + 2 \sin \alpha)$$

$$= a(a + \sin \alpha) \quad \text{--- i,}$$

Now

$$3 = x + \frac{(y + y_1^2)}{y_2}$$

$$= a(1 - \cos \alpha) + \frac{(1 + \cos^2 \alpha/2)}{\frac{1}{4a \sin^4 \alpha/2}}$$

$$a(1 - \cos \alpha) - 4a \sin^4 \alpha/2 \cdot \sec^2 \alpha/2$$

$$a(1 - \cos \alpha) - 4a \sin^2 \alpha/2$$

$$a(1 - \cos \alpha) - 2a \cdot 2 \sin^2 \alpha/2$$

$$= a(1 - \cos \alpha) - 2a(1 - \cos \alpha)$$

$$= a(1 - \cos \alpha - 2 + 2 \cos \alpha)$$

$$= a(\cos \alpha - 1) \quad \text{--- ii,}$$

there are equal to cycloid.

$$\textcircled{1} \quad x^{2/3} + y^{2/3} = a^{2/3}$$

its parametric Eqs are

$$x = a \cos^3 \alpha \quad ; \quad y = a \sin^3 \alpha$$

$$\frac{dy}{dx} = -3a \cos^2 \alpha \sin \alpha \quad ; \quad \frac{dy}{d\alpha} = 3a \sin^2 \alpha \cos \alpha$$

$$\frac{dy}{dx} = \frac{dy}{d\alpha} \cdot \frac{d\alpha}{dx}$$

$$= 3a \sin^2 \alpha \cos \alpha \cdot \frac{1}{-3a \cos^2 \alpha \sin \alpha}$$

$$y_1 = \frac{\sin \alpha}{-\cos \alpha}$$

$$y_1 = -\tan \alpha$$

$$y_2 = -\sec^2 \alpha \cdot \frac{d\alpha}{dx}$$

$$y_2 = -\sec^2 \alpha \cdot \frac{1}{-3a \cos^2 \alpha \sin \alpha}$$

$$y_2 = \frac{1}{3a \cos^4 \alpha \sin \alpha}$$

Let  $(\alpha, \beta)$  be the coordinates of center of curvature

$$\alpha = x - \frac{y_1(1 + y_1^2)}{y_2}$$

$$\alpha = a \cos^3 \alpha + \frac{\tan \alpha(1 + \tan^2 \alpha)}{\frac{1}{3a \cos^4 \alpha \sin \alpha}}$$

$$\alpha = a \cos^3 \alpha + \frac{\sin \alpha}{\cos \alpha} \cdot 3a \cos^4 \alpha \sin \alpha \cdot \sec^2 \alpha$$

$$\alpha = a \cos^3 \alpha + \frac{\sin \alpha}{\cos \alpha} \cdot 3a \cos^4 \alpha \sin \alpha \cdot \frac{1}{\cos^2 \alpha}$$

$$\alpha = a \cos^3 \alpha + 3a \cos \alpha \sin^2 \alpha \quad \text{--- i,}$$

now,

$$\beta = y + \frac{(1 + y_1^2)}{y_2}$$

$$= a \sin^3 \alpha + \frac{(1 + \tan^2 \alpha)}{\frac{1}{3a \cos^4 \alpha \sin \alpha}}$$

$$= a \sin^3 \alpha + 3a \cos^4 \alpha \sin \alpha \cdot \sec^2 \alpha$$

$$\beta = a \sin^3 \alpha + 3a \cos^2 \alpha \sin \alpha \quad \text{--- ii,}$$

Adding i & ii

$$\alpha + \beta = a \cos^3 \alpha + 3a \cos \alpha \sin^2 \alpha + a \sin^3 \alpha + 3a \cos^2 \alpha \sin \alpha$$

$$\alpha + \beta = a(\cos^3 \alpha + 3 \cos \alpha \sin^2 \alpha + \sin^3 \alpha + 3 \cos^2 \alpha \sin \alpha)$$

$$\alpha + \beta = a(\cos \alpha + \sin \alpha)^3$$

$$(\alpha + \beta)^{2/3} = a^{2/3}(\cos \alpha + \sin \alpha)^2 \quad \text{--- iii,}$$

Sub i & ii

Similarly

$$(\alpha - \beta)^{2/3} = a^{2/3}(\cos \alpha - \sin \alpha)^2 \quad \text{--- iv,}$$

Adding iii & iv

$$(\alpha + \beta)^{2/3} + (\alpha - \beta)^{2/3} = a^{2/3} \left( \frac{(\cos \alpha + \sin \alpha)^2}{+ (\cos \alpha - \sin \alpha)^2} \right)$$

$$= a^{2/3} \left( \frac{\cos^2 \alpha + \sin^2 \alpha + 2 \sin \alpha \cos \alpha}{+ \cos^2 \alpha + \sin^2 \alpha - 2 \sin \alpha \cos \alpha} \right)$$

$$\frac{2/3}{(\alpha + \beta)} + \frac{2/3}{(\alpha - \beta)} = a^{2/3} (1 + 1)$$

$$= 2a^{2/3}$$

Change  $(\alpha, \beta)$  into current coordinates  $(x, y)$

$$(x + y)^{2/3} + (x - y)^{2/3} = 2a^{2/3}$$

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$$\begin{aligned} \textcircled{8} \quad x &= a (\cos t + \log \tan t/2) \\ \frac{dx}{dt} &= a \left( -\sin t + \frac{\sec^2 t/2 \cdot 1/2}{\tan t/2} \right) \\ &= a \left( -\sin t + \frac{1}{2 \cos^2 t/2 \cdot \frac{\sin t/2}{\cos t/2}} \right) \\ &= a \left( -\sin t + \frac{1}{\sin t} \right) \\ &= a \left( \frac{-\sin^2 t + 1}{\sin t} \right) \end{aligned}$$

$$\frac{dx}{dt} = a \frac{\cos^2 t}{\sin t}$$

$$y = a \sin t$$

$$\frac{dy}{dt} = a \cos t$$

$$\begin{aligned} y_1 &= \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} \\ &= a \cos t \cdot \frac{\sin t}{a \cos^2 t} \\ &= \frac{\sin t}{\cos t} \end{aligned}$$

$$y_1 = \tan t$$

$$\begin{aligned} y_2 &= \sec^2 t \cdot \frac{dt}{dx} \\ &= \sec^2 t \cdot \frac{\sin t}{a \cos^2 t} \end{aligned}$$

$$y_2 = \frac{\sin t}{a \cos^4 t}$$

Let  $(\alpha, \beta)$  be the coordinates of Centre of curvature.

$$\alpha = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$\begin{aligned} &= a(\cos t + \log \tan t/2) - \frac{\tan t(1+\tan^2 t)}{\frac{\sin t}{a \cos^4 t}} \\ &= a(\cos t + \log \tan t/2) - \frac{\sin t \cdot a \cos^4 t}{\cos t} \cdot \sec^2 t \\ &= a \cos t + a \log \tan t/2 - a \cos t \end{aligned}$$

$$= a \log \tan t/2$$

replace  $\alpha$  by  $x$

$$x = a \log \tan t/2$$

$$\frac{x}{a} = \log \tan t/2$$

$$\frac{e^{x/a}}{e} = \tan t/2 \quad \text{--- } i_1$$

$$\frac{1}{e^{x/a}} = \frac{1}{\tan t/2}$$

$$\frac{e^{-x/a}}{e} = \frac{1}{\tan t/2} \quad \text{--- } ii$$

Adding  $i_1$  &  $ii$

$$\frac{e^{x/a}}{e} + \frac{e^{-x/a}}{e} = \tan t/2 + \frac{1}{\tan t/2}$$

$$\frac{e^{x/a} + e^{-x/a}}{2} = \frac{\tan^2 t/2 + 1}{2 \tan t/2}$$

$$\cosh x/a = \frac{\sec^2 t/2}{2 \frac{\sin t/2}{\cos t/2}}$$

$$\cosh x/a = \frac{1}{\sin t}$$

$$\cosh x/a = \operatorname{cosec} t \quad \text{--- } iii$$

Now  $\beta = y + \frac{(1+y_1^2)}{y_2}$

$$= a \sin t + \frac{(1+\tan^2 t)}{\frac{\sin t}{a \cos^4 t}}$$

$$= a \sin t + \frac{a \cos^4 t}{\sin t} \cdot \sec^2 t$$

$$= a \sin t + \frac{a \cos^2 t}{\sin t}$$

$$= \frac{a \sin^2 t + a \cos^2 t}{\sin t}$$

$$= \frac{a(\sin^2 t + \cos^2 t)}{\sin t}$$

$$= \frac{a(1)}{\sin t}$$

$$\beta = a \operatorname{cosec} t$$

replace  $\beta$  by  $y$

$$y = a \operatorname{cosec} t \quad \text{--- } iv$$

from  $iii$   $\operatorname{cosec} t = \cosh x/a$

$$y = a \cosh \frac{x}{a}$$

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$$x^3 + y^3 = 3axy$$

$$f = x^3 + y^3 - 3axy$$

$$f_x = 3x^2 - 3ay, \quad f_y = 3y^2 - 3ax$$

$$y_1 = -\frac{f_x}{f_y}$$

$$= -\frac{3x^2 - 3ay}{3y^2 - 3ax}$$

$$= -\frac{(x^2 - ay)}{(y^2 - ax)}$$

$$y_1 = \frac{ay - x^2}{y^2 - ax}$$

$$\text{at } \left(\frac{3a}{2}, \frac{3a}{2}\right)$$

$$= \frac{a \cdot \frac{3a}{2} - \left(\frac{3a}{2}\right)^2}{\left(\frac{3a}{2}\right)^2 - a \cdot \frac{3a}{2}}$$

$$= -\frac{3a^2/4}{3a^2/4}$$

$$y_1 = -1$$

$$y_2 = \frac{(y^2 - ax)(ay - 2x) - (2xy - a)(ay - x^2)}{(y^2 - ax)^2}$$

$$\text{at } \left(\frac{3a}{2}, \frac{3a}{2}\right)$$

$$= \frac{\left(\frac{9a^2}{4} - \frac{3a^2}{2}\right)(-a - 3a) - (-3a - a)\left(\frac{3a^2}{2} - \frac{9a^2}{4}\right)}{\left(\frac{9a^2}{4} - \frac{3a^2}{2}\right)^2}$$

$$= \frac{\left(\frac{3a^2}{4}\right)(-4a) - (-4a)\left(-\frac{3a^2}{4}\right)}{\left(\frac{3a^2}{4}\right)^2}$$

$$= \frac{-3a^3 - 3a^3}{\frac{9a^4}{16}} = -6a^3 \cdot \frac{16}{9a^4}$$

$$y_2 = -\frac{32}{3a}$$

Let  $(\alpha, \beta)$  be the center of curvature.

$$\alpha = x - \frac{y_1(1 + y_1^2)}{y_2}$$

$$= \frac{3a}{2} - \frac{(-1)(1+1)}{-\frac{32}{3a}}$$

$$= \frac{3a}{2} - \frac{6a}{32} = \frac{48a - 6a}{32} = \frac{42a}{32} = \frac{21a}{16}$$

$$\beta = y + \frac{(1 + y_1^2)}{y_2}$$

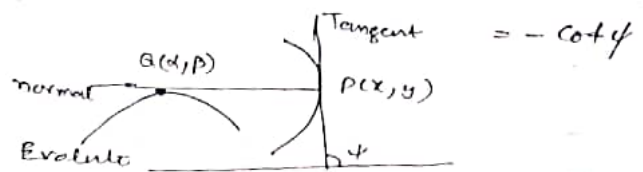
$$= \frac{3a}{2} + \frac{(1+1)}{-\frac{32}{3a}} = \frac{3a}{2} - \frac{6a}{32} = \frac{48a - 6a}{32} = \frac{42a}{32} = \frac{21a}{16}$$

$$\text{Center of curvature } \left(\frac{21a}{16}, \frac{21a}{16}\right)$$

(10) Let  $P$  be the pt along the curve.

$$\text{Slope of tangent at } P = \frac{dy}{dx} = \tan \psi$$

$$\text{Slope of normal at } P = -\frac{dx}{dy} = -\frac{1}{\tan \psi}$$



$$\text{Slope of tangent to evolute} = \frac{d\beta}{d\alpha}$$

$$\text{we are to prove } \frac{d\beta}{d\alpha} = -\cot \psi$$

i.e. tangent to evolute is normal to the curve.

Sol:- The coordinates of center of curvature corresponding to the point  $P$  are

$$\alpha = x - \rho \sin \psi, \quad \beta = y + \rho \cos \psi$$

$$\frac{d\alpha}{dx} = 1 - \rho \cos \psi \frac{d\psi}{dx} - \frac{d\rho}{dx} \sin \psi$$

$$\text{As } \rho = \frac{ds}{d\psi}$$

$$\tan \psi = \frac{dy}{dx}$$

$$\frac{\sin \psi}{\cos \psi} = \frac{dy/ds}{dx/ds}$$

$$\sin \psi = \frac{dy}{ds}, \quad \cos \psi = \frac{dx}{ds}$$

$$= 1 - \frac{d\rho}{d\psi} \cdot \frac{dx}{ds} \cdot \frac{d\psi}{dx} - \frac{d\rho}{dx} \sin \psi$$

$$= 1 - 1 - \frac{d\rho}{d\psi} \sin \psi$$

$$\frac{d\alpha}{dx} = -\frac{d\rho}{d\psi} \sin \psi \quad \text{--- i}$$

$$\text{And } \beta = y + \rho \cos \psi$$

$$\frac{d\beta}{dx} = \frac{dy}{dx} + \rho \sin \psi \frac{d\psi}{dx} + \cos \psi \frac{d\rho}{dx}$$

$$\frac{d\beta}{dx} = \frac{dy}{dx} - \frac{d\rho}{d\psi} \frac{dy}{ds} \frac{d\psi}{dx} + \cos \psi \frac{d\rho}{dx}$$

$$\frac{d\beta}{dx} = \cos \psi \frac{d\rho}{dx} \quad \text{--- ii}$$

Dividing ii by i

$$\frac{d\beta}{d\alpha} = \frac{d\beta/dx}{d\alpha/dx}$$

$$= \frac{\cos \psi \frac{d\rho}{dx}}{-\sin \psi \frac{d\rho}{dx}}$$

$$\frac{d\beta}{d\alpha} = -\cot \psi$$

so proved.



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## Exercise No. 7.9

①

$$y = mx + \frac{a}{m}$$

$$my = m^2x + a$$

$$m^2x - my + a = 0 \quad \text{--- i}$$

$$f(x, y, m) = m^2x - my + a$$

$$\frac{\partial f}{\partial m} = 2mx - y$$

$$\frac{\partial f}{\partial m} = 0$$

$$2mx - y = 0 \quad \text{--- ii}$$

To get envelope, eliminate 'm' from

$$\text{i) \& ii) } \Rightarrow 2mx = y$$

$$m = \frac{y}{2x}$$

Put in i,

$$\left(\frac{y}{2x}\right)^2 x - \left(\frac{y}{2x}\right)y + a = 0$$

$$\frac{y^2}{4x} - \frac{y^2}{2x} + a = 0$$

$$-\frac{y^2}{4x} + a = 0$$

$$-y^2 + 4ax = 0$$

$y^2 = 4ax$  is the eq. of envelope.

$$\text{2(a)} \quad f(x, y, \alpha) = A\alpha^2 + B\alpha + C = 0 \quad \text{--- i}$$

$$\frac{\partial f}{\partial \alpha} = 2A\alpha + B$$

$$\frac{\partial f}{\partial \alpha} = 0$$

$$2A\alpha + B = 0 \quad \text{--- ii}$$

To get envelope, we eliminate  $\alpha$

$$\text{from i \& ii} \quad 2A\alpha = -B$$

$$\alpha = -\frac{B}{2A}$$

Put in i,

$$A\left(-\frac{B}{2A}\right)^2 + B\left(-\frac{B}{2A}\right) + C = 0$$

$$A \cdot \frac{B^2}{4A^2} - \frac{B^2}{2A} + C = 0$$

$$B^2 - 2B^2 + 4AC = 0$$

$$-B^2 + 4AC = 0$$

$$B^2 - 4AC = 0$$

So the envelope is discriminant.

$$\text{2(b)} \quad f(x, y, \alpha) = A\alpha^3 + B\alpha^2 + C\alpha + D = 0 \quad \text{--- i}$$

$$\frac{\partial f}{\partial \alpha} = 3A\alpha^2 + 2B\alpha + C = 0$$

$$3A\alpha^2 + 2B\alpha + C = 0 \quad \text{--- ii}$$

Multiplying i by 3 & ii by  $\alpha$

$$3A\alpha^3 + 9B\alpha^2 + 9C\alpha + 3D = 0$$

$$3A\alpha^3 + 6B\alpha^2 + 3C\alpha = 0$$

$$3B\alpha^2 + 6C\alpha + 3D = 0$$

$$B\alpha^2 + 2C\alpha + D = 0 \quad \text{--- iii}$$

$$\text{ii) } A\alpha^2 + 2B\alpha + C = 0 \quad \text{--- iv}$$

By cross-multiplication.

$$\frac{\alpha^2}{2C \quad D \quad | \quad - \quad B \quad D \quad | \quad B \quad 2C} = \frac{\alpha}{2B \quad C \quad | \quad - \quad A \quad C \quad | \quad A \quad 2B} = \frac{1}{B \quad 2C \quad | \quad A \quad 2B}$$

$$\frac{\alpha^2}{2C^2 - 2BD} = \frac{\alpha}{-(BC - AD)} = \frac{1}{2B^2 - 2AC}$$

$$\frac{\alpha^2}{2C^2 - 2BD} = \frac{1}{2B^2 - 2AC}$$

$$\alpha^2 = \frac{2C^2 - 2BD}{2B^2 - 2AC} = \frac{C^2 - BD}{B^2 - AC} \quad \text{--- v}$$

$$\frac{\alpha}{-(BC - AD)} = \frac{1}{2B^2 - 2AC}$$

$$\alpha = \frac{-(BC - AD)}{2B^2 - 2AC}$$

$$\alpha = \frac{BC - AD}{2(AC - B^2)}$$

Squaring both sides

$$\alpha^2 = \frac{(BC - AD)^2}{4(AC - B^2)^2} \quad \text{--- vi}$$

Equating v, & vi

$$\frac{C^2 - BD}{B^2 - AC} = \frac{(BC - AD)^2}{4(AC - B^2)^2}$$

$$4(C^2 - BD)(AC - B^2)^2 = (B^2 - AC)(BC - AD)^2$$

$$-4(BD - C^2)(AC - B^2)^2 = -(AC - B^2)(BC - AD)^2$$

Dividing both sides by  $-(AC - B^2)$

$$4(BD - C^2)(AC - B^2) = (BC - AD)^2$$

is req. envelope.

$$\text{③} \quad y = x \tan \alpha - \frac{gx^2}{2u^2 \cos^2 \alpha}$$

$$y - x \tan \alpha + \frac{gx^2}{2u^2 \cos^2 \alpha} = 0$$

$$2u^2y - 2u^2x \tan \alpha + gx^2 \sec^2 \alpha = 0$$

$$2u^2y - 2u^2x \tan \alpha + gx^2(1 + \tan^2 \alpha) = 0$$

$$2u^2y - 2u^2x \tan \alpha + gx^2 + gx^2 \tan^2 \alpha = 0$$

$$gx^2 \tan^2 \alpha - 2u^2x \tan \alpha + gx^2 + 2u^2y = 0$$

$$a = gx^2 \quad b = -2u^2x \quad c = gx^2 + 2u^2y$$

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This is quadratic in  $\tan \alpha$ .

So envelope is  $b^2 - 4ac = 0$

$$(-2u^2x)^2 - 4(8x^2)(2u^2y + 8x^2) = 0$$

$$4u^4x^2 - 4(28x^2u^2y + 8^2x^4) = 0$$

$$u^4x^2 - (28x^2u^2y + 8^2x^4) = 0$$

is req. envelope.

$$y = mx + \sqrt{a^2m^2 + b^2}$$

$$y - mx = \sqrt{a^2m^2 + b^2}$$

Squaring both sides

$$(y - mx)^2 = (a^2m^2 + b^2)$$

$$y^2 + m^2x^2 - 2mxy - a^2m^2 - b^2 = 0$$

$$(x^2 - a^2)m^2 - 2xym + y^2 - b^2 = 0$$

$$A = x^2 - a^2, B = -2xy, C = y^2 - b^2$$

This is quadratic in  $m$ .

So envelope is  $B^2 - 4AC = 0$

$$(-2xy)^2 - 4(x^2 - a^2)(y^2 - b^2) = 0$$

$$4x^2y^2 - 4(x^2y^2 - x^2b^2 - a^2y^2 + a^2b^2) = 0$$

$$x^2y^2 - x^2b^2 + x^2b^2 + a^2y^2 - a^2b^2 = 0$$

$$x^2b^2 + a^2y^2 = a^2b^2$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

is req. envelope.

Given ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Let CP and CQ be

semi-conjugate diameters

of ellipse with center C.

Then the coordinates of P and Q are  $(a \cos \alpha, b \sin \alpha)$ ,  $Q(-a \sin \alpha, b \cos \alpha)$

Now, Eq of line joining extremities of P and Q is

$$\frac{y - b \sin \alpha}{b \cos \alpha - b \sin \alpha} = \frac{x - a \cos \alpha}{-a \sin \alpha - a \cos \alpha}$$

$$y - b \sin \alpha = \frac{(b \sin \alpha - b \cos \alpha)(x - a \cos \alpha)}{(a \sin \alpha + a \cos \alpha)}$$

$$y - b \sin \alpha = \frac{b}{a} \frac{(\sin \alpha - \cos \alpha)(x - a \cos \alpha)}{(\sin \alpha + \cos \alpha)}$$

$$y(\sin \alpha + \cos \alpha) - ab \sin \alpha (\sin \alpha + \cos \alpha) =$$

$$bx(\sin \alpha - \cos \alpha) - ab \cos \alpha (\sin \alpha - \cos \alpha)$$

$$ay(\sin \alpha + \cos \alpha) - ab \sin^2 \alpha - ab \sin \alpha \cos \alpha =$$

$$bx(\sin \alpha - \cos \alpha) - ab \sin \alpha \cos \alpha + ab \cos^2 \alpha$$

$$bx(\sin \alpha - \cos \alpha) - ay(\sin \alpha + \cos \alpha) + ab \cos \alpha$$

Dividing both sides by  $ab$

$$\frac{x}{a}(\sin \alpha - \cos \alpha) - \frac{y}{b}(\sin \alpha + \cos \alpha) + 1 = 0$$

$$\frac{x}{a}(\sin \alpha - \cos \alpha) - \frac{y}{b}(\sin \alpha + \cos \alpha) = -1$$

Diff. partially w.r.t.  $\alpha$

$$\frac{x}{a}(\cos \alpha + \sin \alpha) - \frac{y}{b}(\cos \alpha - \sin \alpha) = 0$$

Squaring and adding (1) & (2)

$$\frac{x^2}{a^2}(\sin \alpha - \cos \alpha)^2 + \frac{y^2}{b^2}(\sin \alpha + \cos \alpha)^2$$

$$- 2 \frac{x}{a} \cdot \frac{y}{b}(\sin \alpha - \cos \alpha)(\sin \alpha + \cos \alpha) +$$

$$\frac{x^2}{a^2}(\cos \alpha + \sin \alpha)^2 + \frac{y^2}{b^2}(\cos \alpha - \sin \alpha)^2$$

$$- 2 \frac{x}{a} \cdot \frac{y}{b}(\cos \alpha + \sin \alpha)(\cos \alpha - \sin \alpha) = 1$$

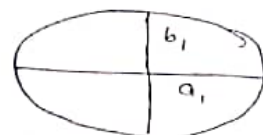
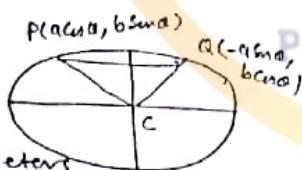
$$\frac{x^2}{a^2}((\sin \alpha - \cos \alpha)^2 + (\cos \alpha + \sin \alpha)^2) + \frac{y^2}{b^2}((\sin \alpha + \cos \alpha)^2 + (\cos \alpha - \sin \alpha)^2) = 1$$

$$\frac{x^2}{a^2}(2(\sin^2 \alpha + \cos^2 \alpha)) + \frac{y^2}{b^2}(2(\sin^2 \alpha + \cos^2 \alpha)) = 1$$

$$\frac{2x^2}{a^2} + \frac{2y^2}{b^2} = 1$$

(6) Eq of ellipse with axes as coordinate axes

$$\frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = 1 \quad \text{--- (1)}$$



Sum of axes = 2

$$2a_1 + 2b_1 = 2a$$

$$a_1 + b_1 = a \quad \text{--- (2)}$$

Eq has two parameters

Taking differentials of

(1) & (2)

$$\textcircled{1} \Rightarrow -\frac{2x^2}{a_1^3} da_1 - \frac{2y^2}{b_1^3} db_1 = 0$$

$$\frac{x^2}{a_1^3} da_1 + \frac{y^2}{b_1^3} db_1 = 0$$

$$\textcircled{2} \Rightarrow da_1 + db_1 = 0$$

Comparing

$$\frac{x^2/a_1^3}{1} = \frac{y^2/b_1^3}{1} = \alpha \text{ (say)}$$

$$\frac{x^2}{a_1^3} = \alpha, \quad \frac{y^2}{b_1^3} = \alpha \quad \text{--- (3)}$$

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$$\frac{x^2}{a_1^2} = \alpha a_1, \quad \frac{y^2}{b_1^2} = \alpha b_1$$

Adding

$$\frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = \alpha a_1 + \alpha b_1$$

$$= \alpha (a_1 + b_1)$$

from (1) & (2)

$$1 = \alpha \cdot a$$

$$\frac{1}{a} = \alpha$$

$$\textcircled{3} \Rightarrow \frac{x^2}{a_1^3} = \frac{1}{a}$$

$$a_1^3 = \alpha x^2$$

$$a_1 = \alpha^{1/3} x^{2/3}$$

$$\frac{y^2}{b_1^3} = \frac{1}{a}$$

$$b_1^3 = \alpha y^2$$

$$b_1 = \alpha^{1/3} y^{2/3}$$

adding

$$a_1 + b_1 = \alpha^{1/3} x^{2/3} + \alpha^{1/3} y^{2/3}$$

$$a_1 + b_1 = \alpha^{1/3} (x^{2/3} + y^{2/3})$$

$$a = \alpha^{1/3} (x^{2/3} + y^{2/3})$$

$$\alpha^{2/3} = x^{2/3} + y^{2/3}$$

$$x^{2/3} + y^{2/3} = \alpha^{2/3}$$

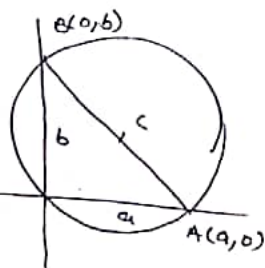
is req. envelope.

⑦

Let A(a, 0)

B(0, b)

be the coordinates of extremities of sliding line.



Center of AB  $(\frac{a}{2}, \frac{b}{2})$

$$\text{also } a^2 + b^2 = c^2$$

So Eq. of circle with 'c' as diameter is

$$(x - \frac{a}{2})^2 + (y - \frac{b}{2})^2 = (\frac{c}{2})^2$$

$$x^2 + \frac{a^2}{4} - ax + y^2 + \frac{b^2}{4} - by = \frac{c^2}{4}$$

$$x^2 + y^2 - ax - by + \frac{a^2}{4} + \frac{b^2}{4} = \frac{c^2}{4}$$

$$x^2 + y^2 - ax - by + \frac{a^2 + b^2}{4} = \frac{c^2}{4}$$

$$\text{as } a^2 + b^2 = c^2$$

$$x^2 + y^2 - ax - by + \frac{c^2}{4} = \frac{c^2}{4}$$

$$x^2 + y^2 - ax - by = 0$$

$$\therefore x^2 + y^2 - ax = by$$

$$(x^2 + y^2 - ax)^2 = b^2 y^2$$

$$\text{as } a^2 + b^2 = c^2$$

$$b^2 = c^2 - a^2$$

$$(x^2 + y^2 - ax)^2 = (c^2 - a^2) y^2$$

$$(x^2 + y^2 - ax)^2 + (a^2 - c^2) y^2 = 0$$

— (1)

Here 'a' is a parameter.

Diff. w.r.t 'a'

$$2(x^2 + y^2 - ax)(-x) + y^2 \cdot 2a = 0$$

$$-x^3 - xy^2 + ax^2 + y^2 a = 0$$

$$-x(x^2 + y^2) + a(x^2 + y^2) = 0$$

$$(x^2 + y^2)(-x + a) = 0$$

$$\Rightarrow -x + a = 0$$

$$a = x$$

Put in (1)

$$(x^2 + y^2 - x^2)^2 + (x^2 - c^2) y^2 = 0$$

$$y^4 + (x^2 - c^2) y^2 = 0$$

$$y^2 + (x^2 - c^2) = 0$$

$$x^2 + y^2 = c^2$$

is req. equation of envelope

B(i) Given

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{--- (1)}$$

$$a + b = c \quad \text{--- (2)}$$

Taking differentials of (1) & (2)

$$-\frac{x}{a^2} da - \frac{y}{b^2} db = 0$$

$$\frac{x}{a^2} da + \frac{y}{b^2} db = 0$$

$$\textcircled{2} \Rightarrow da + db = 0$$

compare

$$\frac{x/a^2}{1} = \frac{y/b^2}{1} = \alpha \text{ (say)}$$

$$\frac{x}{a^2} = \alpha, \quad \frac{y}{b^2} = \alpha \quad \text{--- (3)}$$



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$$\frac{x}{a} = \alpha a, \quad \frac{y}{b} = \alpha b$$

Adding

$$\frac{x}{a} + \frac{y}{b} = \alpha a + \alpha b$$

$$\frac{x}{a} + \frac{y}{b} = \alpha(a+b)$$

from (1) & (2)

$$1 = \alpha \cdot c$$

$$\frac{1}{c} = \alpha$$

(3)  $\Rightarrow$

$$\frac{x}{a^2} = \frac{1}{c}, \quad \frac{y}{b^2} = \frac{1}{c}$$

$$a^2 = cx, \quad b^2 = cy$$

$$a = \sqrt{c} \sqrt{x}, \quad b = \sqrt{c} \sqrt{y}$$

Adding

$$a + b = \sqrt{c} \sqrt{x} + \sqrt{c} \sqrt{y}$$

from (2),

$$c = \sqrt{c} (\sqrt{x} + \sqrt{y})$$

$$\sqrt{c} = \sqrt{x} + \sqrt{y}$$

is req. envelope.

(ii) Given

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{--- (1)}$$

$$a^n + b^n = c^n$$

--- (2)

Taking differentials w.r.t (1) & (2)

$$\text{①} \Rightarrow -\frac{x}{a^2} da - \frac{y}{b^2} db = 0$$

$$\frac{x}{a^2} da + \frac{y}{b^2} db = 0$$

$$\text{②} \Rightarrow n a^{n-1} da + n b^{n-1} db = 0$$

$$a^{n-1} da + b^{n-1} db = 0$$

Comparing

$$\frac{x/a^2}{a^{n-1}} = \frac{y/b^2}{b^{n-1}} = \alpha (\text{say})$$

$$\frac{x/a^2}{a^{n-1}} = \alpha, \quad \frac{y/b^2}{b^{n-1}} = \alpha$$

$$\frac{x/a}{a^n} = \alpha, \quad \frac{y/b}{b^n} = \alpha$$

$$\frac{x}{a} = \alpha a^n, \quad \frac{y}{b} = \alpha b^n \quad \text{--- (3)}$$

Adding

$$\frac{x}{a} + \frac{y}{b} = \alpha a^n + \alpha b^n$$

$$\frac{x}{a} + \frac{y}{b} = \alpha(a^n + b^n)$$

from (1) & (2)

$$1 = \alpha \cdot c^n$$

$$\frac{1}{c^n} = \alpha$$

(3)  $\Rightarrow$

$$\frac{x}{a} = a^n \cdot \frac{1}{c^n}, \quad \frac{y}{b} = b^n \cdot \frac{1}{c^n}$$

$$a \cdot a^n = c^n x,$$

$$b \cdot b^n = c^n y$$

$$a^{n+1} = c^n x$$

$$b^{n+1} = c^n y$$

$$a = c^{\frac{n}{n+1}} x^{\frac{1}{n+1}},$$

$$b = c^{\frac{n}{n+1}} y^{\frac{1}{n+1}}$$

$$a^n = c^{\frac{n^2}{n+1}} x^{\frac{n}{n+1}},$$

$$b^n = c^{\frac{n^2}{n+1}} y^{\frac{n}{n+1}}$$

Adding

$$a^n + b^n = c^{\frac{n^2}{n+1}} x^{\frac{n}{n+1}} + c^{\frac{n^2}{n+1}} y^{\frac{n}{n+1}}$$

$$a^n + b^n = c^{\frac{n^2}{n+1}} (x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}})$$

from (1)

$$c^n = c^{\frac{n^2}{n+1}} (x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}})$$

$$c^{\frac{n}{n+1}} = x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}}$$

is req. Eq of envelope.